

Faculty of Medicine
Artificial Intelligence in Medicine

Master of Science Thesis

Benchmarking Decoders for Directional Intent Control of a Mobile Gait Trainer Using a Dry-Electrode EEG-Based BCI

by

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Biel, September 2026

Abstract

Rehabilitation after spinal cord injury or stroke depends on repeated, active engagement of the brain’s sensorimotor networks and their remaining connections to the spinal motor neurons. Robot-assisted training is standard clinical practice, but often kinematically constrained and unable to reproduce natural movement variability. Decoding a user’s motor intentions offers a patient-driven alternative: a controlled setting where walking is practised under the patient’s own volition. This thesis investigates a non-invasive brain–computer interface, based on a dry-electrode electroencephalography (EEG) headset, that translates decoded intentions into physical assistance through a mobile, gravity-supported gait trainer. By letting users explore their range of motion without imposed patterns, the framework aims to engage the neural pathways underlying activity-dependent neuroplasticity.

Nine decoders were evaluated on a five-class directional movement task, using the DSI-24 EEG headset and trained on a healthy participant who physically executed the cued movements. The architectures comprise a classical machine learning pipeline, four networks trained from scratch, three pretrained EEG foundation models, and a non-physiological control. Decoders were compared under matched tuning constraints: signals were segmented into two-second windows at a half-second stride, and each one received its strongest configuration, including the pretrained models’ native preprocessing. Performance was assessed under three protocols: a within-subject ceiling, a leave-recordings-out deployment scenario, and a cross-person transfer test.

On the leave-recordings-out protocol, TimesNet reached the highest balanced accuracy, $85.5\% \pm 2.1\%$ against the 20% five-class chance level. It nevertheless fell within 2.7 points of five other decoders, including a classical spatial-filtering pipeline and a compact convolutional network. A Friedman test across the nine models was significant, but after correction for multiple comparisons no pairwise difference remained statistically distinguishable. Under favourable adaptation, the best foundation model matched rather than exceeded this group. The non-physiological control reached 39.3% and, on a binary direction metric, only 30.2%, below the 50% chance level that every EEG decoder exceeded. Cross-person transfer remained near chance for all architectures.

These findings establish the feasibility of directional decoding from dry electrodes, given per-participant calibration: every model’s inference runs over tenfold faster than the half-second decode cadence, and TimesNet’s confidence gate lifts retained-window balanced accuracy to 94.1%. The results refine two expectations. First, EEG pretraining achieved parity with, rather than superiority over, the top-performing decoders. Second, peak accuracy did not predict data efficiency: the compact EEGNet, not the highest-scoring model, reached a usable threshold first when calibration data were scarce.

Acknowledgements

This thesis marks the conclusion of my Master of Science in Artificial Intelligence in Medicine at the University of Bern, in collaboration with the ARTORG Center for Biomedical Engineering Research. The research itself was conducted at the neuro-rehaLab, the Laboratory for Neurorehabilitation Engineering of the Institute for Microtechnologies and Medical Engineering at the Bern University of Applied Sciences in Biel. I am deeply grateful to everyone who made this work possible.

First, I would like to thank Prof. Dr. Juan Fang for her ongoing support in the lab. Our regular meetings were a valuable opportunity to present my progress, discuss the project from a clinical perspective, and seek her expert feedback and recommendations on the design of rehabilitation technologies. She provided the main training computer with its RTX 4080 GPU, allowed me to use the lab's dry EEG headset, and prepared both the mini-juAndago and the main juAndago system that she and her students have been developing, ensuring that it was ready for my use within this study.

I thank Prof. Dr. Stavroula Mougiakakou, my supervisor at the University of Bern, for making this project possible by connecting me with her postdoctoral researcher, Dr. Lorenzo Brigato, and for once again taking over the supervision for the final phase of the thesis as he leaves the university.

I owe a profound debt of gratitude to Dr. Brigato for his extensive contributions to the technical core of this project. His guidance is woven throughout the experiments, the protocols, the benchmark design, and the entire artificial intelligence side of this thesis. It was his pivotal suggestion to shift the project from building a new decoder to systematically benchmarking existing models, which gave this research its direction. Furthermore, the careful distinction made throughout this report between what was measured and what can be claimed is a direct testament to the scientific rigour I learned from him.

I also want to thank the neuro-rehaLab for providing the workspace and practical support essential for data collection and my months of work on this master's thesis. A special thank you goes to the second participant, who generously volunteered their time and made the cross-person analysis possible. Without them, a key research question would have remained unanswered.

Finally, I thank my family and friends for their endless patience and encouragement throughout this journey, especially during the demanding writing process.

Declaration of Originality

Version 2 (with use of AI)

Ich erkläre hiermit, dass ich diese Arbeit selbstständig verfasst und keine anderen als die angegebenen Quellen benutzt habe. Alle Stellen, die wörtlich oder sinngemäss aus Quellen entnommen wurden, habe ich als solche gekennzeichnet. Als Hilfsmittel habe ich Künstliche Intelligenz verwendet. Sämtliche Elemente, die ich von einer Künstlichen Intelligenz übernommen habe, werden als solche deklariert und es finden sich die genaue Bezeichnung der verwendeten Technologie sowie die Angabe der «Prompts», die ich dafür eingesetzt habe. Mir ist bekannt, dass andernfalls die Arbeit mit der Note 1 bewertet wird bzw. der Senat gemäss Artikel 36 Absatz 1 Buchstabe r des Gesetzes vom 5. September 1996 über die Universität zum Entzug des auf Grund dieser Arbeit verliehenen Titels berechtigt ist. Für die Zwecke der Begutachtung und der Überprüfung der Einhaltung der Selbständigkeitserklärung bzw. der Reglemente betreffend Plagiate erteile ich der Universität Bern das Recht, die dazu erforderlichen Personendaten zu bearbeiten und Nutzungshandlungen vorzunehmen, insbesondere die schriftliche Arbeit zu vervielfältigen und dauerhaft in einer Datenbank zu speichern sowie diese zur Überprüfung von Arbeiten Dritter zu verwenden oder hierzu zur Verfügung zu stellen.

The detailed disclosure of artificial intelligence use that this declaration requires follows as an appendix.

Biel, 1 September 2026

Ludovico Grabau

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List of Abbreviations

AI	artificial intelligence
BCI	brain-computer interface
CPU	central processing unit
CSP	common spatial patterns
CUDA	Compute Unified Device Architecture (NVIDIA's GPU computing platform)
DSI-24	Dry Sensor Interface 24, the dry-electrode headset (Wearable Sensing)
EDF	European Data Format
EEG	electroencephalography
ERP	event-related potential
FBCSP	filter-bank common spatial pattern
FM	foundation model
fp16	half-precision (16-bit) floating point
fp32	single-precision (32-bit) floating point
GPU	graphics processing unit
HPO	hyperparameter optimisation
ICA	independent component analysis
LDA	linear discriminant analysis
LoRA	low-rank adaptation
MI	motor imagery
MNE	MNE-Python, software for EEG analysis (named for minimum-norm estimation)
PEFT	parameter-efficient fine-tuning (also the library of that name)
ROC-AUC	area under the receiver operating characteristic curve
SCI	spinal cord injury
SD	standard deviation
SHA-256	Secure Hash Algorithm, 256-bit
WSL2	Windows Subsystem for Linux

Chapter 1

Introduction

1.1 Motivation

Recovery of walking after spinal cord injury (SCI) or stroke depends on repeated, intention-driven practice [63, 67]. A rehabilitation robot, whether a wearable lower-limb exoskeleton or a mobile, gravity-supported gait trainer [24], can supply the motion, but the therapeutic value of that movement rests on whether the patient initiates it. A device driven by a fixed schedule moves the legs, whereas one that starts when the patient intends to walk engages the motor system that rehabilitation is trying to retrain. Coupling the device to the patient’s intention requires reading it directly from them, and doing so quickly enough that the machine responds in time.

Electroencephalography (EEG), the recording of the brain’s electrical activity from electrodes on the scalp, is the only non-invasive modality both fast enough to drive a device in real time and inexpensive to place in a clinic. Its cost, however, is signal quality. A gel-based research montage is impractical for routine clinical use: it takes nearly an hour to apply, requires a trained operator, and leaves the patient’s scalp to be washed afterwards. A dry electrode headset removes all three obstacles and can be fitted in minutes, at the price of higher electrode impedance, greater susceptibility to motion, and a lower signal-to-noise ratio. Whether a usable decoder can be built on top of that trade-off is an empirical question, and a clinical deployment depends on its answer.

The control problem is also more demanding than the one the brain–computer interface (BCI) literature usually addresses. Steering a gait trainer requires more than a binary go/stop signal: it needs a direction. This work therefore targets a five-class decoding problem (Idle, Forward, Backward, Left and Right), recorded from a 19-channel dry-electrode headset [73] while the participant physically attempts to perform each movement.

The physical attempt is the decisive difference from the motor-imagery paradigm. The participant generates a genuine, natural motor intent, activating the necessary neural pathways to accomplish the movement based on their residual body perception, rather than merely imagining it. This intent remains authentic regardless of whether physical execution succeeds despite motor impairment. Therefore, this is movement activation, not motor imagery (MI), and the resulting cortical activity differs accordingly. The training data for this work was recorded from a healthy participant. While evaluating the translation of these models to an injured demographic remains future work, the underlying hypothesis is that a natural, automatic attempt to move will elicit cortical patterns analogous to those observed during successful execution in a healthy candidate. Given this fundamental difference in cortical activity, it follows that the numbers reported in this thesis are not comparable with the usual MI benchmarks that dominate the field, of which BCI Competition IV is the canonical example [68], and no such comparison is made. It also follows that no public dataset supplies a suitable benchmark. The standard corpora cover predominantly hand-based imagery [29], not full-body directional gait intent, and public EEG recordings made during exoskeleton use are themselves scarce [54]. The data underlying this work was therefore recorded specifically for it.

1.2 Aims of this work

This work has two aims. The first is to benchmark the candidate decoder families as classifiers of this clinical task under the conditions of a deployment: whole recordings held out of training, calibration data limited to what one patient can supply, an inference latency inside the control cadence of the device, and confidence estimates that a safety mechanism can act on. The second is to establish the preliminary feasibility of the directional-intent decoding stage of the system built for this work, a prototype of one intended for clinical use with the mobile gait trainer. The evidence for that aim is offline: the decoding stage was evaluated on recordings from one healthy participant, and the live system was exercised in unrecorded sessions rather than measured, so the feasibility established here is that of the real-time decoding stage, and the clinical system around it is left to be validated in later work.

The second aim is met when the decoding stage satisfies four criteria. It resolves direction within a movement category above the two-way floor, the 50 % that a decoder which detects movement but guesses its direction would score. It reaches an accuracy well above the chance level, the 20 % that a decoder guessing among five classes would score, under a leave-recordings-out protocol, in which every test window (the two-second segment of signal that the model classifies as one unit) comes from a recording it never trained on. It classifies each window inside the half-second cadence at which the device requests its next command. Its confidence estimates must be trustworthy enough to support a gate that withholds any command scoring below a set threshold and leaves the device idle, so that coverage can be traded for accuracy. The first two criteria are answered under RQ0 and by the benchmark itself, the last two under RQ4 (Section 1.4), and RQ1 to RQ3 select the decoder to which the criteria are applied.

Three of the decoders compared here are general time-series architectures drawn from the Time Series Library [74, 71], through the benchmark code of a re-evaluation study [13] that ranked eight forecasting models, two of these three among them, by prediction error across 14 datasets and found that none consistently exceeds the baselines (Section 2.5). This thesis ranks decoders on the axis that determines a deployment decision: accuracy on recordings held out of training, the calibration data it needs before becoming usable, its latency against the device cadence, and the reliability of its confidence for gating, measured for a single clinical device, one dry montage and executed gait. Neither benchmark ranks on that axis. Both aims are pursued on the dataset recorded for this work.

1.3 Problem statement

Given that data, the engineering question becomes one of model selection: which family of decoder should drive the device?

Three answers are in contention. The first is classical: bandpower features, the signal energy within set frequency ranges, extracted by spatial filtering and classified linearly, in the tradition of the filter-bank common spatial pattern (FBCSP) method [4]. The second is a compact convolutional network built specifically for EEG, of which EEGNet [42] is the standard representative. The third is the transfer of large pretrained models into the domain. It is the answer the benchmark was built to test.

That third answer has two distinct forms, and they are readily conflated. One form borrows *architectures* from general time-series modelling. TimesNet [74] was designed for time-series analysis in general, with classification among the tasks it was built and benchmarked for, and iTransformer [44] and S-Mamba [72] for long-horizon forecasting. All three are used here to classify EEG windows. These models are trained from scratch on the EEG recordings and carry no pretrained weights. What transfers is the architectural *inductive bias*: the assumptions about the signal that an architecture builds in before it sees any data. The benchmark code that supplies them was assembled for a long-term forecasting study [13], and their use for classification is an adaptation.

The other form borrows *weights*. Neuro-GPT [17], CBraMod [70] and LaBraM [35] are EEG foundation models (FMs) pretrained on thousands of hours of clinical recordings, offered on the expectation that such pretraining supplies representations a small dataset cannot learn unaided. Chronos completes the set as a deliberate negative control. It is a forecasting model pretrained on series from domains such as energy, finance, transport and weather, and on no physiological data at all [5], so it establishes what a large pretrained sequence model contributes when its corpus is irrelevant to the signal.

The expectation attached to the third answer is that pretraining should help most exactly where this application sits: a single participant, a few hundred recordings, and a decoder that must be recalibrated per patient. Testing that expectation is the first aim of this thesis, and testing it fairly is the harder requirement. A pretrained model evaluated under preprocessing it never saw, with its encoder frozen, will fail for reasons that have nothing to do with the question being asked. The benchmark therefore evaluates every model in its own optimal configuration: FMs under their native preprocessing and with their encoders fully fine-tuned, meaning that every pretrained weight is updated on the recorded data rather than held frozen. The split machinery (the rule that assigns each recording to the training or the test portion), the fold count (the number of times that assignment is redrawn with a different portion held out), the seeds (the fixed random starting points that make a run repeatable), the metrics and the training budget are held identical across all nine. There is a single disclosed exception, stated and justified in Section 4.5: the control’s training on the deployment protocols was capped.

Under those conditions no large pretrained model is needed for a place in the leading group: decoders that learn from the recordings alone already occupy it, and optimally adapted EEG pretraining places the best of the FMs in that group and not above it. On the deployment protocol the best of them reaches 83.9% balanced accuracy (the mean of the per-class accuracies, so that no class dominates the score) against a chance level of 20%, while TimesNet (trained from scratch and borrowed from general time-series modelling) attains the highest mean at 85.5%. The FBCSP baseline, with no deep learning and no hyperparameter optimisation (HPO), reaches 82.8%, within three points of TimesNet and level with the weakest network in the leading group.

The control pretrained on non-physiological data falls more than 40 points below the leader, at 39.3%, a figure carrying the capped budget noted above. Under the full search and budget of Protocol A the same control reaches 51.2%, still more than 37 points behind the leader there, so the conclusion does not rest on the cap.

With nine models over five folds a Friedman test rejects the hypothesis that the decoders are equivalent, but no pairwise comparison survives correction for multiple testing at this sample size. TimesNet has the highest mean among a group of decoders separated by a few points at most, but it does not statistically separate from any of them: no clear winner emerges. Pretraining admits the best of the FMs to that leading group and no further.

Two further limits frame every claim made here. All results are within-subject: a decoder trained on one participant performs near chance when applied to another, meaning calibration data must be collected from each new user. The participant is also healthy, so generalisation to the patient population the gait trainer is ultimately intended for is not established by this work and is not claimed.

1.4 Research questions

The problem above decomposes into one precondition and four questions.

RQ0 Does the benchmark decode directional intent, or only the presence of movement?

Overall accuracy cannot answer this, because a decoder that identifies the movement *category* without resolving direction within it already scores well above chance. Unless direction is genuinely recovered, no ranking of decoders means anything, so this question is logically prior to the four that follow.

RQ1 Do EEG FMs outperform decoders trained from scratch?

Evaluated on the deployment protocol, with each model in its own best configuration.

RQ2 Does EEG-specific pretraining matter, set against its general time-series equivalent?

Answered against a pretrained forecasting model whose corpus contains no physiological data, which bounds from below what pretraining contributes when its domain is irrelevant.

RQ3 Which paradigm best handles the low-data patient-calibration regime?

Clinically this is the binding constraint: a decoder that needs many calibration trials before it becomes usable will not be used in a clinic.

RQ4 Can the best decoder drive safe real-time control of the gait trainer?

This question encompasses latency, memory, and the reliability of the model's stated certainty, because live operation demands more than mere accuracy. The confidence gate depends directly on this final factor, since an activation threshold is only effective if the decoder's underlying probability scores are well-calibrated.

1.5 Contributions

This thesis makes the following contributions.

1. **A five-class directional movement-intent dataset** recorded on a dry-electrode headset under movement execution, for which no public equivalent exists, together with the acquisition system built to record it.
2. **A controlled comparison of nine decoders** spanning classical spatial filtering, purpose-built EEG convolution, general time-series architectures and pretrained EEG FMs, under a stated comparison contract that grants each model its own optimal configuration while holding all evaluation machinery fixed.
3. **A controlled result on EEG FMs** for this task: pretraining admits the best of them to the leading group and places none ahead of it, established under conditions designed to give pretraining its best case rather than its worst.
4. **A metric that separates direction decoding from movement detection**, applied to every headline result so that decoding claims cannot be satisfied by category detection alone.
5. **A deployment analysis** covering latency, memory, the reliability of the confidence estimates and the behaviour of the gate itself. It identifies a tension between the decoder that scores highest and the one that becomes usable on the least calibration data, and resolves it in favour of accuracy as the governing criterion for a clinical deployment.
6. **Methodological findings on benchmark construction**, including the effect of a grouping variable confounded with the class label, two model-selection errors arising from single-fold evaluation that point in opposite directions, a set of design constraints that outlived the artefact they were introduced to counter, and the consequences of a five-fold design for statistical inference.

1.6 Outline of this document

Chapter 2 reviews EEG-based BCIs for lower-limb rehabilitation, the distinction between movement execution and MI, what a dry-electrode montage costs, and the three decoder families under comparison. It then examines what transfers when a general time-series architecture is repurposed for classification, what large-scale EEG pretraining is expected to provide, and evaluation practice in studies of this size. Chapter 3 describes the acquisition system, the recording protocol, the dataset, the signal conditioning and the command pipeline from decoder output to platform, including the mechanisms that make it safe to operate. Chapter 4 states the comparison contract, the three evaluation protocols, the HPO, the metric panel and the statistical treatment. Chapter 5 reports the results, organised by finding, with the research question each one answers stated in place. Chapter 6 interprets them, proposes an explanation for the pattern observed across architectures, draws out what the exercise implies for benchmark construction, and states the limitations. Chapter 7 concludes.

Appendix A discloses the use of artificial intelligence (AI) in the production of this work, as the declaration of originality requires. Appendix B contains the model cards, the data card and the provenance record. Appendix C contains the Protocol-C regime breakdown, the per-class tables and the supporting studies. Appendix D documents how every reported number can be traced to an archived artefact.

Chapter 2

Background and Related Work

This chapter establishes the clinical rationale for decoded movement intent and details the evaluation framework used to compare models. This focus on methodology is deliberate: because a central finding of this benchmark is the lack of statistical separation among the decoders of the leading group, the statistical threshold for a meaningful advantage must be clearly defined in advance.

2.1 EEG-based brain–computer interfaces for lower-limb rehabilitation

Who regains walking after SCI is neither uniform nor well predicted by the initial injury alone. The factors involved are numerous enough to have their own dedicated and comprehensive reviews [63]. These studies and stroke rehabilitation research share one mechanism: recovery depends on neural plasticity, and activity is what drives it [67]. This is the premise on which intention-driven rehabilitation rests, and it sharpens the contrast drawn in Section 1.1. Movement supplied on a fixed schedule is only movement, whereas the same motion produced when the patient intends it is contingent on their own motor command, and the plasticity argument requires that dependence, not the motion.

Systems of that kind have been surveyed as a class. A recent review of adaptive closed-loop BCIs in neurorehabilitation describes the shared pattern: decode the user’s intention, act on it, and return feedback that reinforces the attempt. It finds that pattern across applications from functional electrical stimulation to robotic orthoses, and reports improvements in motor function in stroke and SCI [36]. The same review is explicit about where the evidence is sparsest: research into brain–computer control of *lower-limb* movement after spinal cord injury it describes as scarce, against a growing interest in pairing such interfaces with wearable exoskeletons. That is the gap this thesis works in.

Two lines of work have pursued the idea for the lower limb directly. Ambulatory exoskeletons have been driven from a brain–machine interface in SCI gait rehabilitation [46], and wearable exoskeletons have been given volition-adaptive control and tested with individuals with incomplete spinal cord injury [61]. Both establish that a pipeline of this kind can be built. Neither settles which decoder should drive it, which is the question this thesis takes up.

Parallel work in the laboratory approaches lower-limb rehabilitation from a different angle: neural interlimb coupling. Volker Dietz and Miriam Schrafl-Altermatt define it as a flexible, task-specific linkage of limbs during complex movement, and demonstrate it experimentally through reflex responses that pass between the limbs [22]. To study this, an arm-and-leg cycling platform was built, motivated by the observation that bilateral arm swing is an integral part of bipedal locomotion and a residual expression of the quadrupedal coordination retained in human walking [21, 25].

The clinical criticism raised by this work bears on how a decoded-intent system should be built, not only on whether it can be. Newer assisted rehabilitation devices tend to direct the patient’s attention to a screen displaying exercise goals, so the brain receives no visual feedback about the impaired limbs themselves. A system that decodes intent and acts on it immediately restores a feedback path that a goal-display does not.

What non-invasive scalp recording gives up is clearest at the other end of the scale. Decoding cortical activity to text has reached word error rates of around 3% with electrocor-

ticography: subdural grids of roughly 250 electrodes, placed for clinical seizure monitoring, decoding sentences drawn from a closed set of a few dozen [48]. That regime is invasive, surgically constrained, limited to a small vocabulary, and open only to patients who already require the surgery. This thesis takes the opposite position: 19 dry electrodes, no gel, no surgery, and a montage fitted in minutes. The performance that is reasonable to expect follows from this position, not from the state of the art in decoding generally.

2.2 Movement execution versus motor imagery

The participant in this study physically performed the movements. That single fact places the work outside the paradigm most BCI benchmarks occupy, and it has consequences that run through every comparison made later. Specifically, when comparing execution to imagery, two distinct classes of neural phenomena are involved: the modulation of sensorimotor rhythms, and the *Bereitschaftspotential*.

The first of these phenomena, the modulation of sensorimotor rhythms, is shared. Event-related desynchronisation and synchronisation occur for both executed and imagined movement [57], and they are the signals most MI interfaces are built upon. A scalp activity recording carries rhythms at many frequencies simultaneously, conventionally summarised by the spectral energy concentrated within five ranges referred to as frequency bands: delta below 4 Hz, theta from 4 to 8 Hz, alpha from 8 to 13 Hz, beta from 13 to 30 Hz, and gamma above that. While these boundaries are historical convention rather than derivations from first principles, the division remains essential because the ranges are not interchangeable: activity in each arises over distinct parts of the scalp and responds to different events. At rest, large populations of neurons over the sensorimotor cortex oscillate in step with one another, and that synchrony is what makes their rhythms visible at the scalp: the μ component over the motor cortex, which occupies the alpha range but is distinct from the activity seen over the back of the head, and the β band above it. Movement breaks the synchrony up, so the amplitude of those rhythms falls as a movement is prepared and carried out and rises again once it is over. Desynchronisation and synchronisation are those amplitude changes, time-locked but not phase-locked to an event.

The second phenomenon, the *Bereitschaftspotential*, separates the two paradigms. This slow negative potential precedes voluntary movement and reflects its preparation and execution [65]. An executed-movement recording therefore contains a component absent from imagery, giving its decoder access to evidence the imagined case cannot supply.

Beyond cortical differences, an executed-movement recording also contains the electromyographic consequences of the movement itself, along with motion artefact. The electrical activity of the muscles executing the movement reaches the scalp electrodes far more strongly than the brain does. Chapter 3 addresses this extensively.

Existing research on the executed case is less extensive than on the imagined one. Prior work in this laboratory, using the same dry headset, observes that microstate analysis (the segmentation of the scalp's electric field into brief periods of quasi-stable topography) is rarely applied in the literature to EEG recorded during movement [25]. That observation is about topographic analysis specifically, not about decoding in general, but the underlying reason applies to both: recording during movement is avoided because the motion contaminates the trace. Topographic analyses have nonetheless been reported to carry subject-specific correlates of motor control [58], which is evidence that the executed case remains tractable.

What is not available, and what this section would otherwise be built upon, is a controlled head-to-head comparison of execution against imagery for the same decoders on the same task. No such study is held in the material assembled for this thesis, and the absence is stated instead of being filled with a weaker one. The nearest evidence is indirect but relevant. EEGNet was validated on four paradigms, including both an executed-movement dataset (self-paced finger taps, with epochs taken around each onset) and an imagined four-class sensorimotor rhythm dataset [42]. The model rankings differ between them: on

the executed dataset both EEGNet variants outperformed every comparison model by a significant margin, whereas on the imagined dataset EEGNet was comparable to a shallow convolutional network and to a classical filter-bank pipeline. The same architecture and preprocessing chain therefore rank the models differently under the two paradigms.

The consequence for this thesis is a constraint on comparison. Balanced accuracies reported here are not comparable with published hand MI benchmarks [68, 29], and no claim in Chapter 5 or Chapter 6 places them side by side. No fair comparison figure exists.

2.3 Dry-electrode acquisition and its constraints

A dry-electrode headset trades signal quality for the time and expertise needed to apply it. This trade-off has been quantified directly: wireless dry headsets have been set against wired wet-electrode systems, both for clinical applications [30] and in a controlled comparison of the two acquisition chains [37]. The consistent picture is that dry systems yield usable but noisier recordings, with markedly higher electrode-skin impedance and greater susceptibility to movement and contact artefact. In exchange, setup takes minutes instead of nearly an hour and requires no gel, no abrasion and no subsequent hair washing. For a rehabilitation device that a patient must tolerate repeatedly, this exchange is the condition of the device being used at all.

The size of the gap has been measured. Kam and colleagues recorded 27 participants with both a wireless dry system and a wired wet one, on different days, and compared the two on resting spectral power, event-related potential (ERP) amplitude and single-trial classification [37]. Their measure of agreement is the rank correlation across participants, which runs from -1 to 1 : a value of 1 means the two systems order the participants identically on a measure, 0 means the two orderings are unrelated, and values above 0.5 count as a strong association by the usual convention [16]. The observed correlations ran from 0.54 to 0.89 . The right reference for those numbers is not 1 , because no EEG measurement reproduces itself exactly: published values for repeating the same wet measurement on another occasion lie between about 0.7 and 0.85 . The upper half of the dry-against-wet range reaches that level, so on the best-agreeing measures the dry system is as consistent with the wet one as that montage is with itself. The lower half does not: a correlation of 0.54 is a strong association but clearly short of repeat reliability, so on those measures the dry montage departs from the wet one by more than that system's own repeat noise. In a clinical population, Hinrichs and colleagues found no significant difference in resting alpha and beta power and comparable ERP latencies and amplitudes between the two montages, with a modest excess of theta and delta power on the dry one and a clear preference of the participants for it [30].

The specific headset used here has an independent track record in the same laboratory. Prior work recorded 19 channels on the same dry system, positioned to the 10–20 standard, keeping electrode-skin impedance below one megohm [25]. This threshold is a thousandfold above what a gel montage would target, a figure that makes the scale of the trade-off concrete. This same work is also instructive regarding data quality on this hardware. Four of the 14 recruited participants were excluded for insufficient signal quality. The study also relied on automated component labelling as a running indicator of how much of each recording was of neural origin, a diagnostic it describes as especially relevant when recording during movement.

One methodological caution follows. The most systematic evidence on which preprocessing steps help comes from clean recordings made under laboratory conditions, and its author says the conclusions may not carry over to noisier data [18]. Dry electrodes worn during executed gait are that noisier case. Section 3.6 applies the evidence anyway, as the best available guide, and states the caveat where it does so.

2.4 Decoder families: classical, purpose-built EEG and general time-series models

The nine decoders compared in this thesis fall into families that differ in what they assume about the signal before processing it, making the comparison more insightful than a simple list of accuracies.

The classical family is represented by the FBCSP pipeline named in Section 1.3 [4]. In its original formulation, it decomposes the signal into nine sub-bands spanning 4 to 40 Hz, learns spatial filters that maximise variance ratios between classes within each band, ranks the resulting log-variance features by their mutual information with the label, and classifies the surviving features with a naive-Bayes Parzen-window model [3]. Every assumption is explicit, and every stage is inspectable. Section 4.2 details three specific areas where this project’s implementation departs from that description.

This pipeline is included not merely as a historical courtesy. FBCSP entries achieved the best mean κ of any submission on both BCI Competition IV datasets, scoring roughly 0.57 on the four-class 2a and 0.60 on the two-class 2b [3]. The method remains the standard reference that new proposals are expected to beat. For example, a 2023 spatial-filter method frames its own advance as achieving 74.6 % against FBCSP’s 63.2 % on dataset 2a, and 81.2 % against 74.1 % on 2b [76]. A pipeline with strong, correct priors and a competitive record is the appropriate control for architectures that claim to learn those priors from data.

One classical family is nevertheless absent. Riemannian methods represent each window by its spatial covariance matrix and classify those matrices on the curved manifold they occupy, either by distance to a class mean or by first projecting them into a flat tangent space. This latter approach raised mean accuracy over common spatial patterns (CSP) with linear discriminant analysis (LDA) from 65.1 to 70.2 % on the same competition dataset [6]. No Riemannian decoder was evaluated here, meaning the classical category of this benchmark relies on a single method where it could have included two. Section 6.6 discusses the consequence of this, and Section 7.2 treats it as one of the most direct experiments left undone.

The purpose-built family is represented by EEGNet [42], the small EEG-specific convolutional network named in Section 1.3, whose design encodes similar reasoning in a learned form. A temporal convolution is applied first, with a kernel length set to half the sampling rate so that the filters can resolve frequencies down to two hertz. A depthwise convolution across channels then learns a spatial filter for each temporal one separately.

The authors describe the result as interpretable in exactly these terms: the temporal stage acts as a bank of narrow-band frequency filters, and the depthwise stage as their band-specific spatial counterpart. They also explicitly note their debt to the filter-bank pipeline. The kernel length serves as a resolution floor rather than a band target, and the original paper indeed halves it for a dataset that was high-passed at four hertz. The entire network contains only a few thousand parameters, making it two orders of magnitude smaller than the convolutional alternatives against which it was compared.

The general time-series family contributes three architectures: two built for long-horizon forecasting, and TimesNet, designed for time-series analysis in general with classification among its benchmark tasks. TimesNet [74] selects dominant periods via Fourier transform, folds the one-dimensional series into a two-dimensional tensor whose axes are position-within-period and period-index, and then convolves over that structure. Periodicity is therefore the first assumption it commits to. iTransformer [44] inverts the usual arrangement, treating each variate (one recorded channel) as a single token whose embedding is its entire series, allowing attention to operate across channels rather than over time. S-Mamba [72] adopts this same inverted embedding but uses a state-space sequence model in place of attention. This recurrent formulation carries a compact hidden state through the sequence instead of comparing all positions pairwise. All three serve here as classifiers, not forecasters, and learn from the recordings alone. The architectural prior transfers, but the learned weights do not. Their model bodies are sourced from a bundled benchmark

library [13, 71].

These decoder families are evaluated side by side because the field already acknowledges a strong interplay between network design and EEG signal types. For instance, the authors of EEGNet observe that architectures tailored to one feature type, such as shallow convolutional networks designed for oscillatory signals, may underperform on tasks involving ERPs, which require decoding time-locked neural voltage peaks rather than ongoing rhythms. By contrast, deeper, general-purpose models offer greater flexibility and are not restricted to a specific feature type [42]. Section 6.3 introduces a hypothesis grounded in this established framework to interpret the findings of this study.

2.5 General time-series architectures repurposed for classification

Borrowing an architecture across tasks assumes that what made it effective for the first task will carry to the second. There is direct evidence that this assumption fails for the specific transition made here.

The clearest evidence comes from the TimesNet paper itself. In a 10-dataset classification comparison, TimesNet achieved the best average accuracy at 73.6%. This placed it just ahead of a strong classical method at 72.5% and a competing deep model at 73.0%, representing a spread of about one percentage point across the top three. In the same table, DLinear [78], a model that performs respectably at forecasting, falls to 67.5%. The authors attribute this drop to its single temporal projection, which is adequate for autoregressive tasks but degenerates badly when higher-level representations are required [74]. Forecasting rank does not predict the classification order, and such a leaderboard is therefore no evidence about performance on that task. The library’s own multi-task leaderboard shows the same separation from the other side: as of March 2024 it places TimesNet first on classification, imputation, anomaly detection and short-term forecasting, and outside the top three when the horizon is long [75].

Recent forecasting benchmarks question their own rankings. A controlled re-evaluation of eight long-term forecasting models across 14 datasets trained roughly 5 000 networks for its hyperparameter searches and evaluated the selected configurations over three seeds. This study concluded that no model consistently outperforms the baselines, and demonstrated the point statistically. A ninth model that the authors built as a proof of concept, iPatch, obtained the best average rank and still did not separate statistically from the eight under a Friedman test on mean absolute error, returning a p -value of 0.31 across the 14 datasets [13]. By resampling the choice of datasets and horizons 5 000 times, the authors found that the winning model depends heavily on which datasets are included. They summarised the situation by noting that experimental variance dominates over architectural advancement.

Two details of that study matter here. First, it notes that no existing forecasting benchmark includes high-frequency data such as EEG or wearable-sensor recordings. Its own selection is the partial exception: two of the 14 datasets, MotorImagery and TDBrain, carry brain activity and sit in the health domain, and the models separate most clearly on the former, so the ranking depends on whether it is included. Both datasets are forecast rather than classified, which leaves the gap this thesis addresses: predicting the next samples of a brain signal is not decoding an intention from it. Second, its conclusion is narrower than it initially sounds: a targeted paired comparison over the same 14 datasets did achieve significance under a sign test. The claim is that aggregate rankings over heterogeneous datasets fail to separate models, not that they can never be separated. Section 4.9 treats this thesis’s own lack of separation in the same way. This study evaluates eight models, and TimesNet is not among them, even though its repository supplies the model body used here.

Opposing evidence exists. Large-scale pretraining on time series does transfer successfully within its own domain. For instance, a generative pretrained model trained on a corpus of roughly one billion time points matched from-scratch performance using only 1 to 5% of the downstream training samples across four separate datasets. It then exceeded

from-scratch performance when all samples were available [45]. Similarly, on a 23-dataset benchmark spanning seven domains and 10 sampling frequencies, evaluated over 97 configurations, FMs outperformed both statistical and deep learning baselines across most areas [2]. Based on this evidence, any expectation that a pretrained time-series model would perform well here was reasonable.

The same 23-dataset benchmark reports that task-trained deep models beat FMs specifically at higher sampling frequencies, as well as on series with high entropy and little trend. Chronos, the pretrained decoder used in this thesis as a non-physiological control, is a univariate forecasting model. Its authors are explicit about its boundaries: their evaluation contains too little data sampled at 15-minute resolution or faster, and transfer to tasks beyond forecasting is offered as a hypothesis rather than a demonstration [5].

The most direct comparison available reinforces this caution. On physiological signals, under a frozen-encoder protocol like the frozen-probe condition this benchmark also ran, an encoder trained from scratch beat a pretrained one on a single task and lost narrowly on another, leaving no consistent winner in either direction. Meanwhile, a language model carrying an incorrect prior collapsed to single-digit accuracy [41]. Pretraining on time series in general is not pretraining on this particular class of them, and Section 6.2 examines that distinction.

2.6 EEG foundation models and the evidence for pretraining

The proposition behind an EEG FM is that self-supervised pretraining on large corpora of unlabelled recordings yields representations that transfer to downstream tasks with little labelled data: precisely the situation of a single-participant calibration session. Three such models are compared here: LaBraM [35], Neuro-GPT [17] and CBraMod [70]. These models were pretrained on corpora of different sizes: roughly 2 500, 5 700 and over 9 000 hours, the last retained from a 27 000-hour clinical archive [35, 17, 70].

A recent review of 14 such models is the most useful single source on how well this proposition has held, and its findings are more equivocal than the field’s own summaries. It reports that many studies find linear probing performs substantially worse than full fine-tuning. In a linear probing setup, the pretrained encoder is kept frozen while only a linear classifier is trained on its outputs, meaning that whatever is decoded must already be present in the representation. The review draws a direct inference from this performance gap: it raises questions about the intrinsic quality of the representations that current self-supervised strategies learn [40]. If the adaptation rather than the pretraining is doing the work, then its quantity is not the relevant variable. Chapter 5 reports a result consistent with this conclusion.

The same review identifies two evaluation problems that make published comparisons difficult to interpret. First, downstream task data is frequently drawn from the pretraining corpus, meaning that what is measured is not true generalisation. Second, studies differ in whether they partition within or between subjects, which the review notes makes performance results incomparable across papers.

Two further observations from that review bear on this thesis specifically. The first is architectural: LaBraM’s training loss failed to converge when reconstructing raw signals directly, and the model was reformulated to predict the Fourier spectrum instead. This is an oscillatory inductive bias arriving by necessity from inside the foundation-model literature, and it is directly relevant to the hypothesis in Section 6.3.

The second observation is an absence. The review’s inventory of downstream evaluation tasks spans abnormality detection, seizure detection and typing, sleep staging, emotion recognition, MI, workload, and several diagnoses, on standard clinical and competition datasets. Nothing resembling gait, locomotion, or directional movement intent appears anywhere in it. While this is an absence from a single survey rather than proof that the literature holds nothing of the kind, it remains the strongest statement the available evidence

supports about how unusual this task is. This also explains why the adaptation procedures published with these models are untuned for it.

Evidence from a neighbouring modality points the other way. In functional magnetic resonance imaging, a pretrained sequence model raises mental-state decoding accuracy from 0.622 to 0.925 on one dataset relative to a linear baseline, by a wide margin the largest single effect in that comparison [50]. The same table also shows the pattern this thesis observes: a representation change that helps a weak decoder a great deal does almost nothing for a strong adapted one. Chapter 6 argues for the second reading without pretending the first is absent.

2.7 Evaluation practice in small-n studies

Two problems make results in this area harder to compare than their presentation suggests. The first problem is analytic flexibility: the number of defensible ways to process the same recording is large enough that the choice can determine the answer. The second problem is that studies of this size rarely have the statistical power to distinguish the models they rank.

This first issue has been quantified thoroughly. A multiverse analysis of a standard ERP pipeline was run on 298 participants across every combination of high-pass and low-pass cutoff, ocular correction method, reference, baseline interval, sensor, and amplitude measure. This produced thousands of scores per person from a single dataset [14]. The choice of reference alone, a binary decision between two entirely conventional options, changed the between-trial standard deviation (SD) of the component score by 56.40%. The choice of amplitude measure shifted the mean score by more than the size of the component itself. The pipeline that optimised data quality was not the one that maximised the between-condition effect, since the two disagreed on the reference, the measure, and the baseline interval.

No chain in that space is uniquely correct, so a paper that reports only one chain describes a single point in a space it has not characterised. The same conclusion is reached from the opposite direction in the systematic comparison of preprocessing steps discussed in Section 2.3. That study finds several routine steps to be neutral or harmful regarding the outcome it measures [18].

The second problem is rarely addressed. The procedure for comparing several models over multiple datasets, which involves a Friedman test over ranks followed by appropriate post-hoc comparisons, has been standard for two decades [20]. Later work has cautioned specifically against reading mean ranks as if they were independent measurements [8]. However, current practice varies more than the age of this procedure would suggest. For example, the 23-dataset forecasting benchmark discussed earlier reports 17 models over 97 configurations without any significance testing [2]. In contrast, the re-evaluation study that does apply a Friedman test finds its apparent champion, a proof-of-concept hybrid the authors added for that purpose, indistinguishable from the other models [13].

When the number of evaluation units is small, the most defensible report is an interval and an effect size. Chapter 4 outlines the choice of five folds as a deliberate decision based on the established analysis of the bias and variance behaviour of cross-validation [39], and not as a default. Effect-size conventions follow Cohen [16].

One further evaluation dimension matters for a system with a confidence gate, yet it is routinely omitted. A model's confidence must be meaningful before it can be used to withhold a command. Modern neural networks are systematically overconfident, a flaw that calibration measurement exposes. This overconfidence can be largely corrected through temperature scaling, which involves rescaling the model's outputs by a single fitted constant so that its stated confidence matches its realised accuracy [28]. Chapter 4 includes calibration in its metric panel for this reason, and Chapter 5 demonstrates that the ranking of models by calibration does not match their order by accuracy.

Chapter 3

System and Dataset

Everything the benchmark measures was recorded, labelled and replayed by a system built for this project [26], and this chapter describes that system before anything is asked of the data. It runs from the headset and the acquisition software, through the recording protocol and what it did not control, to the participants and the resulting dataset. It ends with how the signal is conditioned and how a decoded class becomes a machine command. The order places every property of the signal the decoders receive, including its imperfections, before the models themselves in Chapter 4.

3.1 Acquisition hardware: the DSI-24 dry-electrode headset

All recordings were made with a Wearable Sensing DSI-24 [73], a wireless dry-electrode headset carrying 19 scalp electrodes in the international 10–20 arrangement and two ear-clip sensors, A1 and A2, that serve as the reference, sampled at 300 Hz through a 16-bit converter. Every scalp channel is measured against the linked-ears reference, the mean of the two clips, which the device applies itself, so that Pz, the factory reference, carries signal like any other channel and the 19 scalp positions are all recorded. The headset also offers three auxiliary inputs and a trigger, none of which this work uses. The 10–20 arrangement is the standard electrode layout that places contacts at fixed proportional distances across the scalp, so that a channel name identifies the same cortical neighbourhood on any head. The headset communicates over Bluetooth and has a run time of more than 24 hours.

The choice of a dry montage is the premise of the entire project. Section 2.3 sets out why a clinic accepts this trade-off: the wearer can fit the headset in a fraction of the time, and the recordings pay for it with higher contact impedance and greater sensitivity to movement.

How much signal quality the dry montage gives up has been measured directly, with the figures given in the same section. Across participants recorded with both systems, dry and wet measures agree with rank correlations between 0.54 and 0.89 [37], on a scale where 1 means the two systems rank the participants identically and where repeating the same wet measurement on another occasion reaches roughly 0.7 to 0.85. The upper half of that range therefore matches the wet system’s own repeatability, while the lower half falls short of it, so the dry montage matches the wet one on some measures and departs from it on others. In a clinical population the two montages gave comparable resting power and ERP measures [30]. Neither study recorded walking participants, which is precisely when a dry montage is most vulnerable to artefacts, so they establish that the trade-off is acceptable at rest and not that it holds during gait. Chapter 5 answers empirically whether dry EEG recorded during executed movement supports a usable decoder, and this chapter describes the system built to ask that question.

One naming detail recurs throughout the software. The headset reports four electrodes under the pre-1991 labels T3, T4, T5 and T6. In contrast, both the modern 10–20 convention and MNE-Python (the EEG analysis toolbox used throughout this project [27]) call these same positions T7, T8, P7 and P8. To ensure that montage lookup succeeds and every downstream tool sees the modern names, a single rename map in the configuration module translates these labels on load. Whenever this thesis names an electrode, it uses the modern form.

Figure 3.1 shows where the headset stands in the live decoding pipeline that the rest of this chapter describes.



Figure 3.1: The live decoding pipeline, from headset to platform command. The dry-electrode headset streams 19 channels at 300 Hz (§3.1–3.2). The stream is band-pass filtered and cut into two-second windows at a half-second stride (§3.6). The decoder assigns one of the five class labels, and a confidence gate passes the label on only when the decoder’s confidence reaches 0.80, sending uncertain windows to Idle. The surviving label is mapped to a body velocity (§3.7), and the pipeline repeats at the half-second decode cadence. No signal returns from the platform to the decoder, so the machine is commanded open-loop, and the only feedback is the user’s own perception of the platform’s motion. *Robot* is the wheeled platform the pipeline drove, which stands in for the gait trainer the system targets.

3.2 Acquisition software: streaming, bridging and recording

The acquisition stack spans two operating systems, and the division follows from where the hardware can be reached. The vendor library for the headset, its Bluetooth serial port and the tooling for the machine controller’s EtherCAT industrial fieldbus (Section 3.7) were installed and configured on the Windows host, because Windows Subsystem for Linux (WSL2) cannot reach devices that need raw device or network-interface access. The analysis pipeline, the models and the training harness live in an Ubuntu environment on WSL2, a preference rather than a requirement, since training runs on either platform. A bridge connects the two: WSL2 invokes a PowerShell command that starts a Windows Python process, which pulls samples from the headset through the vendor library and writes them as text to its standard output, where the Linux side collects them. One shared helper builds this launch for the headset and for the machine controller alike.

The vendor library is called directly from Python through `ctypes`, its built-in mechanism for invoking C code, so no third-party wrapper is needed, and an entry point missing from the installed version is recorded as absent instead of stopping the program. During acquisition the library delivers data in batches of about 100 milliseconds (30 samples at 300 Hz), so the reader waits for each batch instead of polling.

Connecting to the headset takes more steps than reading from it. WSL2 finds the serial port through two PowerShell queries, the first enumerating those available and the second listing Bluetooth devices. The headset is found by name in the Bluetooth list, its address is read from that entry, and the serial port carrying the same address is the one to use. The connection then tries a sequence of candidate ports and closes the handle between attempts. This matters because the vendor’s library opens the port before the protocol handshake. A failed handshake therefore leaves the port held open, and a later attempt on it fails because the port is busy, not because the headset is absent. The stack separates these two failures in its messages, so the operator knows whether a port could not be opened or the headset did not respond.

In the free-form recording mode, labels are entered from the keyboard: the operator holds a direction key during a movement, the incoming samples are tagged with it, and no key means Idle. Every file in this thesis comes instead from the timed trial of Section 3.3, whose label is fixed before recording starts, so the keyboard reader never ran during those sessions.

Recordings are saved in the European Data Format (EDF), the standard container for physiological signals. In free-form mode the recorder converts the sample-by-sample labels into events, consecutive identical values becoming one block with a start time and a duration, and writes them as EDF annotations. For a timed trial the label is one value for the whole file, so the recorder removes the annotation track and writes the class into

the patient header, the one field that reliably survives a write-and-reload cycle, with the filename carrying it a third time. On import the class is resolved from the annotations first, the header second and the filename last, so for this corpus it comes from the header, except for three early files that still carry an annotation. The export module uses MNE for channel renaming and montage lookup, the same calls the offline pipeline makes, so a file is ready for analysis as saved.

All of this is reached through a single terminal menu, the operator’s entire interface to the running system, which offers four choices. *Dashboard* opens the live view described below. *Record EEG* begins a trial in one of the two sub-modes of Section 3.3. *System Status* reports the headset’s battery and metadata, then a raw per-channel microvolt table. *Neuro-Drive* runs the live decoding pipeline of Section 3.7. Underneath, the bridge, the port discovery and the sample reader are the same code for all four, which is what makes the rest of this chapter a description of one system. Neuro-Drive in particular reuses the dashboard view, fanning each sample out to both the plot and the model, so the decoding pipeline is the live view with inference attached. The dashboard is kept alongside it because inspecting the stream should not require a checkpoint or a robot. Figure 3.2 shows the menu and the setup step of a timed trial as the operator sees them.

Two visualisation tools support recording. The dashboard is built with PyQt6 and pyqtgraph and shows four panels at once. Three of them are a scrolling 19-channel trace of the last five seconds at about 20 Hz, a scalp topography of the voltage pattern across the head at about 5 Hz, and a power spectrum with an electrode selector. The topography module fetches electrode coordinates from MNE once at start-up and uses precomputed interpolation after that, so no costly computation runs inside the refresh loop.¹ Figure 3.3 shows the dashboard during a live stream.

The fourth panel shows EEG microstates, brief periods during which the voltage pattern across the scalp stays stable before switching to another. Resting EEG is commonly segmented into such states [38], and the panel follows the established procedure [59], fitting the four canonical map topographies at the peaks of the global field power, the SD of the signal across all electrodes at a given instant. It applies an average reference and a band-pass filter to a rolling buffer of raw samples but omits the removal of ocular components that rigorous microstate analysis requires [49], so it meets that standard only in part. The panel is retained because a stable topographic readout gives quick confirmation of signal quality and headset seating during a session, in line with the laboratory’s earlier protocol [25]. It is a monitoring aid only, and no result in this thesis comes from microstate data.

The second tool is an offline viewer for saved recordings. It is a separate command, not a menu entry, and works on existing files only. It shows the data with its annotations under a colour key, orders the channels topographically instead of alphabetically, and applies the same frequency band, filter order and notch settings as the training pipeline.

3.3 Recording protocol and the five-class label scheme

The classification scheme consists of five labels: Idle, Forward, Backward, Left and Right, with each serving a specific functional purpose. The Idle class, in particular, is a system requirement. A decoder controlling a gait trainer must possess the capability to explicitly command an absence of movement. Without a dedicated no-command class, the system would be forced to assign an active motion to every time window. Methodologically, including this fifth class establishes a base chance level of 20 %, ensuring that reported accuracies can be evaluated directly without statistical correction. Additionally, the deployment confidence gate detailed in Section 3.8 relies on this class to handle rejected predictions. If the Idle class were omitted, low-confidence classifications would lack a safe fallback designation, potentially resulting in unintended physical actuation.

¹Three earlier one-view modes (microstates, spectrum and a scrolling microvolt table) remain in the source with their menu entries commented out. Switching between them mid-session was too slow, so their elements were rebuilt inside the single dashboard.

```

+-----+
|           Welcome to the DSI-24 EEG GUI!           |
+-----+

1 Dashboard
  Real-time waveforms, scalp topomap and frequency spectrum in one window.

2 Record EEG
  Free-form or timed-trial recording. Choose sub-mode after selecting this option.

3 System Status
  Battery, headset metadata and settings, then raw EEG stream. Q to return.

4 Neuro-Drive
  Dashboard + live decoded-intent label. The AI model drives the robot from EEG. Needs --model CKPT.

q Quit

-> Enter a choice [1 2 3 q]: 2

Record EEG

1 Free-form      Live W/S/A/D motion labels. Q to stop and save.
2 Timed Trial    Fixed label, auto-stop recording with countdown.
b Back          Return to main menu.

-> [1 2 b]: 2

Timed Trial -- Setup
Set once; shared by every recording until you change settings.
Press ESC at any time to return to the main menu.

Subject ID [sub01]: lg
Task/condition [rest]: normal

Movement label:
1 Idle
2 Forward
3 Backward
4 Left
5 Right

-> Label [1-5]: 2
Trial duration in seconds [15]:
Countdown before recording [3]:

Recording Preset
-----
Subject      lg
Task         normal
Label        Forward
Duration     15 s (default 15)
Countdown    3 s (default 3)
Recording    1 (next, auto-detected from existing files)
-----

Controls during each recording:
ENTER / SPACE Start (triggers countdown)
ESC           Abort and return to menu

-----
Recording 1 | Forward
Press ENTER or SPACE when ready to start the countdown.
Press ESC to return to the main menu.

```

Figure 3.2: Screenshots of the terminal interface. Top: the main menu with its four choices, the Record EEG sub-mode prompt and the movement-label list that follows it. Bottom: the setup step of a timed trial, where subject, task type, movement label, trial duration and countdown are entered once and held for every recording that follows, and the preset is confirmed before the first trial starts.

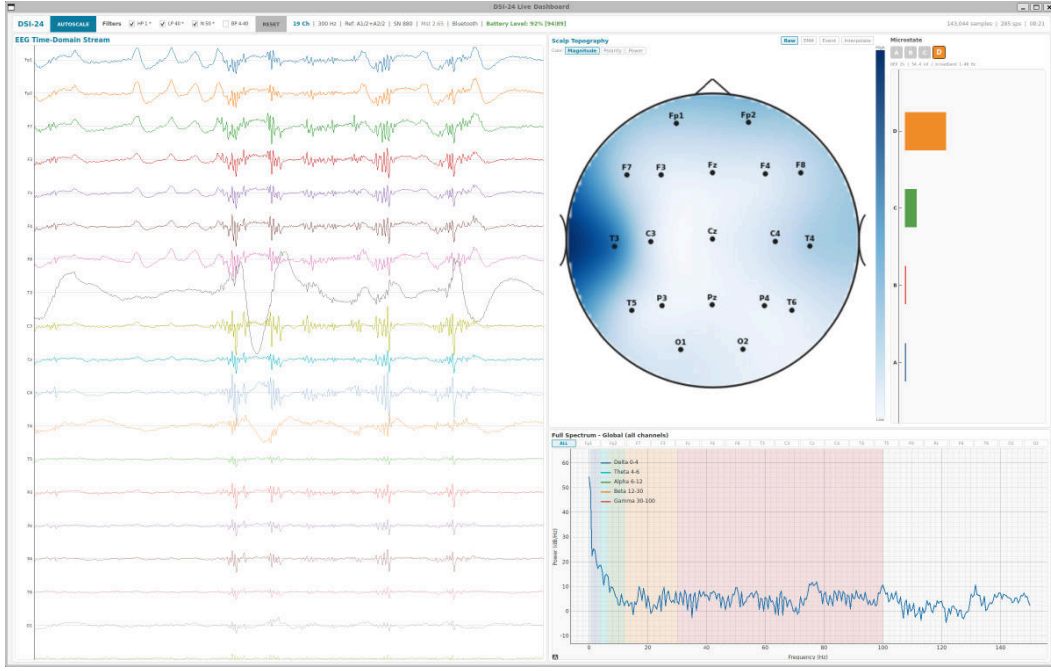


Figure 3.3: Screenshot of the live dashboard: the scrolling 19-channel trace on the left, the scalp topography and the microstate readout on the right, and the power spectrum with its electrode selector below. The header reports the channel count, the sampling rate and the reference the headset applies.

Each recording is a single fifteen-second clip of one intended movement. The physical unit of the protocol was the length of the laboratory. A recording began with the participant’s back to a wall and ended on touching the opposite one, at the pace the task called for. Fifteen seconds was enough to cross the room at a normal walking speed.² Using that distance as the timer is what made the pace approximately repeatable across sessions without any instrumentation: the participant could not walk much faster or slower and still finish at the wall.

The straight-line classes were recorded under a range of gait styles, not one canonical walk, so that the decoder would not learn a single motor pattern. Normal walking dominates, but the set also includes long, short, wide, crossed and shuffling steps, deliberately unsteady and silent gaits, jumps and skips, and walks with intermittent stops. Idle was recorded both standing and seated. The turning classes were recorded at graded radii, paced out in the participant’s own foot-lengths against markings on the laboratory floor and therefore approximate: roughly 0.3 m, 0.9 m and 1.5 m. Two further tasks have no radius at all: turning on the spot, and rotating while seated in a chair. Section 3.5 gives the counts.

No fixed order of tasks was prescribed in advance, and none emerged. The primary participant recorded on eight days between 10 April and 8 May 2026, the second on one only, and what was captured on a given day depended on which classes and gait styles still needed recordings. Some days covered a single direction and others several. Each recording day began the same way. The headset was fitted and the channels inspected, in particular for an electrode seated over a superficial vessel, which appears as a regular pulse-locked deflection on that channel alone and was corrected by reseating.

The recording chain was then checked. The participant made a few check recordings of the first direction planned for the day while deliberately blinking or clenching the jaw, and

²The laboratory measures approximately 15 m wall to wall along the traverse.

each was reopened in the offline viewer, or watched on the live dashboard, to confirm that an artefact made at a known moment appears at that point in the saved data. Without that check there was no way of knowing that the system was recording what the participant was actually doing. The check was repeated whenever the headset was put on again. It was also repeated, without re-inspecting the data, for each further direction recorded that day, so that every class carried some recordings with induced blinks and clenches and the artefacts would not be a feature of one class only. A check that had gone wrong was occasionally deleted, but most were kept and later renamed into the basic task type of their direction, and Section 3.5 states what that means for the corpus. Fitting and checking took roughly 15 minutes before the first ordinary recording of a day.

The recordings themselves proceeded one task type and gait style at a time, moving to the next once the intended repeats were done. Sessions ended for physical reasons, not on reaching a count: fatigue, discomfort from the compression band that holds the front of the headset in place over the frontal electrodes, and, in sessions containing turning tasks, dizziness. Because the stopping rule was the participant’s condition and not a quota, the realised counts vary by class and by task type, and Section 3.5 reports what was obtained rather than what was aimed at.

The terminal menu offers two recording sub-modes, free-form and timed trial. In the first the label follows the keyboard and the operator ends the recording manually, while in the second it is set beforehand and the recording stops automatically. This thesis uses timed trials only, because a fixed label and an automatic stop suit sessions of repeated tasks. The class is chosen once in a terminal setup step and persists across trials. The operator presses a key to start each one, the file index advances by itself, and the settings hold until they are changed or the main menu is reopened.

Timed trials also bypass the majority rule that gives a free-form file its single label from the dominant annotated class, since theirs is known in advance. They are checked by length instead: a recording well short of its expected duration is set aside and one substantially over it is cropped.

The filename encodes the complete description and structure of a recording. The naming convention includes the participant, task type, class, a sequential index, duration, time of day, date, and a **raw** marker indicating the processing state (which applies to all files in the corpus). For example, `lg_cross_steps_forward_002_15s_23-00-52_15-04-2026_raw` represents the second forward recording of cross-steps by participant `lg`, lasting 15s, and captured at 23:00:52 on 15 April 2026. The index counts within one combination of participant, task type and class, and resets when the combination changes, so it marks a position in a series rather than a global count. The names were brought to this form in one renaming pass after recording had ended. Fourteen early recordings, all of the **test** task type, kept the recorder’s original form, with the same fields in a different order, and were set aside in a directory outside the class folders, to which Section 3.5 returns. The task-type breakdown detailed in that section is derived directly from these filenames.

Uncontrolled variation during recording

A single-participant gait recording made with a dry headset in a working laboratory contains much that the protocol does not control. This section documents that variation, while its associated drawbacks and potential benefits are explored in Chapter 6.

The word *artefact* is used here in its signal sense: any component of the recording that did not originate in the process under study. (Chapter 4 uses the word in a second sense, for the concrete trained object that was evaluated.) In this corpus that means blinks and eye movement, jaw clenches and other facial muscle activity, and stray electrical events from the environment or the hardware, alongside the postural, motor and environmental sources catalogued below.

Ocular and facial sources were present throughout every session. Eyes dried and itched over a session, walking generated enough airflow to provoke blinking, and fatigue late in a session produced eyelid twitching. Blinks, jaw clenches and the occasional yawn occur

throughout the corpus. Additionally, a small number of recordings in every class contain blinks and jaw clenches produced on purpose (Section 3.5).

Were these events random with respect to the task, they would average out across many epochs and could be set aside. This assumption does not hold in general. Blinking during a demanding task follows a pattern instead of spreading evenly. In a study of Formula car driving [51], three expert drivers blinked at the same points on the circuit from lap to lap and blinked less when the combined acceleration on the car was highest, both effects holding against a null model that shifts each driver’s own blinks in time.

That study differs from this one in several respects. It involves three drivers in a steering task, its authors attribute the effect to visual demand, and it records no EEG at all, so it transfers here as a caution rather than as a measurement. The caution remains the right one: whatever pattern the blinks and clenches in this corpus follow, it cannot be assumed independent of what the participant was doing at the time.

Postural and motor variation was equally uncontrolled: small sideways sway and momentary losses of balance, inconsistent arm swing between and within recordings, unintended variation in walking style, occasional slipping between shoe and floor, and abrupt starts after which the participant corrected the pace mid-traverse. Head position and gaze varied freely, since neither was constrained, and occasional finger twitches and involuntary small movements were not excluded.

The headset itself contributes. Its nasal bridge support becomes uncomfortable over a session, and hair length changes electrode contact. The participant was buzz-cut for the recordings, which limits but does not remove this. The pulse-locked channel described in the session routine is the third headset-borne source. Cognitive state was not controlled either: intrusive thoughts, music running in the head, and emotionally salient recollection all occurred during recordings and none was excluded. The environment was a working building, not a shielded room, and contributed church bells, sirens, ventilation from machines left computing overnight, plumbing, traffic, birdsong and muffled voices through the walls.

None of these sources was removed from the data. Section 3.6 explains why independent component analysis (ICA), the usual instrument for at least the ocular components, is not applied: the method assumes sources that remain fixed relative to the electrodes, which walking violates. These sources are therefore present in the training data, and a decoder that performs well on this corpus has done so in their presence. This is the correct condition under which to evaluate a system intended for a clinic instead of a shielded laboratory, but it also means the decoders may have learned to use some of them, a possibility Chapter 6 takes up.

3.4 Ethics, consent and data protection

Two people contributed recordings. The author, identified throughout as **lg**, performed all five classes and supplied every recording used for training. The second participant, **ac**, is an adult acquaintance of the author who volunteered for a single recording day and contributed Forward and Backward only, in six task types, every one of them also among the primary participant’s. Those recordings are never trained on and exist solely as the held-out participant of Protocol C, the cross-person transfer test that Chapter 4 defines.

The second participant agreed, before any recording was made, to the recordings being used in this thesis. The agreement was verbal. The author explained the procedure and the intended use of the data, and remained present throughout that session as the operator of the system. Nothing was signed. A participant information and informed-consent form was drafted, by reworking the recording protocol into a participant-facing document, but it was never used and no part of it was put to anyone. It is mentioned here only because the draft survives in the project record and its existence should not be mistaken for a study that was run.

No formal ethics approval was sought or obtained, and none is claimed. These recordings were not conducted as a participant study. They are recordings of the author and of one consenting adult, made with a non-invasive dry-electrode EEG headset, with no intervention beyond voluntary movement that either person could perform unaided and stop at any point.

The recordings are coded, not anonymous. The two-letter codes are the only identifiers in the dataset and the archive, no other personal information was recorded, and the link between the codes and the two people is known to the author. The recordings are not released. Releasing them would require returning to the second participant for consent on those terms, which has not been done.

The consequences of this arrangement are stated as limitations. Age, sex and clinical screening were not recorded for either participant, so the dataset cannot support any claim conditioned on them and no demographic description of the participants appears in this thesis. Both participants are adults with no known neurological or motor condition, self-reported and not screened, so nothing here establishes that the results extend to the SCI or stroke population the system is ultimately intended for. Any work with that population, or with any participant recruited as such, would also require a separate study under formal ethics approval, which this work neither has nor substitutes for. The scope of what the thesis claims is bounded accordingly throughout, and Chapter 6 returns to it.

3.5 Dataset composition, task types and known imbalances

Two dataset sizes appear in this thesis and they are not interchangeable. The archive holds 774 recordings, 664 from the primary participant and 110 from the second, and every one of them parses into the corpus. Training data comes from the primary participant only. The benchmark trains on a pool of 500 of that participant’s recordings, what remains after a cap of 100 per class, and Protocols A and B, the within-participant evaluations that Chapter 4 defines, use all 500. Protocol C, the cross-person test, draws its training data from the pool’s 200 Forward and Backward recordings, holds one fold of those back as a within-participant reference, and adds the second participant’s 110 recordings, which serve as test data only. Table 3.1 gives the corpus by class and participant, and Figure 3.4 shows the same distribution graphically. Where this thesis quotes a dataset size, it is one of these two, named.

The shape of that distribution matters everywhere downstream, in two respects. Left and Right are the data floor at exactly 100 recordings each: so few that the per-class cap does not touch them, and the balanced pool is built by cutting the other classes down to meet them. Within a class, the task types are also unevenly represented. Of the primary participant’s 211 Forward recordings, 140 are ordinary walking, 60 are spread across 11 other gait styles, and 11 carry the `test` task type of the first two recording days. The cap-selection rule takes one recording of every task type first and fills the remaining quota by seeded sampling, so the rare styles survive into the training pool, but it cannot make them common. A decoder trained on this corpus has seen the unusual gaits, but not often.

The corpus spans 20 task types in all, and the straight-line families dominate by volume. The turning classes consist entirely of the tasks described in Section 3.3: 80 recordings at the widest radius, 40 each at the intermediate and the tightest, 30 turning on the spot and 10 rotating in a chair, split evenly between Left and Right in every case.

Two further groups of recordings are in the corpus. The first are the check recordings of Section 3.3, made with deliberate blinks and jaw clenches while performing the labelled movement and later renamed into the basic task type of their direction, so they cannot be told apart from the recordings that followed them. By the author’s estimate they number about 10 for each of Idle, Left and Right and 10 to 15 for each of Forward and Backward in the primary participant’s recordings, and about five per class in the second participant’s, so an estimated 5 to 10% of the recordings contain an induced blink or jaw clench in addition to whatever occurred naturally. The estimate cannot be verified from the archive. The second are the 14 recordings that kept the `test` task type, 11 Forward and 3 Idle made

Table 3.1: Composition of the recorded dataset, by class and participant.

Class	Participant lg	Participant ac	Total	Task types
Idle	103	–	103	3
Forward	211	55	266	13
Backward	150	55	205	7
Left	100	–	100	5
Right	100	–	100	5
Total	664	110	774	20

Each recording is fifteen seconds at 300 Hz over nineteen channels. Participant lg is the primary participant and supplies all training data, while participant ac contributed only Forward and Backward and is used solely as the held-out person in Protocol C. The counts are recordings, not analysis windows: this project also quotes window counts and the 500-recording benchmark pool, and Section 3.5 says which is which. The task-type column counts the types recorded under each class by either participant, and 13 of the 20 types were recorded under more than one class, so the column does not sum to the total row, which counts distinct types.

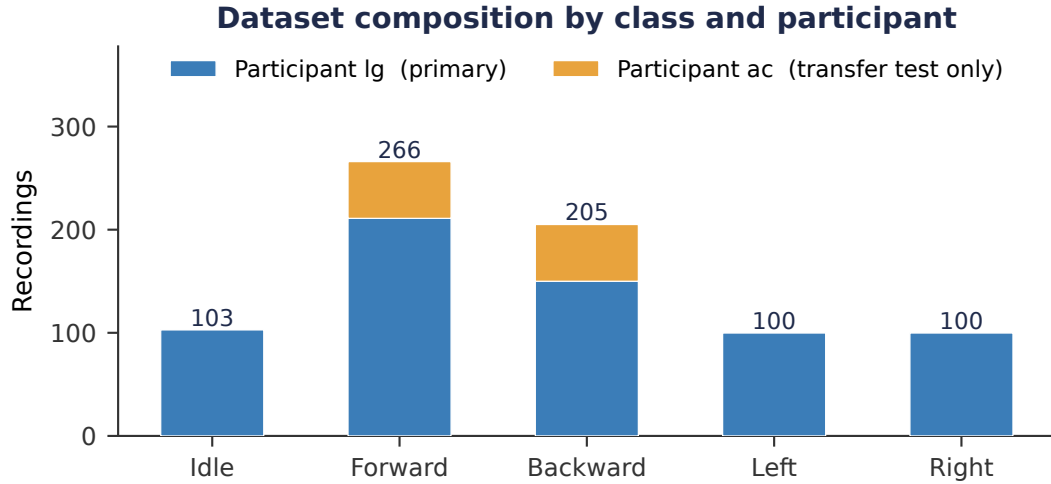


Figure 3.4: Recordings per class and participant. Left and Right, at 100 recordings each, set the per-class cap, and the second participant contributed only the two classes used for the cross-person test.

on the first two recording days while the acquisition software was being set up and left in a directory outside the class folders, in the recorder’s earlier filename form. The loader included them, and nine of them (six Forward, three Idle) are in the 500-recording pool, 216 of its 12 000 windows. The effect of either group on the results was not measured, and Section 6.6 lists both among the open measurements.

Public EEG corpora were also considered. Four datasets were selected as candidate comparison material: PhysioNet EEGMMIDB [62], the Graz BCI Competition IV dataset 2a [68], a mobile scalp- and ear-EEG corpus recorded during locomotion [43], and a continuous two-dimensional trajectory-decoding set from attempted movement [60]. Adapters were written to convert each into this project’s format with the appropriate resampling and label mapping, but none of them was used in the end. The thesis rests on the recordings described in this chapter, and no result reported anywhere in it derives from external data. The adapter code remains in the repository, reachable only through a manual command.

3.6 Signal conditioning and the anti-leakage rules

Preprocessing here is deliberately minimal, and the reason is evidential. Delorme’s systematic comparison of EEG preprocessing choices [18] evaluates each step by whether it

Table 3.2: The preprocessing decisions of this pipeline. Each ground is argued in the text, and “Delorme” is the systematic comparison of [18].

Step	Decision	Ground
High-pass, 1 Hz	applied	dry-electrode drift, Delorme’s most reliable gain
Low-pass, 40 Hz	applied	motor rhythms below it, gait muscle activity above
Notch, 50 Hz ($Q = 30$)	applied	mains, of which the band-pass leaves a tenth
Per-epoch z-score	applied (all nine)	training stability, no cross-window statistics
ICA	omitted	sources move against the montage during gait
Re-referencing	omitted	electrode positions only approximate
Bad-channel interpolation	omitted (untested)	channels controlled at acquisition, routines never called
Epoch rejection	omitted	too little data to afford discarding any

improves statistical power on real data. Two of the steps he tested reliably improve it, a further pair reduce it, and the rest make no dependable difference either way. Work on the analytical multiverse of ERP pipelines [14] reaches a compatible conclusion about how much a chain of defensible choices can move a result. Read together, they turn an omitted step into a design decision that must be justified. Each step below was therefore decided individually. Table 3.2 summarises the decisions, and the rest of this section argues them.

The pipeline applies a Butterworth band-pass from 1 to 40 Hz together with a 50 Hz notch against the mains interference no working laboratory avoids. Both filters are applied to each recording as a whole, before any window is cut, and each runs forward and then backward over the signal, which cancels the phase distortion either pass alone would introduce. The band-pass is designed at order four per edge, an eighth-order filter in total, built as cascaded second-order sections in SciPy [69], while the notch is a single one with a quality factor of 30, a width of about 1.7 Hz. Running each filter twice squares its magnitude response, which sharpens the roll-off but also pulls the passband inward: the half-power points of the response as applied sit near 1.1 and 36.4 Hz rather than at the design corners. Throughout all of this the recordings stay on the linked-ears hardware reference they were acquired against (Section 3.1).

The low-pass keeps the motor rhythms, which all lie below it, while discarding a band dominated by jaw and neck muscle activity during gait. The notch then does work the band-pass cannot: at 50 Hz its doubled response is down by only about 20 dB, leaving a tenth of the line’s amplitude in place, and mains pickup can stand orders of magnitude above the EEG. The notch removes what that roll-off leaves, taking the whole chain below -30 dB everywhere within half a hertz of the line. Figure 3.5 shows the response as applied and its effect on a constructed signal.

The high-pass removes the baseline drift that dry electrodes are prone to, and its corner is the one parameter where the evidence and the choice point in different directions, so it needs to be stated explicitly. In Delorme’s comparison the high-pass is the single most reliably beneficial step, but the optima he reports lie between 0.1 and 0.75 Hz and his working default is 0.5 Hz, all below the 1 Hz used here. The band below 1 Hz carries known signal: the movement-related cortical potential is a low-frequency negative shift that precedes voluntary movement, and a body of movement-intention detection is built on exactly that band [64], including a BCI that detects gait initiation from single trials [34]. The corner therefore surrenders a known signal family, and it does so deliberately, for two reasons. First, on a dry montage worn during executed gait, the sub-hertz band is where electrode drift and motion-driven baseline wander concentrate, so the region that literature reads is, under this protocol, the band dominated by artefacts. Delorme’s caveat that his optima come

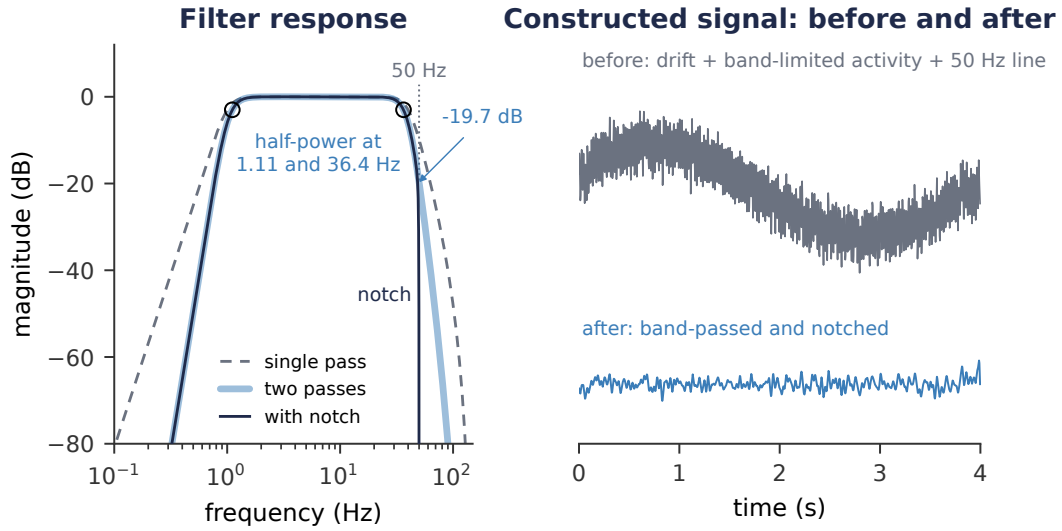


Figure 3.5: The signal-conditioning chain. *Left*: magnitude response of the band-pass and notch as applied, both zero-phase, with a single forward pass shown for comparison. The two-pass application pulls the half-power points to 1.1 and 36.4 Hz, and at 50 Hz the band-pass alone leaves a tenth of the line’s amplitude, which the notch removes. *Right*: the chain applied to a constructed demonstration signal: band-limited activity plus baseline drift and a 50 Hz line. The demonstration signal is synthetic, and no recorded EEG is shown.

from clean recordings therefore argues for a higher corner here, not a lower one. Second, the decoders do not depend on the neighbourhood of the cutoff: a control reported in Section 5.2 later shows that two of the leading architectures, EEGNet and TimesNet, keep most of their direction accuracy from the sensorimotor rhythms alone. The task matters too, since this thesis classifies sustained directional states and not the movement onsets around which the pre-movement potential is defined.

Per-channel, per-epoch normalisation follows. An *epoch* is EEG terminology for one analysis window, here two seconds of 19-channel signal, and this thesis uses the word in that data sense only, while a pass over the training set is always called a *training epoch*. Normalising means each channel of each window is shifted and scaled to zero mean and unit variance, using that window’s own samples and nothing else. It is applied to all nine decoders, with one qualification. The source code contains a comment warning that per-epoch scaling should not be applied before FBCSP, the classical baseline, whose spatial patterns are built from covariance ratios that the rescaling perturbs: z-scoring flattens each channel’s broadband amplitude before the filter bank splits it, so between-channel power differences survive only relative to each channel’s total. The exemption was never implemented. FBCSP was therefore trained and evaluated on the same normalised windows as every other decoder, its results stand as measured under that input, and the direction of the effect is unknown until the rerun listed in Section 6.6 is performed. The intent recorded in the comment and the pipeline as executed differ, and the latter is what the numbers describe.

Two-second windows are extracted from the signal at a stride of 0.5 seconds, so consecutive ones share 75 % of their samples. Each is classified independently: the decoder takes a single window as input and predicts one of the five classes. Neighbouring windows are highly correlated because of the overlap, but they are separate inputs to the classifier and never serve as prediction targets for one another. Figure 3.6 traces one of them from the recording down to its samples and its prediction.

The omissions begin with ICA [7], the standard instrument for handling the ocular and muscular artefacts catalogued in Section 3.3. It decomposes the multichannel recording into

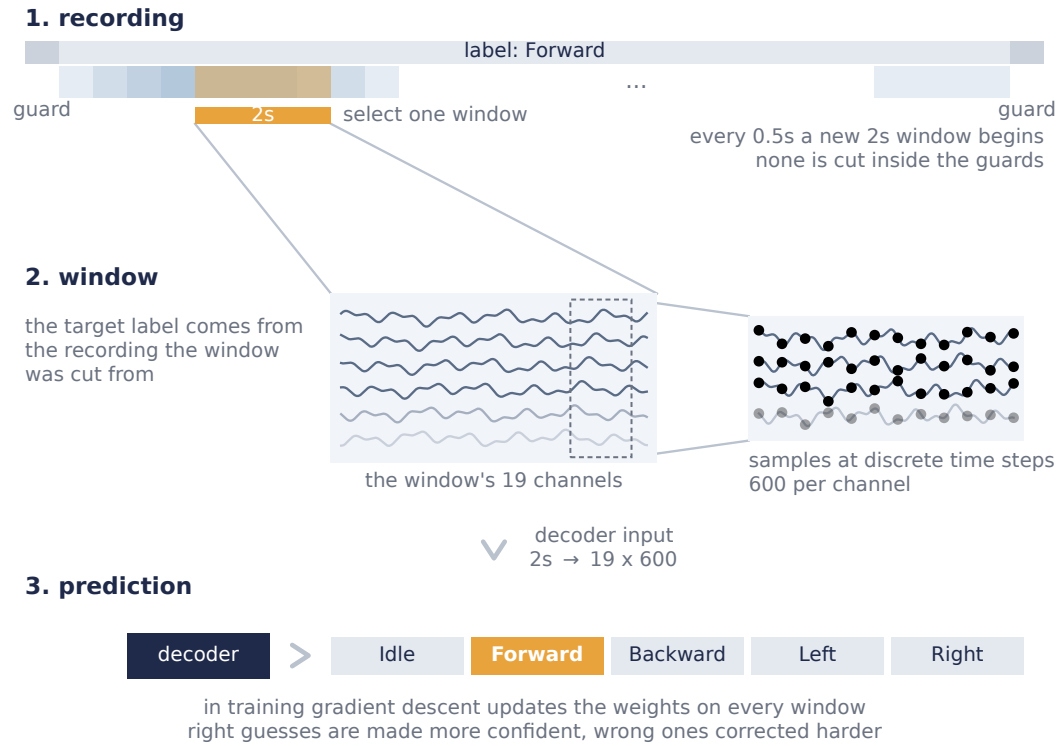


Figure 3.6: From a recording to one prediction, in three steps read top to bottom. A fifteen-second, single-label recording yields two-second windows at a half-second stride, none inside the half-second guard at either end, a margin justified below. Recording and windows are drawn to scale: the windows form the shaded band directly beneath the recording, each half-second stride visible as a step in the shading, and the last window meets the far guard. One window holds two seconds of all 19 channels, z-scored per channel. The inset to its right magnifies the stretch marked by the dashed outline and shows the signal as it is stored, 600 discrete samples per channel. The whole window, every channel and every sample together, is the decoder's input, and the decoder emits a single class. The target is the window's own label, inherited from the recording it was cut from, and in training every window updates the weights by gradient descent whether its prediction was right or wrong. Consecutive windows are separate inputs to the same classifier, never each other's targets, and nothing is forecast. The waveforms drawn are schematic, not recorded data.

statistically independent source signals so that artefactual components can be removed and the rest recombined, and in doing so it assumes that each source remains spatially stationary throughout the recorded data [19]. That assumption is exactly what this protocol breaks. The postural sway and head movements described earlier continuously shift the electrode-to-source geometry within a recording, and estimating a fixed mixing matrix for a mixture that does not hold still is a known cause of decomposition failure. The omission follows from the physical realities of this recording protocol.

The general literature, taken by itself, would not support the exclusion. Delorme concluded only that the automated rejection of eye and muscle components failed to increase performance *reliably*. Within his results, the method showed a trend advantage for ocular components and a significant one for muscle [18]. The multiverse study mentioned earlier is similarly mixed: of the two components it examined, decomposition yielded better data quality than the regression-based alternative for one, and no significant difference for the other [14]. On the general evidence the step is neutral to mildly beneficial. The exclusion

here therefore rests on the stationarity argument, and not on the literature.

Re-referencing, which recomputes every channel against a new zero point, is excluded, so the linked-ears reference of acquisition is also what the decoders see. None of the usual replacements is applied: not the common average reference, not the reference electrode standardisation technique, not a surface Laplacian. The dry montage’s electrode positions are only approximate, which the position-sensitive schemes assume away, and re-referencing reduced statistical sensitivity throughout Delorme’s evaluation.

Automatic epoch rejection is also omitted, to preserve statistical power. In Delorme’s evaluation, removing noisy trials reduced the number available, and the cleaner signal that remained rarely compensated for the statistical power lost with them. Given that the benchmark pool is limited to approximately 100 recordings per class, that penalty would weigh at least as heavily here. Of the omissions, this one is the most exposed to the first concession below. This corpus contains more genuinely corrupted windows than the recordings that evidence rests on, so here the benefit of removing them could outweigh the loss of data. The decision to retain every window therefore rests on the scarcity of data in this study, not on the general evidence. Retaining them also keeps the training data representative of the conditions a clinical system will meet, the argument Section 3.3 makes, at the cost noted there that the decoders may learn from the artefacts as well as from the signal.

Two concessions qualify these decisions.

First, Delorme evaluated these preprocessing steps on relatively clean recordings acquired in laboratory conditions, and he explicitly noted that his conclusions might not hold for noisier data collected in less controlled environments. The data here, recorded with a dry headset during executed gait, falls squarely into that noisier category, so his recommendations do not translate seamlessly to every step of the pipeline. For the high-pass filter, the elevated noise argues for a corner at least as high as the one selected. For epoch rejection, the added noise could instead invert his recommendation, as noted above. The ICA omission rests on the stationarity argument alone. Ultimately, his comparison is relied upon because it remains the most systematic study available and because its findings hold across all three of his datasets, not because their transfer to this specific protocol is guaranteed.

The second concession is practical rather than theoretical. Delorme identified bad-channel interpolation as one of only two steps that reliably improve outcomes, yet this pipeline does not apply it: the necessary detection and interpolation routines are fully written in the codebase, but they are never called. The omission was a choice, and its grounds were operational. Channel quality was controlled at acquisition instead of being repaired in software afterwards. Section 3.3 describes the fitting at the start of each recording day, in which every channel was inspected and a badly seated electrode was corrected by reseating it. The signal-quality monitoring of Section 3.2 then kept each channel in view while recording proceeded. Under that regime a failing channel is caught and corrected before data is committed, leaving little for a downstream repair stage to act on. Interpolation also carries a risk the working system had no need to accept. It replaces a flagged channel with an estimate assembled from its neighbours under assumed electrode positions. On a dry montage those positions are only approximate, the same ground on which position-sensitive re-referencing is declined above, so a false detection would silently substitute a fabricated signal for a genuine one. This reasoning was never tested, however: no run measured what interpolation would have gained or cost on this data. The thesis therefore cites Delorme’s evidence for two beneficial steps and implements only one, and Section 6.6 lists the untested step among the measurements a future repetition of this work should make.

The live pipeline must condition its signal as well, and an asymmetry follows. In training the pipeline applies zero-phase filters over the whole recording, forward and backward. That is acausal: each output value depends on samples after it, so the same operation cannot run on a stream one sample at a time. The common remedy is to switch to a causal filter at deployment. This pipeline does not, because such a filter changes the signal’s phase and the decoder would then see live data unlike anything it had learned from.

The deployment path instead applies the *same* zero-phase filters to a rolling buffer. It holds the current two-second window plus one second of the signal immediately before it. The filter runs over all three seconds, the history is discarded and only the conditioned current window goes to the classifier.

That history absorbs the transient that filtering a finite segment produces at its leading edge. The trailing edge has no such padding, since the newest samples are by definition the end of the buffer. One second is enough for the edge it protects: the impulse response of the whole zero-phase chain has decayed to about one part in a thousand of its peak at that distance from its centre.

Because the filter is identical, whatever difference remains between training and deployment is an edge effect of the finite buffer, not a change of algorithm, and the codebase measures it instead of assuming it is small. A parity test takes an interior window of a recording and conditions it both ways, over the whole file as in training and through the rolling buffer as in deployment, and the live path passes only if the median per-channel correlation between the two exceeds 0.97.

The cost is memory, not latency: one second of stored signal, not of added delay. The buffer is updated continuously, so the history is already in place when a new window arrives, and only the first few after start-up are decoded against an incomplete one.

The 75 % overlap between consecutive windows multiplies the amount of training material, and it is also the main way information could cross a split. Three guards apply: one against label ambiguity within a recording, two against leakage between splits.

First, label ambiguity at the start and end of a recording. A half-second margin is applied at each end of the labelled segment, and windows are taken only from its interior. The signal is never altered or spliced, and a position that overlaps the margin is not extracted. Every recording in this corpus is a single-label timed trial of fifteen seconds (Section 3.3), so the labelled segment is the whole recording and the margins are its two ends, where the participant is still standing at the first wall or slowing to a stop at the far one. Discarding them removes the windows most likely to hold a physical state that does not match the label.

Second, near-duplicate windows across the training, validation and test sets. At every boundary between them, two seconds of positions are discarded, one full window length or four strides. The closest retained windows on either side therefore share no samples, and at least half a second of signal between them remains unused. The filtering must also be accounted for here. Zero-phase filtering runs over the whole recording before windowing, so in principle it couples samples across a split in both directions. The gap is wide enough for that coupling to be negligible: by the filter’s impulse response, the interference is below half of one percent of the peak at the nearest edge and below one part in ten thousand two seconds further out.

Third, shared normalisation statistics. A scaler fitted on data that includes the held-out sets leaks their distribution into training. This pipeline standardises every window on its own mean and variance and computes no dataset-wide statistic, so the normalisation cannot carry information across a split.

The first guard addresses label ambiguity within a recording, the other two keep information from crossing a dataset split. Protocol A, the per-class split that Chapter 4 defines, relies on the split gap to keep its sets free of window-overlap leakage, and that wording is deliberate: the overlap is the one mechanism the gap removes. Nothing in its construction keeps recording-level structure, such as electrode seating, impedance drift and fatigue, from crossing a boundary, since the recording it falls in contributes windows to both sides. Protocol B, which never lets a single recording span a split, removes both mechanisms by design, as Section 4.4 sets out, and that is one reason the deployment claims of this thesis rest on Protocol B.

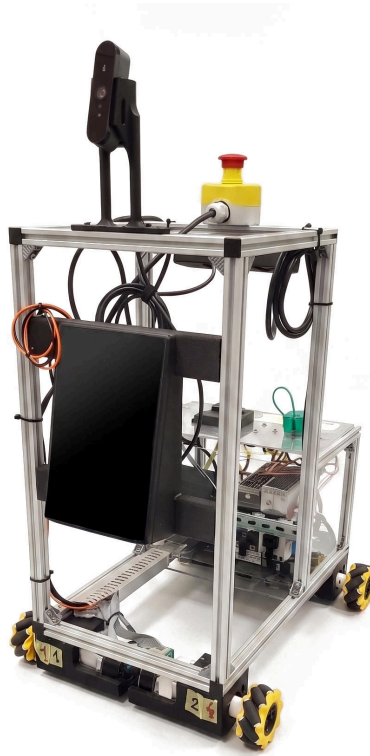


Figure 3.7: The smaller wheeled surrogate on which the pipeline was demonstrated, distinct from the full-size gait trainer of Figure 3.8 that it stands in for. It is a four-wheel Mecanum chassis at a much smaller scale, whose on-board controller runs the wheel kinematics and accepts the same body-velocity contract as the full machine, so the host drives either one after a change of configuration. The photograph shows the electronics in the lower bay, a display on one side, a camera on a short mast and an emergency-stop button on the top plate. It provides no body-weight support and serves as the endpoint of the command pipeline only.

3.7 From decoder output to gait-trainer command

The deployment stack, meaning the inference and machine-control packages, completes the pipeline by translating a windowed EEG stream into a motion command on a wheeled prototype platform (Figure 3.7). This prototype serves as a surrogate for the final clinical gait trainer. The pipeline runs in one direction only. Nothing measured on the platform is returned to the decoder, so in control-engineering terms the machine is commanded open-loop, and the only feedback is the user’s own perception of the platform’s motion.

The target device is a gait trainer of the Andago class [31], shown in Figure 3.8: a wheeled frame that moves with the patient, takes part of their weight from above, and leaves their legs free to move. The specific machine is the research gait trainer developed by Fang and colleagues, who built it as a gravity-supporting chassis on Mecanum wheels that follows the user by camera-based pose estimation [24]. This thesis reuses that platform for a different purpose: steering it from decoded EEG. Assistance on such a device takes the form of support and directed motion, not the imposed joint patterns of a wearable, joint-actuated exoskeleton, and that changes the decoding task: what the system needs from the EEG is a high-level direction. The same five commands would serve a wearable exoskeleton as well, since its controller already generates the gait and needs only the intended direction, though a new device would still require its own calibration and command mapping.

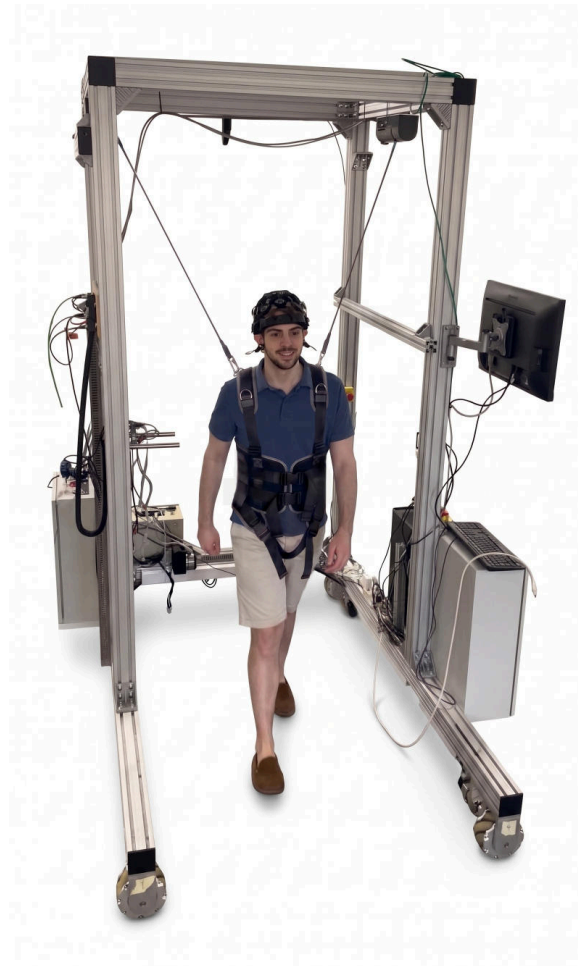


Figure 3.8: The mobile, gravity-supported gait trainer the decoder is meant to steer: a Mecanum-wheeled frame that rolls with the user while an overhead harness carries part of their body weight, leaving the legs free. The dry-electrode headset of §3.1 is worn in the photograph. The image documents the platform and nothing more. No recording in this thesis was made inside the frame, as the corpus was recorded walking the length of the laboratory (§3.3). The pipeline described in this section was demonstrated on the smaller wheeled prototype of Figure 3.7, not on this machine.

The interface between the decoding and driving modules is deliberately minimal. A decoded class becomes one velocity: Forward and Backward a straight-line speed, Left and Right a rate of turn in place, Idle zero. One command is issued per window, so a path is a sequence of them, never a blend of two. The command enumeration is bound to the label map when the module loads, and any mismatch halts the program with an explicit error, so the decoder’s output index serves directly as a driving instruction.

The host does not compute individual wheel speeds. Instead, it writes the three velocity components into as many sub-entries of one index in the motor controller’s object dictionary, the addressable table of parameters that a device exposes on an EtherCAT industrial network, and reads the achieved values back from three neighbouring ones. The controller itself runs the Mecanum kinematics, translating the single body velocity into individual speeds for all four wheels. This keeps the chassis geometry out of the research code altogether, meaning a differently built platform that accepts the same velocity contract requires no changes to the host.

Three stages run at different rates, and the slowest sets the response time. The decoder emits one command every 0.5 s, at the window stride. That command is held constant until the next decision: the controller re-sends the current velocity every 20 ms (50 Hz), and the Windows-side agent, which communicates over EtherCAT, writes it as a setpoint every 5 ms (200 Hz), so the motor drive always holds a valid target and a lost command decays to zero through the watchdog of Section 3.8 rather than persisting. The agent runs as a separate process on Windows for the same reason the acquisition bridge does: the fieldbus requires raw access to a network interface, which WSL2 does not provide, so it cannot host the master. The two processes exchange a two-verb text protocol over the agent’s standard input: one verb sets the velocity, the other stops the machine and exits.

The robot’s own controller was not modified. The host writes only the body velocity described above, and the pre-existing firmware converts it into wheel speeds, so the platform’s control algorithm, gains and ramps are those of the machine as delivered. The delay between a change in the user’s movement and the platform’s response is therefore set by the EEG side, not by the robot. The two-second analysis window must first fill with the new movement, which takes up to 2.5 s including the stride, and the start hysteresis of Section 3.8 then requires three agreeing windows, which adds 1.0 s. A start can therefore follow a change of intention by up to about 3.5 s, whereas a stop takes about 1 s and a safety stop one frame of 0.5 s. The two fast stages add at most 25 ms. This response time is a property of decoding intent from two-second windows, and it is accepted by design for a rehabilitation trainer that moves slowly and should start deliberately.

The live engine refuses anything but an explicit path to a weights file, so a physical demonstration is never driven by whichever checkpoint a search happened to leave behind. No quantified test of a decoder driving the platform was run, and none is reported. The pipeline was exercised live in two unrecorded sessions, both reported as qualitative observations in Section 6.5: a demonstration to two visitors on the wheeled surrogate, and a session by the training participant on the complete pipeline with the machine endpoint in the shallower of the two simulation modes described below. The participant’s session ran with the artefact-rejection stage of Section 3.8 disabled, because at its configured threshold it forced Idle too often for the decoder to be observed at all. The archived evidence for the pipeline consists of the replay and dry-run modes described below, which exercise the software chain at its live cadence and record the inference latency without the machine.

Three reasons kept the platform out of the measured results. No recording of the headset exists while the machine moves, so the artefact gate of Section 3.8 is uncalibrated for that condition. A session on a person would be a participant study, which the consent arrangement of Section 3.4 does not cover. The deployment-readiness criteria that the project set for itself were not met.

The full-size gait trainer itself was never driven, for practical reasons of a different kind. The frame is large, the laboratory is small and shared with other equipment, and a misread command followed by the response time described above leaves the operator little chance to intervene by hand before the machine moves, so a wrong input carried a real risk of

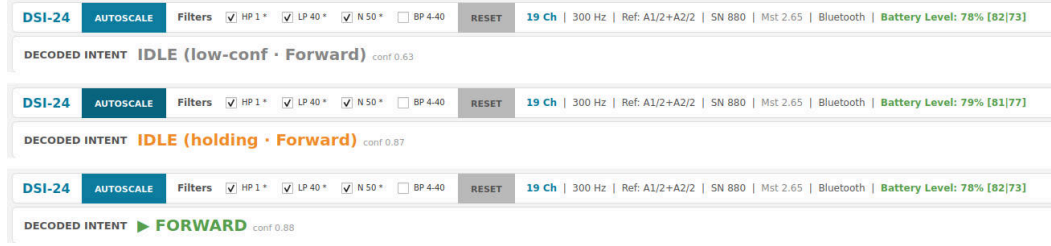


Figure 3.9: The three display states of the intent line, cropped from three screenshots of the running dashboard during the author’s self-test (§6.5). Top, “low-conf”: the window’s maximum probability lies below the gate, so Idle holds. Middle, “holding”: the window passed the gate and the three-window start count is not yet complete, so Idle still holds. Bottom, the green state: the count is complete and a Forward command is being sent. The confidence values are those visible in the interface at capture and illustrate the states rather than reporting a result. The header of each strip reports the reference the headset applies.

damaging the frame or its surroundings. Driving it would also have added no evidence: both machines accept the same body-velocity contract over the same fieldbus, so the host code drives either one after a change of configuration and nothing else, and the surrogate demonstrations exercise every part of the pipeline that the full frame would. A supervised and recorded session with the platform is therefore the next experiment (Section 7.2), not a result of this thesis, and no scientific claim rests on either session.

The software chain can be tested without the machine, at two simulation depths. The shallower one replaces the machine and prints the three velocity components it would have sent, which confirms that each class produces its intended motion. The deeper one is a dry-run flag on the agent itself: started manually, it exercises the text protocol and the watchdog without loading the EtherCAT library, though no launcher selects it automatically. A replay mode drives the entire chain from a recorded file at its original pace, allowing both mapping and timing to be checked without powering anything on.

3.8 Engineering for clinical operation

Decoder accuracy alone does not make a machine that moves a person safe. The layer between the decoder and the machine enforces one invariant: doing nothing is always the safe state, and every failure path leads to Idle.

A decoded window must pass four gates before it becomes a command. First, it is rejected if it looks like an artefact rather than a signal, determined by either a peak-to-peak ceiling or the fraction of band power sitting above the muscle-band edge. Second, it is rejected if the decoder’s maximum class probability falls below 0.80. Third, if the fraction of the last five windows that failed either of those two gates is strictly above 0.6, so that four of them trigger it and three do not, the controller forces Idle regardless of what the current one says. Fourth, a movement command is only issued once three consecutive windows agree confidently. Conversely, two disagreeing ones are enough to drop it.

The operator sees these gates as three display states of the intent line at the top of the dashboard (Figure 3.9). A window whose maximum probability falls below the gate is shown as Idle with the rejected class and its confidence beside it, in grey, the state the interface labels “low-conf”. A window above the gate that has not yet completed the three-window start count is shown as Idle with the pending class, in amber, the state the interface labels “holding”. Once the count is complete the class itself is shown, in green, and a command is being sent. Idle never reaches green, because nothing commands the safe state.

Given the half-second decode stride, this asymmetry has a concrete meaning: roughly one and a half seconds of sustained agreement to start moving (three windows, the last two of which add the 1.0s counted in the response time of Section 3.7), once the analysis

window has filled with the new movement, and one second to stop. Hard failures, such as detecting an artefact or triggering the low-confidence fail-safe, bypass this release counter and stop the machine in a single frame.

A rolling majority vote over recent windows is computed and recorded, but it is deliberately kept out of the control path. Feeding this vote into the state machine would add a full window’s lag to every decision, including the one to stop. A stop decision must not be delayed. Instead, the vote is retained in the per-window audit record, one row per stride, for traceability.

Two independent watchdogs sit beneath all of this. The controller commands Idle if no fresh instruction arrives within 0.4 s, or if the source raises an exception. Simultaneously, the machine-side agent runs its own timer on a separate input, set to the same 0.4 s by the launcher, and zeroes the setpoints if the host goes quiet. This second watchdog matters most architecturally, because it fires even if the host process is killed outright. A third guard makes the fail-safe design’s own failure mode visible. A decoder that never receives data would sit silently in Idle, which is indistinguishable from a decoder correctly deciding not to move. To prevent this ambiguity, a buffer that has not filled after five seconds raises a warning instead of continuing quietly.

This layer has three limitations. First, the artefact threshold is a placeholder. The gate tests the largest peak-to-peak swing of any channel within a two-second window, measured in microvolts after the band-pass and notch of Section 3.6 and before normalisation. On this montage during gait that quantity is large in ordinary operation. Computed over all 18 576 windows of the corpus with the training filter chain, its median is $401\ \mu\text{V}$ and its upper quartile $1709\ \mu\text{V}$, with the same statistic reading $185\ \mu\text{V}$ for Idle and $483\ \mu\text{V}$ for the movement classes, and isolated excursions up to $43\ 105\ \mu\text{V}$. These values are large because a dry montage has a far higher electrode-skin impedance and a greater susceptibility to movement and contact artefacts than a gel montage (Section 2.3), and every recording here was made under the protocol of Section 3.3, most of them while walking or turning, with the sources catalogued in Section 3.3. The ceiling of $150\ \mu\text{V}$ that the deployment code first carried would reject 90.3 % of the corpus’s windows, and even the $2500\ \mu\text{V}$ now configured rejects 21.1 %. A gate meant for saturation and electrode pops therefore already removes ordinary motion at this setting. Because a rejected window forces Idle at once and resets the start counter, the platform would halt on roughly one window in five, more often while walking, and need three clean consecutive windows to move again. The gate must be calibrated per montage from the amplitude distribution of a replay session before any use on a person. In the author’s self-test (Section 6.5) the stage was set aside for that reason, since at its configured threshold it forced Idle too often for the decoder to be observed, and in the author’s account its predictions held up despite sudden spikes in the trace. The artefact handling this system needs before clinical use is therefore open: a threshold calibrated from replay, a preprocessing step that removes the spikes, or reliance on the decoder’s own tolerance of them.

Second, no training data was recorded while the machine was moving, so the artefact environment of a live session with the platform in motion remains unmeasured.

Third, the classical baseline cannot be deployed in this pipeline at all. The live decoder requires a model with a PyTorch inference path, so the single decoder in the benchmark that needs no graphics processing unit (GPU) is also the one the deployment stack cannot run.

Chapter 4

Benchmark Design

Rather than proposing a novel decoder, this thesis focuses on comparing existing models. This approach addresses a recognised weakness in the literature: new architectures for sequential data are introduced at a rate that outpaces independent, standardised re-evaluation. The clearest evidence comes from long-term forecasting models. When eight of them were re-trained under one unified harness across 14 datasets, none consistently outperformed the baselines, and a ninth model the authors built as a proof of concept, which obtained the best average rank, did not separate statistically from them under a Friedman test [13]. This thesis differs from that setup in three distinct ways. First, the input is high-frequency physiological EEG data recorded during executed movement, a signal those datasets did not cover. Second, the task is five-class classification instead of forecasting, so the decoders built for the latter carry a classification head. Third, the evaluated approaches span classical, EEG-specific, general time-series and pretrained families, and only the three general ones are drawn from the library those authors released (Section 2.5). Introducing an additional architecture without first establishing a reliable reference for this task would yield a result that cannot be meaningfully interpreted. The primary contribution of this work is therefore to evaluate the existing model families under standardised conditions, with each decoder in the strongest configuration it can be granted, thereby providing a reference against which future architectures for gait rehabilitation BCIs can be measured.

This chapter details the complete experimental methodology prior to presenting the results. It first defines the comparison contract, which fixes what each decoder may and may not be given, and presents the nine decoders in the form that contract accepts, since Section 2.4 introduced them only by family. It outlines the three evaluation protocols and provides the rationale for the chosen deployment split. Finally, it specifies the experimental pipeline used to generate the results in Chapter 5, including HPO, training procedures, data augmentation techniques, evaluation metrics, and statistical methods. Several elements of this design were revised during the study to resolve limitations encountered in an earlier version, as detailed in Section 4.4.

4.1 The comparison contract

The nine decoders of this benchmark do not share a natural common configuration. A filter-bank spatial filter has no learning rate to tune and no seed to vary. A model pretrained at 200 Hz on a clinical corpus expects an input the deployment pipeline does not produce. A forecasting model pretrained on non-physiological series has no EEG-specific knowledge. Any single configuration imposed on all nine would favour the family it most resembled, and the resulting ranking would measure the choice of configuration rather than the decoders.

The benchmark therefore fixes a contract with two halves. On the *inside*, each model is granted the configuration that makes it strongest. On the *outside*, the machinery that turns a trained model into a number is held identical for all nine. Anything that cannot be equalised is reported with a flag rather than absorbed into the comparison.

The inner half of the contract permits only two deviations, both reserved for the pretrained models. The four networks trained from scratch (TimesNet, S-Mamba, iTransformer and EEGNet) run on the deployment preprocessing with hyperparameters found by their own search. The three EEG FMs are granted exactly two deviations. The first is their own *native preprocessing*: the sampling rate, passband and amplitude scale each was pretrained under, stated in full in Table 4.1, with the notch kept at the recording’s real 50 Hz line

Table 4.1: Native preprocessing granted to each foundation model.

Model	Sampling rate	Passband	Notch	Amplitude scale
CBraMod	200 Hz	0.3–75 Hz	50 Hz	volts → units of 100 μ V
LaBraM	200 Hz	0.1–75 Hz	50 Hz	volts → units of 100 μ V
Neuro-GPT	250 Hz	0.1–75 Hz	50 Hz	volts → units of 100 μ V

Read at build time from `MODEL_NATIVE_PREPROCESS` in the codebase, so this table cannot drift from the configuration that ran. The notch is set to the recording’s real 50 Hz line frequency rather than the 60 Hz of the (largely North American) pretraining corpora: the native passbands reach 75 Hz, so an unremoved line component would hand the encoder a large artefact it never saw in pretraining. The amplitude scale replaces the deployment pipeline’s per-epoch z-scoring for these runs.

frequency, not the 60 Hz of the largely North American pretraining corpora. The second is *full fine-tuning* of the pretrained encoder in place of a frozen probe. These two concessions are the ones a pretrained model requires, and no others are granted. Alongside the headline configuration, Chapter 5 also reports a three-arm study for these models (encoder frozen, adapted through low-rank adaptation (LoRA) [33], or fully fine-tuned), so that the contribution of the adaptation itself can be measured rather than assumed. FBCSP+LDA, the filter-bank spatial-pattern pipeline of Section 2.4 with an LDA classifier, is a deterministic classical fit with no search loop. Chronos, the non-EEG control, receives the full search on Protocol A so that its position cannot be attributed to under-tuning.

The outer half of the contract fixes every remaining element: the same participant pool under the same per-class cap, the same label maps, the same five seeds, the same three protocol definitions, the same training budget of 80 epochs with early stopping at patience 15 (a run halts after 15 epochs without validation improvement), the same metric panel, and the same half-second decode budget. Section 4.6 describes this experimental setup in sufficient detail for the claim of identical conditions to be checked. The budget contains one exception to this standardised scheme: the training for the control model on Protocols B and C was capped. Section 4.5 details this cap, its recorded rationale, and its impact on the comparability of the results.

Despite standardising these external variables, some inherent methodological differences remain. For instance, it is impossible to align a non-EEG control or a deterministic classical model perfectly with a tuned deep neural network across all parameters. To manage these unavoidable discrepancies, the evaluation framework explicitly flags the results they affect using three distinct designations. A result may be considered *fully comparable*. It might instead serve as a *control*, establishing a baseline bound rather than acting as a direct competitor. Alternatively, a result can carry *limited coverage* if specific components of the experimental pipeline could not be applied. To prevent misinterpretation and ensure readers do not mistake a baseline control for a competing model, these designations are integrated directly into the main results table (Table 5.1) rather than placed in footnotes.

One terminology conflict is present in the underlying documentation of the project. The original model documentation categorises decoders into tiers based on their *architecture family*, while the validity analysis uses the same term to grade individual *results* based on their adherence to the standardised framework. To avoid ambiguity, this thesis uses the term “tier” for neither scheme and refers instead to *decoder families* for architectural groupings and to *comparability flags* for the evaluation criteria.

4.2 The nine decoders and their integration

Before presenting the decoders individually, this section explains why these nine were selected. This selection is not a survey of the entire field. Instead, each decoder represents one approach a model can take to this task, so that performance differences can be attributed to specific design choices. The classical machine learning approach relies on fixed frequency bands, spatial filters computed from class covariances, and a linear classifier, with no learned

representation in between. The purpose-built category includes a network designed exclusively for EEG. The general time-series component evaluates three architectures borrowed from that field. They are included as a group because each relies on a different assumption about where the information in a sequence lies: TimesNet assumes it lies in periodic structure, iTransformer in the dependencies between channels, and S-Mamba shares that channel-wise view but replaces attention with state-space recurrence (Section 2.4). Two of these, iTransformer and S-Mamba, were directly compared in the previously discussed study, which also provides the architectures for all three models [13]. Therefore, the time-series models in this benchmark have already been evaluated under a shared experimental setup, although on a forecasting task.

The pretrained subset opens with the three EEG FMs whose weights could be obtained when the benchmark was built. Availability alone determined this selection, and any loss of fidelity compared to the originally published models is declared below. A ninth decoder, a non-physiological pretrained control, completes the set. Its role is to establish a lower bound for the pretrained track: whatever accuracy such a model reaches here is attainable without any EEG-specific knowledge. Thus, an EEG FM must exceed that baseline before its pretraining can be considered beneficial.

One classical family, the Riemannian decoders of Section 2.4, is absent from this selection. Its value as a second representative of the classical approach became clear only later in the project, through the ongoing analysis of the results, after the experimental scope had been finalised and the benchmark was already underway. Because the major decoder roles and architecture families were judged to be well covered, adding another model to the evaluation was deemed unnecessary. The absence is instead acknowledged as a limitation: Section 6.6 discusses the consequence, namely that the competitiveness of classical methods is demonstrated here by one representative of a single family, and Section 7.2 lists the missing baseline among the most direct experiments left open. However, given that the focus was the evaluation of decoders on a classification task, the need for a second classical baseline was not considered critical to the main conclusions of the thesis.

The nine decoders, drawn from four distinct approaches, must present a uniform interface to the training pipeline. Every neural decoder is therefore wrapped so that it exposes an encoder returning a fixed-width embedding, together with a forward pass that composes this encoder with a swappable classification head, operating on tensors of 19 channels by 600 samples. This classification head is deliberately minimal: by default, it is exactly a linear layer, so attaching it in place of the originally published classifiers changes nothing about what is being compared. Consequently, the only component that genuinely differs across the evaluated models is the encoder.

Five decoders, drawn from three families, are fitted from scratch on this corpus.

FBCSP+LDA [4] is the classical baseline. It consists of five Butterworth sub-bands covering 4 to 40 Hz (theta, alpha, low and high beta, and gamma), one-versus-rest CSP at four components per band, log-variance features, and LDA with automatic shrinkage, implemented using scikit-learn [56]. The pipeline is deterministic, has no gradient-trained parameters (its spatial filters and linear discriminant are solved in closed form), and is fitted once per fold, so there is no seed axis to vary and its Protocol-A SD is zero. The SD reported for Protocol B measures fold-to-fold variation instead. Like every other decoder, it receives per-epoch normalised windows. The source code records an intention to exempt it from this normalisation, because CSP filters are built from covariance ratios that per-epoch scaling can distort, but that exemption was never enacted: the classical runner passes no preprocessing override, so the model inherits the pipeline’s global default. Refitting only this single deterministic model would have been feasible, but the issue was found after the analysis was complete, and reworking it was not undertaken. The effect was never measured, but the distortion described above predicts a handicap rather than a benefit. Therefore, no reported conclusion depends on the correction, which is deferred to future work. Section 3.6 discloses the discrepancy, and Section 6.6 records what the refit would measure.

This implementation departs from FBCSP as published in three respects, and since this chapter declares every deviation for the pretrained models, the same standard applies to

the baseline. First, the published method uses nine Chebyshev type II sub-bands, whereas this implementation takes five Butterworth ones over the same 4–40 Hz span. Second, the published method ranks features by mutual information and discards the weakest, whereas this implementation performs no feature selection at all and passes the full concatenated set to the classifier. Third, the published method classifies with a naive-Bayes Parzen-window model [3], whereas this implementation uses shrinkage LDA. Only the third deviation is visible in the name this thesis uses for the pipeline. None of the three deviations constitutes a defect, and the pipeline remains a competent instance of the FBCSP scheme. They are declared because this baseline finishes inside the leading group in Chapter 5, its strength is an argument the thesis relies on repeatedly, and its exact relation to the published method must therefore be clear.

EEGNet [42] is the purpose-built EEG convolutional network, implemented directly from the paper without recourse to an existing repository. At 3541 parameters it is the smallest network in the roster, nearly 60 times lighter than the next in size and four orders of magnitude below the largest.

TimesNet [74], **iTransformer** [44] and **S-Mamba** [72] are general time-series architectures, the first designed for analysis across tasks that include classification, the other two for long-horizon forecasting. Their model bodies come unmodified from the benchmark library that supplies them [13, 71], and only the project-side wrapper is original.

The remaining four decoders are pretrained, and each carries a declared deviation from the model as published. These deviations are stated in the main text because a reader assessing the central negative result of Chapter 5, that the pretrained models at best match the networks trained from scratch, needs to know exactly which artefact was evaluated.

Neuro-GPT [17] pairs two components in its published form. An encoder of temporal and spatial convolutions followed by self-attention layers, the EEG-Conformer design [66] ported from the reference implementation, compresses each fixed-length chunk of the multi-channel signal into an embedding, and a GPT-style decoder-only transformer is pretrained to predict that of a masked chunk from the ones preceding it. Only the encoder serves classification, so the model is loaded *encoder-only*: the decoder and its output projection, some 80 tensors and 280 MB, are dropped at load. Its encoder expects 22 channels, so the 19 recorded ones are zero-padded to fit. Its embedding is only 40 dimensions wide, the narrowest in the roster: the next-narrowest embeddings, the 200 dimensions of CBraMod and LaBraM, are already five times as wide.

CBraMod [70] has its encoder body ported verbatim from the reference implementation under Apache-2.0, and only the project-side adapter is original. Among the code ported directly into the project’s own model files, it is the one component that carries an explicit third-party licence, while the vendored forecasting library ships with its own (MIT, Appendix B).

LaBraM [35] carries the most consequential deviation. The weights published with the paper sit behind an academic-licence request, so the braindecode community checkpoint is used instead. It is not the checkpoint the paper reports, its key names differ, and the file carries a four-way classification layer that is discarded on load. This layer indicates that the file is itself a fine-tuned artefact and not the pretrained model as released. Chapter 5 interprets LaBraM’s position in the light of this provenance.

CBraMod and LaBraM share a further mismatch, which concerns the *patch*: the fixed-length chunk of samples that these models cut the signal into, each chunk becoming one token of their input sequence. Both were pretrained at 200 Hz, where their 200-sample patch spans exactly one second of signal. The deployment pipeline runs at 300 Hz, so the same 200 samples span only 667 ms, and every temporal relationship the pretraining learned per patch covers two thirds of the interval it was learned on. The patch length cannot be adjusted, because the pretrained weights require it, and no amount of fine-tuning removes the compression. It is a property of the shared harness and not of the models themselves: under the native preprocessing granted above, both models are resampled to 200 Hz, the patch recovers its full second, and their reported accuracies come from this regime.

Chronos [5] is the non-EEG control. It encodes each of the 19 channels *sequentially*, as an independent univariate series, before averaging their embeddings, which is why it is the slowest decoder measured. It is also the largest model in the roster.

Two reported quantities, TimesNet’s parameter count and the inference costs of the foundation models, require context for correct interpretation. TimesNet’s 28.5 million parameters are a consequence of its configuration, not of a reimplement: its embedding is the sequence length times the model width, 38 400 values, and the whole of it feeds a single linear head. The inference-cost measurements of Section 5.6 were taken on the shared deployment harness, where the three FMs run as frozen probes, whereas their reported *accuracies* come from the native fine-tuned configuration. For those three rows, the cost and accuracy columns therefore describe different regimes, and the tables state this explicitly.

4.3 Three evaluation protocols and what each measures

An evaluation protocol is the rule that decides which recordings a decoder may learn from and which it is tested on, and that rule alone decides what the resulting accuracy means. Every protocol in this benchmark classifies one two-second window at a time, exactly as the live pipeline does (Section 3.7). The protocols therefore differ only in which unit of the system is withheld from training: the window, on which the live decoder acts, the recording, to which the label attaches and on which the calibration workflow rests, a user supplying single-movement clips for the decoder to train on, or the person, to whom the calibration applies.

A benchmark can test generalisation only along the axes its data contain, and this dataset offers two. The first is internal to the primary participant and separates a *window* from the *recording* it was cut from: that is, whether a decoder has seen other windows of the same fifteen-second clip. The second is the *person*: a decoder trained on one participant is asked to work on another. Recording date is not a third axis, for reasons Section 4.4 sets out, and the benchmark does not treat it as one.

Three protocols are defined below, each by the question it answers.

Protocol A asks how well the movements can be decoded at all. It holds out windows, the unit the live decoder acts on. Within the primary participant, the two-second windows of each class are placed in a fixed seeded recording order (one file of every task type first, then a seeded shuffle of the remainder, as Section 3.5 describes) and, within each recording, in time order. That sequence is cut once into 80 % training, 10 % validation and 10 % test, with a two-second gap discarded at each boundary so that no test window overlaps a training one. Every test window is then classified on its own, as it is in live use. This is the kind of hold-out the original single-participant pipeline was built with, at different proportions, before the benchmark existed, and the protocol under which the main hyperparameter search of Section 4.5 was run. Because the cut is made once per class, almost every recording falls entirely on one side of it. The construction offers no guarantee at the recording level, however: a boundary falling inside one splits it between the two neighbouring sets. With this corpus’s counts the training boundary falls between recordings, while the validation–test cut lands inside one, so a single recording per class reaches model selection and the test set alike. The test set is therefore one fixed draw of roughly 10 recordings per class, and because the pinned representative of each task type is placed at the front of the order, in training, any such type with a single recording in the pool is never tested under this protocol, a limit Section 6.6 records.

Protocol A is therefore read as a ceiling rather than as an estimate of deployed performance: it establishes that the participant’s movements are learnable and serves as the within-participant reference, while any claim about unseen recordings requires the guarantee that only Protocol B provides. The configuration the deployed checkpoint carries is the one found under this protocol (Section 6.5). The task is five-class against a chance level of 20 %, run at five seeds (fixed random starting points, so that run-to-run variation is measured).

Protocol B asks whether a decoder generalises to a recording it has never seen. It holds out recordings, the unit the label attaches to. Every recording is a single movement of one class, so a decoder that labels a held-out recording’s windows has classified that movement, which makes the recording the natural unit of a classification benchmark. It is also the unit of the calibration workflow the system is built around, in which the question is what the decoder does on the user’s next clips, and a live session presents signal it has never seen, so the same number estimates what the calibrated decoder does in use. Folds are formed by stratified group k -fold with $k = 5$ (the recordings are divided into five groups with every class equally represented, and each in turn is held out for testing while the other four supply the training and early-stopping validation sets), the grouping unit being the individual recording identified by its file path. Each fold’s held-out recordings become the test set, and an inner stratified group split of the remainder supplies the early-stopping validation set. The inner split is drawn by recording as well, so no window of a file used for model selection appears in training or test either, so early stopping, and not only the final evaluation, is free of leakage, the passage of held-out information into training. Five folds by five seeds give 25 measurements per model (one per fold for the deterministic classical baseline), on the same five-class task against 20 % chance. Protocol B constitutes the deployment axis, and every ranking claim in this thesis rests on it.

Protocol C asks whether a decoder transfers to a new person. It holds out a person, the unit calibration attaches to. It is a two-class problem, Forward against Backward, the two classes the second participant recorded, and every model is trained afresh on those two classes rather than reduced from a five-class decoder.

The primary participant’s Forward and Backward recordings are split once by whole recording, with the same leave-recordings-out routine as Protocol B: most recordings train the model, one recording fold supplies the early-stopping validation set, and one further fold is held out as a within-participant (the same condition as within-subject) reference test set. All recordings of the second participant are then added to the test set, and none of them ever reaches training or validation. The run is otherwise identical to the other protocols, with the same training budget over five seeds (the control’s cap of Section 4.5 excepted), but on a single split rather than five folds.

The split routine audits itself and raises an error if any epoch of the held-out participant reaches training or validation, and equally if the transfer set is empty, so a silently absent one cannot pass. The task is two-class against a chance level of 50 %.

Because the transfer task is two-way and the other two are five-way, a Protocol C number cannot be placed beside a Protocol A or B one, and this thesis never does so. Protocol C is also reported without confidence intervals, because none were computed for it, and it is therefore used to establish that transfer fails rather than to rank the models by the size of the drop. The held-out reference fold is what makes the transfer figure interpretable at all, because it supplies a matched within-participant accuracy on the same two classes from the same trained model. Without that reference a weak decoder’s smaller absolute drop would appear as better transfer.

Table 4.2 places the three protocols side by side, and Figure 4.1 draws what each one holds out.

Two differences between the protocols carry information of their own, and Chapter 5 uses both. The gap between Protocol A and Protocol B is the accuracy lost when a decoder must handle a recording it has never seen: the clinical case, in which a patient calibrates on some clips and then produces fresh ones. The gap between Protocol C’s reference and its transfer figure is the cost of changing person, isolated from the fact that a two-way problem is easier than a five-way one.

4.4 The deployment axis: why recordings are held out

The grouping variable of a cross-validation split defines the interpretation of the resulting metric, and it is the one design choice in this benchmark that had to be corrected after the

Table 4.2: The three evaluation protocols side by side, as design facts, with the results in Chapter 5.

	Protocol A	Protocol B	Protocol C
Unit held out	Windows	Recordings	A person
What that unit is	The unit the live decoder acts on.	The unit the label attaches to, and the unit of the calibration workflow.	The unit calibration attaches to.
Question	Can this participant’s movements be learned at all?	After calibration on a set of clips, does the decoder classify the next ones?	Does the decoder work on another person without calibration?
Trained on	The primary participant, the first 80 % of each class’s window sequence.	The primary participant, most of the recordings of four folds, the rest forming the validation set.	The primary participant’s Forward and Backward recordings.
Tested on	Roughly the last 10 % of each class’s window sequence, one fixed draw.	The recordings of the held-out fold, each fold in turn, so every recording is tested once.	Every recording of the second participant, with one held-out fold of the primary participant as reference.
Classes and chance	Five, 20 %	Five, 20 %	Two, 50 %
Repetitions	Five seeds	Five folds by five seeds	One split by five seeds
How the number is read	Within-participant ceiling.	Estimate for the calibrated decoder, and the axis that carries every ranking.	A boundary, not a ranking.

fact. The correction is reported here because the size of the error is the clearest argument for the design that replaced it.

The unit of the experiment is the recording. Each recording is a fifteen-second clip of one movement (Section 3.3), and no decoder sees it whole: the pipeline cuts it into two-second windows at half-second steps, so that each clip yields 24 windows, each serving as a single training example. Adjacent windows share one and a half of their two seconds. A split drawn at the level of the window therefore fills the test set with near-duplicates of training examples. Protocol A avoids this failure with its single cut per class and the discarded gap at each boundary, but its unit remains the window position, so nothing prevents a boundary from falling inside a recording, and in this corpus the validation–test boundary of every class does: one recording in each contributes windows to both model selection and testing. Protocol B makes the recording itself the unit, so that none can belong to two sets. This difference in unit is what separates the two protocols, and it is why Protocol A is treated as a ceiling.

The first version of the deployment protocol grouped by session instead, defined as participant and date. That choice failed for two independent reasons. First, the recordings were captured ad hoc, so the date is an incidental timestamp rather than a designed factor, and “cross-session generalisation” would test an axis the experiment does not contain. Second, and more seriously, session is aliased with the class label. Because the recordings were collected one movement block at a time, each of the primary participant’s nine date groups (eight recording days, one of them logged under two date strings) spans only one to three of the five classes. Counting the majority class of each group gives Idle four, Forward two,

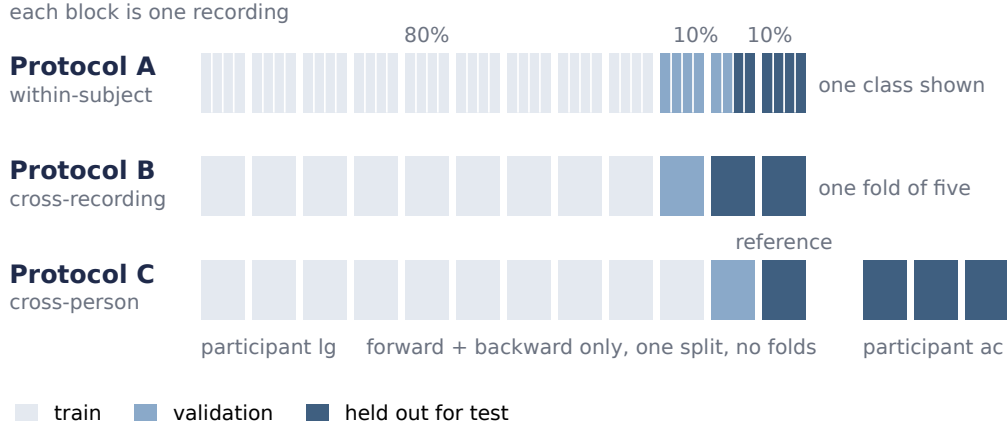


Figure 4.1: What each protocol holds out. Each block is one recording, and its fill, keyed in the legend, assigns all 24 of its windows to the same set: the split decides membership only, every window is still trained and classified on its own, and nothing is forecast (Figure 3.6). The first row shows one class under Protocol A, its roughly 100 recordings compressed into 12 blocks: its windows, in a fixed seeded recording order, are cut once into 80/10/10, and the cut falls inside one recording, the part-coloured block, whose windows are divided between validation and test. The first row alone is drawn with window slices, because Protocol A assigns membership by window position, whereas the other two protocols move whole recordings and their blocks are solid. The second row shows one fold of Protocol B, where every seam falls between recordings: whole recordings are held out and none is ever divided, which is why it and not Protocol A is the deployment axis. Protocol C trains once on the first participant’s Forward and Backward recordings, split by whole recording as in Protocol B, holds out one recording fold of that participant as a within-participant reference, and holds out the second participant entirely as the transfer test.

Backward two, Right one and Left none: Left is never the majority class of any session and exists only as a minority within groups dominated by other classes. Holding out a session therefore meant, for the most part, holding out a single class. No association statistic was computed for this and none is claimed. The alias is established by the coverage counts themselves.

The consequences of that aliasing were visible during training. Validation folds covered two or three of the five classes, and their training sets were nearly disjoint from the classes those validation sets contained. Training accuracies of roughly 75 to 82 % stood against validation figures of 9 to 25 % and losses between 3.6 and 11.6, which is the signature of a decoder assigning high confidence to classes it has never been shown. The held-out test sets were severely imbalanced even when greedily class-covered, in one case placing 24 Forward windows against 1248 Backward. These figures measured an ill-posed split rather than generalisation of any kind. The loader printed a fold-coverage warning on every degenerate fold, so the defect was surfaced during the runs.

The figures produced by the session-grouped split are retired. They are quoted here once, to measure the size of the confound. Under session grouping the five from-scratch and classical decoders scored 55.5, 50.7, 53.3, 51.6 and 56.7 % balanced accuracy respectively, whereas under recording grouping the same five architectures, unchanged, score 85.5, 84.0, 82.9, 82.8 and 82.6 %.¹ They appear nowhere else in this thesis.

Three conclusions of the earlier analysis are reversed by that comparison. The collapse

¹The corrected figures in this sentence are all taken under the frozen Protocol-A configuration, so that the grouping rule is all that differs between the two sets. EEGNet’s headline figure of 83.9 % in Table 5.1 comes from its own Protocol-B search (Section 4.5) and is not the number to compare against its retired value here.

of roughly 30 points between protocols was produced by the grouping variable and not by the architectures, which remained identical across both evaluations. The “calibration cliff” it was taken to demonstrate does not exist: the true gap between Protocol A and Protocol B lies within about three points for every decoder trained under the full budget. The conclusion that no decoder reached 75 % on the deployment axis is also inverted: the five architectures named above exceed it.

The replacement strategy groups by the individual recording and uses stratified group k -fold cross-validation. Each recording carries exactly one class, and each enters the split with 100 recordings after the per-class cap (Section 3.5), so the splitter can distribute them evenly across folds. The degeneracy disappears by construction: every fold’s training and test sets carry all five classes, and no recording contributes windows to more than one of training, validation and test.

Five folds is a deliberate choice, and the reasoning is the one the cross-validation literature sets out: the fold count trades the variance of each estimate against the amount of data each model sees [39]. Each class has 100 recordings, so $k = 5$ leaves about 20 of them for testing per fold, enough for a stable per-fold estimate, whereas $k = 10$ would halve that number. Leave-one-recording-out, the most exhaustive option, would mean some 500 folds each testing a single clip, at roughly 100 times the compute, with far higher per-fold variance and no better estimate of the mean. Five folds crossed with five seeds give 25 measurements per model, from which confidence intervals are bootstrapped, that is, estimated by resampling the observed values with replacement. Section 4.9 sets out the claims that this sample size supports.

The repair is enforced by four safeguards. The superseded splitter was deleted and a note left in its place recording why, so that it cannot be reinstated by someone unaware of the defect. The reporting path asserts that no fold can be degenerate and fails with an error. The results assembler refuses, by assertion, to admit any pre-repair Protocol-B row. The cross-person audit is itself tested against a deliberately injected leak, proving the guard actively fires.

4.5 Hyperparameter optimisation: where the search was run

Every decoder with a search loop received the same Bayesian HPO: 120 trials, each training the model for 20 epochs, proposed by a tree-structured Parzen estimator [9, 1], a strategy that models the distribution of high-scoring trials and draws new candidates from it, with validation balanced accuracy as the objective. Two budgets must be distinguished. The *search budget* of 20 epochs per trial is what makes 120 of them affordable. The *training budget* of Section 4.1, 80 epochs with early stopping at patience 15, is what the winning configuration is then trained under for every reported run. The budget was binding for a minority of runs: across the seven networks trained under it, 75 of 175 on Protocol B reached the 80-epoch ceiling without early stopping, iTransformer most often (20 of 25) and EEGNet least (6 of 25), so for those runs the reported accuracy is the value at the cap rather than at convergence. The search budget is matched across the roster, so no model’s position can be attributed to having been searched harder than another. It is also the dominant cost of the project: the Protocol-A search across the roster ran for 12 days of continuous computation on a single consumer GPU. No pruner is used, because the objective reports no intermediate value and a pruner depending on one would be inert. This detail is stated because it means that no trial was cut short. The search varies the learning rate, the weight decay, the class-imbalance strategy, the data-augmentation regime (which synthetic perturbations of the training windows are applied, and how often, from the library of Section 4.7) and the mixing probabilities, with conditional groups so that the choice of regime governs which further parameters are sampled at all. The full space and the winning configurations are tabulated in Appendix C.

One parameter in that space requires a caveat, which rests on a principle governing the whole benchmark. The principle is that the decoders are compared as built: everything

around a model may be tuned, that is, the data, the preprocessing, the training configuration and the hyperparameters, but the network itself is not altered. No pruning, no structural ablation and no architectural dropout were applied to any decoder, and the model bodies come unmodified from their sources (Section 4.2). Any such change would have produced a variant of the published architecture, and a benchmark of those would no longer say anything about the architectures it names. The one structural element the benchmark adds is the shared linear classification head, which is declared in Section 4.2 and is identical for all nine. The caveat concerns head dropout, which the search sampled and carried through the configuration. A later audit of the codebase established that the value was filtered out before the model was built and so reached no decoder: every run reported in this thesis trained with the head’s default of zero. The omission was not designed but found after the runs, and Appendix A records it as such. Its effect, however, is consistent with the principle above, since a tuned dropout layer inside the head is exactly the kind of structural change the benchmark otherwise avoids. The parameter is listed here because it was part of the specified search, and it is disclosed so that its scope is not overstated. It has been left in the code, because deleting it would put the source at odds with both the archived trial records and the signed declaration.

The scope of the search differs between the three protocols and is therefore stated protocol by protocol. On **Protocol A**, every model with a search loop was explored in full: the four from-scratch networks, the three FMs as frozen probes on the shared harness, and the control. FBCSP+LDA is a deterministic classical fit with no search loop, so it has no winning configuration to find.

On **Protocol B**, five of the nine models were searched a second time, from their base form and under the same budget, on the deployment split itself (the first fold, at 20 epochs per trial): the four from-scratch networks under the shared harness, and Neuro-GPT under native preprocessing with the encoder fully fine-tuned, the regime its headline row uses. Each Protocol-B winner was then retrained across all five folds and five seeds, exactly as the Protocol-A winner had been. For these five models the headline row of Table 5.1 is therefore the better of two fully searched configurations, the Protocol-A and Protocol-B winners (for Neuro-GPT, the Protocol-A configuration carried into the native fine-tuning regime at a tenth of its learning rate), chosen by the higher five-fold mean. That choice is made on the Protocol-B test folds, and it is stated here so that it is not mistaken for a choice made on validation data. It is a two-way selection, applied by the same rule to all five models, and it can move a row by no more than the difference between its two candidates, which in this benchmark is at most about two points.

On **Protocol C**, no search of any kind was run. Every model, EEGNet included, runs the hyperparameter configuration found on Protocol A, that alone and not its split, and the reason is structural rather than economic: a search maximises validation accuracy, and in Protocol C that data comes from the training participant while its test set is a different person. Tuning would optimise within-person accuracy, which this benchmark shows predicts cross-person transfer poorly, and the only signal reflecting the actual objective is the test set itself, which tuning may not touch.

The Protocol-B search produced a finding of its own about the stability of the winning configurations. For TimesNet, S-Mamba and iTransformer the Protocol-A winner remained the better configuration on the deployment split: the Protocol-B winner scored lower by under a point for two of them and by two for the third. For EEGNet the Protocol-B winner scored higher by more than a point (83.9 against 82.6%) and was adopted, which is why its Protocol-A and Protocol-B rows come from different configurations and its within-model comparison in the headline table is the one that is not configuration-matched. The retention of the Protocol-A configuration by three of the four from-scratch networks is a positive result about the tuning rather than an omission: it shows that the winning configuration is not specific to the split it was found on. A general warning also follows from it. The Protocol-B search ran on a single fold at 20 epochs, and so short a run overfits its own validation split, so a peak obtained that way is not evidence of a reachable test ceiling. Section 5.3 encounters the same failure mode from the opposite direction.

Three models received no Protocol-B search, and the consequence is stated for each. **CBraMod** and **LaBraM** run under their native harness but reuse the configuration found for them on Protocol A, a search conducted with the encoder frozen, where their validation scores were 0.42 and 0.39. The omission was a scheduling decision, and the choice of which model to search was made on evidence. The three-arm adaptation study of Chapter 5 had already been run on the first Protocol-B fold, and under full fine-tuning it placed NeuroGPT above CBraMod and further above LaBraM. The question the fine-tuning search had to answer was whether any pretrained model could surpass the networks trained from scratch, and if the strongest of the three could not, the other two had little prospect of doing so. With the analysis deadline approaching, the search was therefore given to NeuroGPT alone. Two of the three FMs are therefore fine-tuned on hyperparameters chosen for a different regime, and their results should be read as lower bounds.

Chronos received the full search on Protocol A and then ran under a *three-epoch training cap* on Protocols B and C, whereas the other learned decoders trained for between roughly 20 and 80 epochs on average under the shared budget, with per-model means on Protocol B ranging from 19.6 (CBraMod) to 78.3 (iTransformer). This is a budget cap, not a reduced search: none was run for the control on Protocol B, and the search tooling skips it by construction. The cap dates from the regeneration of the deployment rows after the repair of the grouped split (Section 4.4), late in the schedule, when the remaining compute went to the candidates. It was set at three epochs on the recorded judgement that the control’s classification head converges within a few hundred optimisation steps, so that further epochs of the roster’s slowest model were expected to add nothing. The archived Protocol-A runs do not support that judgement: there the same model, under the same configuration, trained for 45 to 70 epochs before early stopping at patience 15 halted it, so its validation accuracy was still improving well past the third epoch. The control’s Protocol-B and Protocol-C figures are therefore lower bounds, and on this evidence loose ones, rather than configuration-matched entries, and comparisons built on them should be read accordingly. The configuration-matched statement of the control is its Protocol-A result, where the full search and budget produced 51.2%, far below every EEG-native decoder on that protocol. That result is what establishes that the control’s weakness is a property of the model on this task rather than of under-tuning.

One reporting choice requires justification, because the most direct summary of the search is misleading. The table of winning configurations is nearly uninformative: across the eight searched models the augmentation regimes split almost evenly, and two won with no augmentation at all. The distribution over the strongest 10% of *all* trials says something different: roughly four in five of them use augmentation, and nineteen in twenty apply some imbalance strategy. Read alone, the winners table would invert the conclusion. The search shows instead that augmentation and imbalance handling help in general, while the specific choice is architecture-dependent, and that is the argument for tuning each model separately rather than imposing one shared recipe.

A final observation concerns the validation scores observed during the search. Under the shared harness, the validation scores of the FMs sat between 0.39 and 0.49 while the from-scratch networks reached 0.80 to 0.88. This was visible *during* HPO, before any test set was touched. The shared harness was already off-distribution for the pretrained models at the search stage, which is what prompted the native-preprocessing study of Section 5.3.

4.6 The training procedure

Section 4.1 establishes that the infrastructure surrounding each model is identical across all nine decoders. This section details that shared infrastructure to ensure transparency and to allow the integrity of the comparison to be verified. Every evaluation protocol invokes the exact same training routine, implemented in PyTorch [55], and no element described below varies by model.

The AdamW optimiser [47] is applied exclusively to parameters requiring gradients. This

condition enables a frozen-encoder probe (Section 4.1) to be trained using the same optimiser and loop as a fully fine-tuned model, as frozen parameters are inherently excluded from the update step. The learning rate is halved if the validation balanced accuracy does not improve for five consecutive epochs, down to a minimum floor of 10^{-6} , and gradients are clipped to a global norm of 1.0 at every step. Training proceeds for a maximum of 80 epochs, employing an early stopping mechanism triggered after 15 epochs without improvement. Both the learning-rate schedule and the early stopping criterion are evaluated against validation balanced accuracy rather than its loss, aligning directly with the primary metric of the benchmark. Test set evaluation uses the weights from the best-performing validation epoch rather than the final one, ensuring that models are assessed at their optimal state prior to any potential late-stage overfitting. To capture seed-to-seed variability, each configuration is trained across five random seeds: 42, 123, 456, 789, and 1234.

The training loop enforces two constraints on class imbalance and data augmentation. First, a class-weighted loss and a balanced sampler are mutually exclusive during any single run, because both mechanisms address the identical class imbalance and applying them simultaneously would result in an over-correction. Second, data augmentation is exclusively applied to the training loader, ensuring that validation and test windows preserve the unaltered recorded signals.

Ensuring that a seed governs the outcome of a run requires controlling all sources of stochasticity, which necessitates four levels of seeding. Initially, the process is seeded at start-up. Each training run subsequently reseeds PyTorch, NumPy, the Python standard library, and CUDA. A dedicated random-number generator controls both the sampler draws and the loader shuffle, guaranteeing that the batch order is reproduced for a given seed even if the global generator state is altered elsewhere. Finally, each data-loader worker is seeded individually. This granular seeding is critical because PyTorch natively assigns each worker a distinct seed for its internal generator but leaves the NumPy one untouched. Since the data augmentations are implemented via NumPy, omitting this per-worker initialisation would result in all workers drawing identical noise, leading to correlated augmentations within a batch. The worker initialiser also iterates through each transform in the augmentation pipeline, rebuilding its cached generator using a combination of the worker seed and the transform’s specific identity. This guarantees that distinct transforms within the same pipeline draw from independent random streams.

The default batch size is set to 32 windows and is dynamically increased where GPU memory permits. Chronos is the sole exception, constrained to the default 32 windows because its architecture processes the 19 channels sequentially through its backbone, exceeding available memory limits at larger batch sizes. Mixed precision is implemented using bfloat16 on compatible hardware and float16 otherwise. A gradient scaler is applied exclusively in the float16 configuration to prevent potential underflow caused by its narrower numeric range. Furthermore, graph compilation is globally disabled for all models. While graph compilation fails for only three of the nine models (S-Mamba, Neuro-GPT, and Chronos) due to implementation-specific limitations unrelated to the underlying decoders, compiling only a subset of the models would introduce an inequality in training conditions contrary to the experimental design. Disabling compilation universally reduces computational throughput but ensures standardised evaluation, and it is not expected to alter the final performance metrics.

A final consideration concerns two hardware-level optimisations that limit reproducibility. Convolution autotuning is left enabled, and reduced-precision matrix multiplication is permitted on supporting hardware. Consequently, training runs are reproducible in distribution (an identical seed yields the exact same configuration and statistically equivalent results) but are not strictly bitwise reproducible. In instances where a run exhausts GPU memory, it is explicitly recorded as a skipped placeholder rather than being silently omitted. The statistical analysis excludes these placeholder rows. No run in the reported results was skipped for this reason.

4.7 Data augmentation

Given the constraint of a few hundred recordings per class, data augmentation functions as a necessary training component rather than an optional refinement, and the HPO treats its configuration as one of the searched axes. A custom augmentation library developed for this project implements 23 per-sample transforms. Each transform operates on a single channels-by-time window and maintains an independent, seeded random-number generator. Five of these transforms were added later, modelled on the catalogue of the tsai time-series library [53] at the supervisor’s recommendation: magnitude warp, time jitter, window warp, time shift and multiplicative noise. These additions introduce smooth gain drift, temporal jitter and sub-window warping, three perturbation modes absent from the original set. Of the 23 transforms, 18 are exposed to the search. The remaining five are withheld because they generate implausible perturbations of a recording. Segment permutation, for instance, preserves the frequency spectrum but destroys the temporal order of an EEG window.

The dataset’s item access method applies augmentation dynamically at the moment a training window is loaded. Every training epoch therefore processes a freshly perturbed version of each window, and the system stores no augmented copies. Transforms are restricted to the training loader, so validation and test windows retain the unaltered recorded signal. Batch-level mixing operates one level higher, within the collate function that assembles a complete batch from individual windows.

Exposing 18 heterogeneous transforms to a single search requires a control parameter that carries the same meaning across all of them. A single magnitude defined on the interval $[0, 1]$ drives every registered transform, where 0 leaves the window untouched and 1 applies the maximal perturbation. For most transforms, the magnitude scales a perturbation upward from zero. Two mappings require inversion, because tightening an amplitude clip and shrinking a random crop constitute stronger interventions as their native parameter values decrease. These inversions ensure that a higher magnitude denotes a stronger perturbation uniformly across the search space.

The library further groups the transforms into three pools of increasing breadth, containing 7, 14 and 18 respectively. The HPO selects one of four augmentation regimes: none at all, a single transform with an independent magnitude (0.1 to 1.0) and application probability (0.2 to 0.9), a pair of transforms governed by separate magnitudes and probabilities, or a multi-transform policy that draws between one and four distinct transforms from a designated pool and applies them at a shared magnitude (0.1 to 0.9). The selected regime dictates which subsequent parameters the search samples, forming the conditional groups established in Section 4.5.

Two batch-level mixing operations, MixUp [79] and CutMix [77], constitute a separate search axis and are sampled independently of the augmentation regime. Each operation is assigned an application probability between 0 and 0.5 and a shared mixing parameter between 0.1 and 0.5. MixUp blends two windows convexly and assigns the target label in identical proportion. CutMix conventionally replaces a rectangular region of one training example with the matching part of another. Applied to a channels-by-time array, a rectangular region would mix data across channels as well as over time. Such an operation lacks physiological validity for scalp EEG: a montage in which a subset of electrodes captures one movement while the rest record another corresponds to an impossible physical state.

To preserve spatial integrity, the CutMix variant implemented here replaces a contiguous slice of time across all 19 channels simultaneously and never mixes a subset of them. The collate function recomputes the mixing coefficient from the integer length of the temporal slice actually cut, so the soft label matches the applied mask exactly. The pipeline applies at most one of the two mixing operations to any given batch, and bypasses both where fewer than two windows are present, where pairing is impossible.

The search manages class imbalance by selecting one of four mutually exclusive strategies: no correction, a class-weighted loss, a balanced sampler, or a targeted augmentation strategy that perturbs only the under-represented classes. A class is under-represented when its total window count within the training split falls strictly below the median class

count. This median rule is evaluated exclusively on the training split, after the dataset has been partitioned, because evaluating it across the entire dataset would leak characteristics of the test set into the training recipe.

4.8 Metrics, including within-category direction accuracy

Balanced accuracy serves as the primary metric of the benchmark. It is defined as the mean of the per-class recalls, each being the fraction of a class’s windows that the decoder labels correctly. Idle is over-represented in the raw recordings, and an unweighted accuracy would disproportionately reward a decoder for predicting the majority class. Balanced accuracy removes that reward, because every class contributes one fifth of the score regardless of its size. The reference points are the chance level, 20 % on the five-class Protocols A and B and 50 % on the two-class transfer test of Protocol C, and the ceiling of 100 %.

Three further aggregate metrics accompany the primary score, so that the argument does not rest on a single number. Macro F_1 is the per-class harmonic mean of precision and recall, averaged over the classes, and therefore weighs false positives and false negatives equally. Cohen’s κ rescales the observed agreement between prediction and truth against what chance would produce. It is 0 for a decoder at chance and 1 for perfect agreement, so it expresses performance independently of the chance level. Macro one-versus-rest area under the receiver operating characteristic curve (ROC-AUC) treats each class in turn as positive against the remaining four, measures how well the decoder’s confidence for that class separates its windows from all others across every possible decision threshold, and averages the result. It is 0.5 at chance and 1 for perfect separation. Because it is threshold-independent, it describes separability across the whole confidence range, not only at the threshold enforced by the deployment gate of Section 3.8.

Three per-class quantities are reported, because they answer different clinical questions. Recall is usability: a low Forward value means the device does not respond when the user intends to move. Specificity, the fraction of windows belonging to other classes that are correctly not assigned to a given class, is safety: low specificity means the device moves when the user did not intend it. Precision, the fraction of issued commands that name the true class, is trust: it states whether a command, once issued, can be acted on. A single aggregate accuracy conceals all three, which is why Section 5.7 reports them separately.

Calibration in the probabilistic sense, which is distinct from the calibration data a new user records, is reported alongside accuracy, because the decision layer of the deployment pipeline rejects any prediction whose confidence falls below a threshold of 0.80 (Section 3.8). A gate is only meaningful if the confidences it reads are trustworthy, and in a device that moves a patient’s legs a confidently wrong prediction is more dangerous than an uncertain one. Two distinct quantities measure this reliability. Expected calibration error is the average gap between the confidence a model states and the accuracy it realises. Computed in its top-label form, the metric groups windows into 15 equal-width confidence bins and averages the gap between mean confidence and observed accuracy over these bins, weighted by bin size [28]. A perfectly calibrated model achieves a score of 0. The Brier score is the squared distance between the full predicted class distribution and the true one-hot label, summed over the five classes and averaged over windows. It is 0 for a perfect model, 2 at worst, and 0.8 for one that always predicts the uniform distribution, the value an uninformative decoder approaches.

Temperature scaling, the correction proposed in the same work, divides the raw pre-softmax outputs (the softmax being the final step that turns the network’s outputs into probabilities summing to one) by a single scalar fitted on validation data and is implemented in the deployment decoder. It was fitted only within a deployment-targeted improvement track whose results are retired, and no fitted temperature accompanies the archived checkpoints. All calibration figures in this thesis therefore describe raw softmax confidences, the same confidence the deployment gate reads.

Within-category direction accuracy, the metric on which this thesis relies most, is not part of the standard panel. A decoder can score well above chance on five-way balanced accuracy by resolving only the movement *category* (Idle, or the forward–backward and left–right axes) while guessing the direction within a category at random. Such a decoder recalls Idle perfectly and each of the four directions at 50 %, yielding a balanced accuracy of 60 % without resolving any of them at all. Overall accuracy therefore cannot establish that direction is being decoded. Within-category direction accuracy restricts evaluation to the windows whose true label belongs to one pair, forward and backward or left and right, and scores the decoder’s *full five-way* prediction as correct only when it names the right member of that pair. A true Forward window predicted as Idle, Left or Right counts as wrong. Because the decoder is never given the two-way subproblem, a model that only detects the presence of movement remains at the 50 % floor by construction.

One shared implementation computes this metric for every neural decoder, so the figures cannot drift between studies, and the result accompanies every Protocol-B headline. The classical pipeline is evaluated through a separate path that does not call this implementation, and its values are recovered from its stored confusion matrices, as Section 5.2 states where they appear. Protocol C requires no separate directional figure, because the transfer task there *is* a single direction pair, forward against backward, so its headline is already a within-pair direction accuracy. Section 5.2 treats this metric as the validity precondition for everything that follows.

The evaluation omits every metric that lacks statistical validity for a given protocol. Protocol-A calibration (expected calibration error and Brier score) comes from the same re-score restricted to the primary participant as the other Protocol-A metrics, and Appendix C tabulates it beside them. ROC-AUC is omitted for Protocol C, for which the evaluation loop stored no value. Because the task is two-class, Protocol C carries no left–right figure at all, and its forward–backward one is the headline itself. Such metrics are omitted rather than reported as zero, so that an absent value is not misread as a measurement.

4.9 Statistical treatment and the consequences of five folds

This section states how the benchmark’s figures are compared. It first fixes the unit of statistical analysis, the fold rather than the individual training run, because that choice determines the width of every confidence interval and the differences the tests can detect. It then describes the tests applied, a Friedman test followed by Holm-corrected Wilcoxon signed-rank tests, shows why a design with five folds cannot yield a significant pairwise difference, and sets out the effect sizes and confidence intervals that Chapter 5 reports instead, together with the comparative statements this evidence permits.

The unit of statistical analysis is the fold. Protocol B generates 25 measurements per stochastic decoder, resulting from the five folds detailed in Section 4.4 crossed with five random seeds. Treating all 25 runs as independent samples would be a statistical error. The five seeds within a given fold train and test on the same data split, so their variance measures initialisation stochasticity rather than data variability. The repeated-measures tests applied below require independence between evaluation blocks, because they assess whether performance differences remain consistent across data subsets. The analysis therefore averages the seed results within each fold, yielding one mean apiece and five matched blocks per model. Protocol A contains no cross-validation folds, so each seed operates as its own block.

The choice of unit changes the width of every confidence interval, and Table 5.1 shows the consequence. Bootstrapping over all 25 individual runs yields a narrow confidence interval for the leading model, whose lower bound lies above every rival’s mean. Bootstrapping over the five fold means produces a 95 % interval of [83.9, 86.6] % for the leader, whose lower bound falls below the second-placed fold mean of 84.0 %. Under this evaluation, the leader no longer separates from the group beneath it. Both calculations are arithmetically valid, but they answer different questions, and only the fold-level aggregation addresses the one

being asked, namely whether an advantage holds across data splits. Chapter 5 defines the unit underlying any reported interval, and every comparative claim relies on the fold-level aggregation.

The statistical analysis employs the standard non-parametric procedure for comparing multiple classifiers [20], whose background Section 2.7 gives. A Friedman test ranks the decoders from best to worst within each block and assesses whether the mean ranks across all blocks diverge more than chance would predict. As an omnibus test, a rejected null hypothesis indicates that at least one decoder differs from another, without identifying the pair. Pairwise Wilcoxon signed-rank tests then isolate two decoders at a time, ranking the absolute paired differences across the blocks to determine whether their signs are consistent. To address the multiple-comparisons problem, the analysis applies a Holm correction [32]. This procedure sorts the raw p -values and scales each by the number of remaining unexamined comparisons, which controls the family-wise error rate, the probability of at least one false positive across the entire set.

The methodology deliberately departs from common practice in one regard: post-hoc tests based on mean ranks are excluded. The significance such tests assign to a given pair depends on which *other* decoders are present in the comparison [8], and their use would make every pairwise result contingent upon the inclusion of the non-EEG control. The Wilcoxon signed-rank test preserves that independence, as its calculation depends only on the two decoders under comparison.

Applying the Friedman test across the nine decoders over five folds rejects the null hypothesis of equivalent performance, yielding $\chi^2 = 33.12$ and $p = 5.86 \times 10^{-5}$. The corresponding test for Protocol A corroborates this finding ($\chi^2 = 31.00$, $p = 6.22 \times 10^{-5}$) across the eight stochastic decoders, excluding the deterministic classical baseline, which has a single run and hence no variance to test. The practical consequence of this omnibus rejection remains limited. No pairwise comparison survives correction: every Holm-corrected Wilcoxon test returns a corrected p -value of one. This null result persists when the leading decoder is compared against the non-physiological control, despite a mean difference of 46.2 percentage points.

This null result reflects a floor on the attainable p -value, a limitation of the experimental design rather than of the decoders. With five matched blocks, the Wilcoxon statistic can assume only one of $2^5 = 32$ equally likely sign patterns under the null hypothesis, so even a decoder that wins on all five folds yields a minimum two-sided p -value of $2/32 = 0.0625$. This floor exceeds conventional significance thresholds before any correction, and applying the Holm procedure across the 36 comparisons of Protocol B (28 on Protocol A) removes what remains. A non-significant p -value in this context denotes an *underpowered* design rather than the absence of an effect, and the analysis interprets it accordingly.

Because null-hypothesis significance testing cannot discriminate with so few blocks, the analysis reports effect sizes and confidence intervals, the practice Section 2.7 presents as the defensible standard when independent evaluation units are scarce. The effect size is paired Cohen’s d , calculated as the mean of those differences across the matched blocks divided by their standard deviation. Under this formulation, a value of 1 indicates that the leader’s advantage equals its own block-to-block variability. The metric maps to conventional interpretive bands: below 0.2 is negligible, 0.2 to 0.5 small, 0.5 to 0.8 medium, and 0.8 or above large [16]. Cohen’s d retains its interpretive validity at any sample size, a property that p -values lose at this one. A bootstrap confidence interval accompanies each reported effect size.

The evaluation constructs confidence intervals with a percentile bootstrap. The procedure resamples the matched blocks with replacement 10 000 times, recomputes the statistic on each resample, and reads the interval from the 2.5th and 97.5th percentiles of the resulting distribution. The intervals presented in the main results table (Table 5.1) rely on 2 000 resamples computed over all 25 individual runs, and they describe seed-induced initialisation variability rather than the fold-level uncertainty governing the comparisons. The pipeline computes fold-level intervals and Holm-corrected pairwise tests for both the within-

category direction axes and the overall balanced accuracy, whereas effect sizes are derived for the five-way axis only.

The evidence described above permits a precise set of comparative claims and excludes others. Decoders whose fold-level confidence intervals overlap the leader’s mean are defined as statistical ties and reported accordingly. The analysis identifies the decoder with the highest mean and quantifies the magnitude of performance differences alongside their uncertainties, but it refrains from claiming statistical separation between the leader and the cluster beneath it, because at this sample size such separation remains undemonstrated. All rankings report the rank, the confidence interval and any statistical ties. Balanced accuracies and other percentages are given to a single decimal place, as the confidence intervals span roughly an order of magnitude wider than a second decimal could resolve.

Chapter 5

Results

5.1 Overview of the benchmark results

Table 5.1 presents the complete benchmark. Each row corresponds to one decoder in the strongest configuration permitted by the comparison framework of Chapter 4. The networks trained from scratch use the deployment preprocessing with the superior of their Protocol-A and Protocol-B configurations (Section 4.5), and the three EEG FMs use their own native preprocessing with fully fine-tuned encoders. All other experimental variables remain constant across the decoders: the split procedure, the folds, the seeds, the training budget and the metric panel.

The three columns address distinct questions and assume different chance levels. Protocol A trains and tests within the same pool of recordings, establishing the within-subject performance ceiling. Protocol B holds out entire recordings, the unit of the calibration workflow, and is the deployment scenario, the condition a calibrated decoder operates under. Both are five-class problems against a chance level of 20 %. Protocol C evaluates transfer to a second participant whose recordings were excluded from training. As a two-class problem with a chance level of 50 %, its results are not directly comparable with those of the other two protocols, and they are analysed separately in Section 5.8. Figure 5.1 shows the outcomes across all three protocols.

Under Protocol B, the deployment scenario, the decoders are closely grouped. TimesNet [74] achieves the highest mean accuracy at 85.5 %. The next five decoders (S-Mamba [72] at 84.0 %, Neuro-GPT [17] at 83.9 %, EEGNet [42] at 83.9 %, iTransformer [44] at 82.9 %, and the classical FBCSP+LDA baseline [4, 3] at 82.8 %) all lie within 2.7 percentage points of the leader. This narrow spread constitutes the central finding of the benchmark. The leading group encompasses a purpose-built EEG convolutional network, three general time-series architectures trained from scratch, an optimally adapted EEG FM, and a classical spatial-filter pipeline with no deep learning and no HPO. Below this group, CBraMod [70] scores 80.1 %, followed by LaBraM [35] at 65.4 %. Chronos [5], the control whose pretraining contains no physiological data, reaches 39.3 % against a chance level of 20 %, placing it 46.2 points below TimesNet. That row was produced under the control’s capped training budget (Section 4.5) and is interpreted accordingly in Section 5.4. Section 5.3 examines whether the leading model’s margin is statistically supported. This section makes no separation claim, and Figure 5.2 shows the interval against which any such claim must be read.

Two further properties of the table are established before the detailed analysis. First, for most decoders the results of the two five-class protocols lie within a few percentage points of each other, and two rows (Neuro-GPT and the classical baseline) score *higher* under the more demanding Protocol B, which indicates that the within-subject ceiling does not result from data leakage that Protocol B then removes. Second, the full metric panel for the deployment scenario (Table 5.2) preserves the structure of the ranking under macro F_1 , Cohen’s κ [15] and ROC-AUC: κ reproduces the balanced-accuracy order exactly, whereas F_1 and ROC-AUC exchange only adjacent members within the leading group, preserving the standing of the top model, of that group and of the three rows below it. The conclusions of this chapter are therefore independent of the selection of balanced accuracy as the primary metric.

Two properties that Table 5.1 cannot show are established in the sections that follow. Every EEG decoder, LaBraM only narrowly, resolves direction within a movement category above the 50 % two-way floor, TimesNet at 85.9 % (RQ0, Section 5.2). The same decoder

Table 5.1: Balanced accuracy (%) for all nine decoders across the three evaluation protocols, ranked by Protocol B. Mean $\pm$ standard deviation.

Decoder	A: within-subject	B: cross-recording	C: transfer	Status
TimesNet	88.8 ± 1.4	85.5 ± 2.1	56.2	comparable
S-Mamba	84.7 ± 1.4	84.0 ± 1.5	53.1	comparable
Neuro-GPT	82.1 ± 1.4	83.9 ± 2.0	43.0	comparable
EEGNet	84.1 ± 1.1	83.9 ± 3.0	55.9	comparable
iTransformer	84.0 ± 0.9	82.9 ± 1.2	51.0	comparable
FBCSP+LDA	80.7 ± 0.0	82.8 ± 2.0	50.2	comparable
CBraMod	82.8 ± 1.2	80.1 ± 1.9	51.9	comparable
LaBraM	68.2 ± 3.5	65.4 ± 2.7	49.1	comparable
Chronos	51.2 ± 0.8	39.3 ± 2.6	48.1	control

Chance is 20% for Protocols A and B (five classes) and 50% for Protocol C (two classes), so Protocol C is not comparable with the other two. Each decoder is evaluated in its optimal native configuration: the from-scratch networks under deployment preprocessing with tuned hyperparameters, the three EEG foundation models under their own native preprocessing with the encoder fully fine-tuned. Chronos is a non-EEG negative control, not a competitor. Every value is read from the archived benchmark record at build time rather than transcribed by hand. Appendix D describes that record and the checks that keep this table from drifting from it.

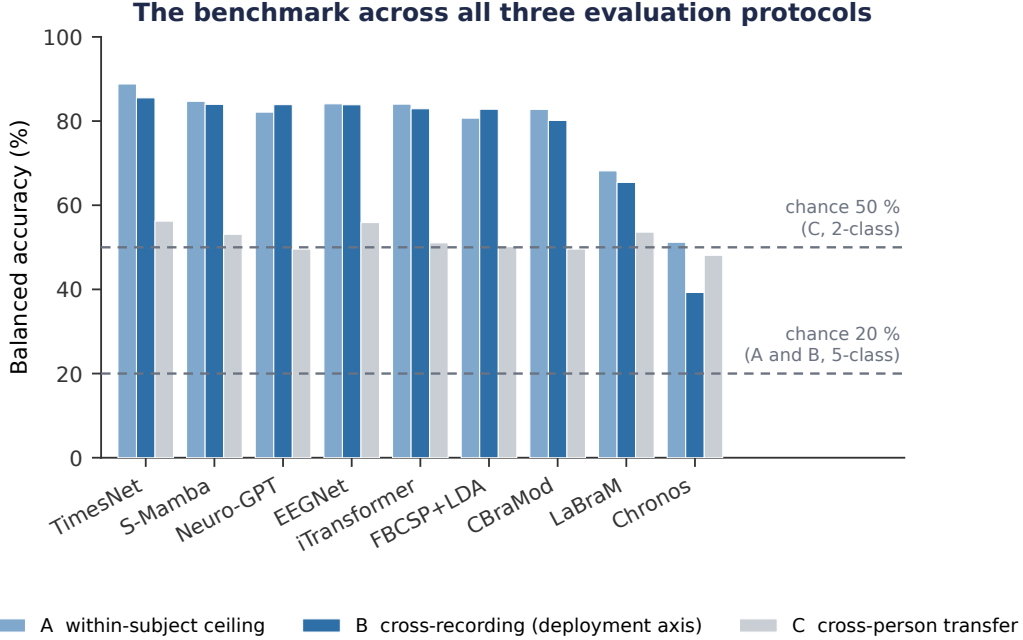


Figure 5.1: Balanced accuracy for every decoder across the three evaluation protocols, ordered by Protocol B performance. The difference between the first two bars in each group is the change produced by whole-recording holdout, and the third bar is the accuracy that remains after a change of participant.

classifies a window with a margin of $16.7\times$ inside the 0.5s decode cadence, where a confidence gate that discards 23.7% of them raises balanced accuracy on those it retains from 85.5% to 94.1% (RQ4, Section 5.6).

Table 5.3 gives the same panel under Protocol A, the within-subject ceiling. Every value comes from the re-score of the archived checkpoints restricted to the primary participant, the two calibration metrics included. For the classical baseline alone the row file carries a stale copy of the pre-rescore values, so its expected calibration error and Brier score are recomputed from the rescored posteriors stored in the same row, with the 15-bin definition

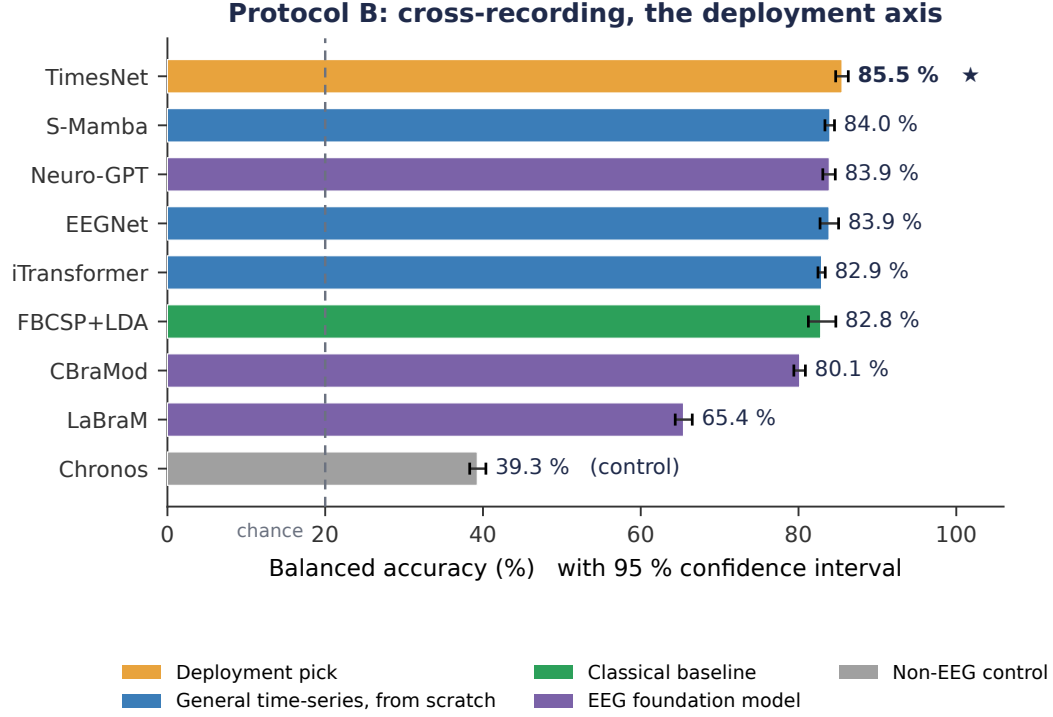


Figure 5.2: Results under Protocol B with 95 % confidence intervals. The intervals shown are bootstrapped [23] over the 25 fold-by-seed runs. The comparative analysis of §5.3 relies on the wider interval computed over the five fold means, under which the leading model does not statistically separate from the cluster beneath it.

of Section 4.8. The assembled benchmark file records Protocol-A calibration as absent for the six shared-harness decoders, a note that is wrong about its own upstream rows and is disregarded here.

The two metrics are the ones the confidence gate depends on. Expected calibration error is the mean gap between the confidence a decoder states and the accuracy it realises, and the Brier score is the mean squared error of its probability vector against the true class,

Table 5.2: Full metric panel on Protocol B, the deployment axis. Balanced accuracy with its 95 % confidence interval, macro F_1 , Cohen’s κ , macro one-versus-rest ROC-AUC, expected calibration error and Brier score.

Decoder	Balanced acc. [95% CI]	macro F_1	κ	ROC-AUC	ECE ↓	Brier ↓
TimesNet	85.5 [84.7, 86.3]	0.855	0.819	0.973	0.031	0.215
S-Mamba	84.0 [83.4, 84.6]	0.839	0.800	0.972	0.052	0.237
Neuro-GPT	83.9 [83.1, 84.7]	0.838	0.799	0.968	0.088	0.256
EEGNet	83.9 [82.7, 85.1]	0.839	0.798	0.972	0.027	0.232
iTransformer	82.9 [82.5, 83.4]	0.828	0.787	0.967	0.025	0.245
FBCSP+LDA	82.8 [81.3, 84.7]	0.829	0.785	0.966	0.084	0.270
CBraMod	80.1 [79.4, 80.9]	0.801	0.752	0.955	0.116	0.314
LaBraM	65.4 [64.4, 66.5]	0.652	0.568	0.876	0.263	0.596
Chronos	39.3 [38.3, 40.4]	0.325	0.241	0.716	0.138	0.754

The interval shown is bootstrapped over the 25 fold-by-seed runs. Comparative claims in this thesis use the interval bootstrapped over the five *fold means* instead, because seeds within a fold share a data split and are not independent. Under that interval TimesNet leads a cluster it does not separate from. Calibration is reported alongside accuracy because the confidence gate of the live pipeline is only meaningful if the probabilities it reads are trustworthy.

Table 5.3: Full metric panel on Protocol A, the within-subject ceiling.

Decoder	Balanced acc. [95% CI]	macro F_1	κ	ROC-AUC	ECE	Brier	Harness
TimesNet	88.8 [87.7, 90.0]	0.887	0.860	0.972	0.028	0.176	lg-only re-score
S-Mamba	84.7 [83.3, 85.7]	0.846	0.809	0.971	0.042	0.230	lg-only re-score
EEGNet	84.1 [83.2, 85.0]	0.841	0.801	0.968	0.027	0.235	lg-only re-score
iTransformer	84.0 [83.2, 84.7]	0.839	0.800	0.967	0.027	0.237	lg-only re-score
CBraMod	82.8 [81.7, 83.6]	0.829	0.785	0.966	0.074	0.261	native (R2-FT)
Neuro-GPT	82.1 [80.8, 83.3]	0.820	0.777	0.967	0.102	0.283	native (R2-FT)
FBCSP+LDA	80.7 [80.7, 80.7]	0.805	0.759	0.960	0.097	0.286	lg-only re-score
LaBraM	68.2 [64.6, 71.1]	0.679	0.602	0.895	0.240	0.550	native (R2-FT)
Chronos	51.2 [50.5, 51.9]	0.498	0.390	0.794	0.088	0.626	lg-only re-score

Every value is the re-score of the archived checkpoints restricted to the primary participant, calibration included. For FBCSP+LDA the row file carries a stale copy of the pre-rescore calibration fields (0.390 and 0.940), so its expected calibration error and Brier score are recomputed here from the rescored posteriors stored in the same row, with the 15-bin definition of the metric panel. Balanced accuracy, macro F_1 , κ and ROC-AUC are read from the rows directly. The three foundation models are shown on their own native preprocessing with the encoder fully fine-tuned, while the other six are on the shared harness. FBCSP+LDA is deterministic, so its interval is degenerate. The formal statistics behind this panel are archived with the results (Appendix D).

so a gate that reads stated confidence is only as safe as these two values. Read beside the deployment panel, they show calibration to be a property of the decoder rather than of the split. The leading decoders keep almost the same expected calibration error under the ceiling split as under the deployment one (TimesNet 0.028 against 0.031, EEGNet 0.027 against 0.027, iTransformer 0.027 against 0.025). LaBraM’s error is an order of magnitude larger under both (0.240 and 0.263), and the classical baseline lies between the leading decoders and LaBraM on both protocols (0.097 and 0.084). TimesNet’s Brier score is the lowest of the nine under both protocols (0.176 and 0.215).

5.2 Evaluation of directional decoding accuracy

Prior to interpreting the rankings in Table 5.1, it is necessary to confirm whether the decoders resolve *direction* or only detect the presence of movement. Overall balanced accuracy is insufficient to address this question. A decoder that consistently identifies the correct movement category (Idle versus the four directional classes) but guesses the direction within that category at random would achieve a balanced accuracy of $(1 + 4 \times \frac{1}{2})/5 = 60\%$, three times the chance level, without resolving direction at all. Executed movement produces broadband, montage-wide signal changes that readily separate “moving” from “still” states, rendering this failure mode a practical possibility rather than a hypothetical scenario. A high overall score is therefore compatible with a decoder that has not learned the directional mapping required for a gait-trainer controller. This section establishes the answer to RQ0, upon which all subsequent analyses rely.

The claim examined here is limited to whether the recorded signal supports *directional decoding*. It makes no assertion regarding the anatomical origin of the discriminative signal, which cannot be localised through the present analysis, and the question of signal origin is addressed in Chapter 6.

Within-category direction accuracy

The primary metric is within-category direction accuracy: among windows whose true class is Forward or Backward, the fraction for which the decoder’s full five-way prediction correctly classifies the target direction, and identically for Left versus Right. Scoring on the five-way argmax, the single class assigned the highest probability by the decoder, is essential. Because the decoder is not presented with a two-way subproblem, a prediction counts as correct only if the true direction receives the highest probability among all five

Table 5.4: Within-category direction accuracy on Protocol B. A window counts as correct only if the full five-way prediction equals the true direction.

Decoder	Forward / Backward	Left / Right	Mean
TimesNet	84.3	87.5	85.9
S-Mamba	82.3	81.7	82.0
Neuro-GPT	82.4	82.7	82.5
EEGNet	81.4	82.2	81.8
iTransformer	80.9	81.2	81.0
FBCSP+LDA [†]	82.5	80.7	81.6
CBraMod	80.6	76.6	78.6
LaBraM	66.1	56.1	61.1
Chronos	33.2	27.2	30.2

The two-way floor is 50 %. A decoder that identified movement *category* without resolving direction inside it would score at that floor while still reaching roughly 60 % overall balanced accuracy, which is why overall accuracy cannot answer this question. [†]FBCSP+LDA’s values are recovered from its per-fold confusion matrices: the scikit-learn path never populated the direction fields. The recovery formula reproduces the stored values exactly for the other eight decoders.

classes. By construction, a decoder that detects movement but fails to resolve its direction remains at the 50 % two-way floor.

Table 5.4 and Figure 5.3 present the results under the deployment protocol. Every EEG decoder exceeds the two-way floor. Within the leading group the mean accuracy across the two directional axes ranges from 81.0 % (iTransformer) to 85.9 % (TimesNet), with the classical baseline achieving 81.6 %. CBraMod reaches 78.6 %, and LaBraM scores lower at 61.1 %, but both remain above the floor, consistent with their placements in the main results table. For five of the eight EEG decoders, performance on both directional axes agrees within two percentage points. TimesNet’s left–right accuracy exceeds its forward–backward score by approximately three points, whereas for CBraMod and LaBraM the left–right figures (76.6 % and 56.1 %, respectively) lag behind the forward–backward ones by several points. Chronos, the non-physiological control model, falls *below* the floor at 30.2 %. This occurs because, for windows routed to the incorrect movement category, the directional information within the true one is lost as well. This performance profile is the signature of a model that detects gross signal changes without resolving which movement occurred. The capped deployment budget of the control (Section 4.5) does not account for this degradation: under the full training budget of Protocol A, its two-way directional accuracies remain below the 50 % floor. The FBCSP+LDA values are derived from its per-fold confusion matrices, because its evaluation path does not store the direction fields, and the table footnote documents the validation of this derivation.

Table 5.4 yields a secondary observation. The left–right axis exhibits the largest advantage for the recommended decoder: TimesNet’s 87.5 % leads the nearest rival by 4.8 points. This margin is wider than the 1.5-point lead it holds in overall deployment accuracy, indicating that the axis on which the decoders differ most is directional resolution itself rather than gross movement detection. The evaluation definition employed here is also the conservative one. A more permissive variant, which restricts scoring to predictions already classified within the correct axis pair, yields higher values for every decoder (90.3 % and 96.2 % for TimesNet). Under this same permissive reading, the control model reaches only 57.7 % and 52.9 %. Because these values lie above the two-way floor, the below-floor signature noted above is specific to the stricter five-way scoring, whereas the substantial margin separating the control from every EEG decoder persists under either definition.

These findings answer RQ0: all eight EEG decoders resolve direction, so the rankings that follow compare directional decoders rather than movement detectors.

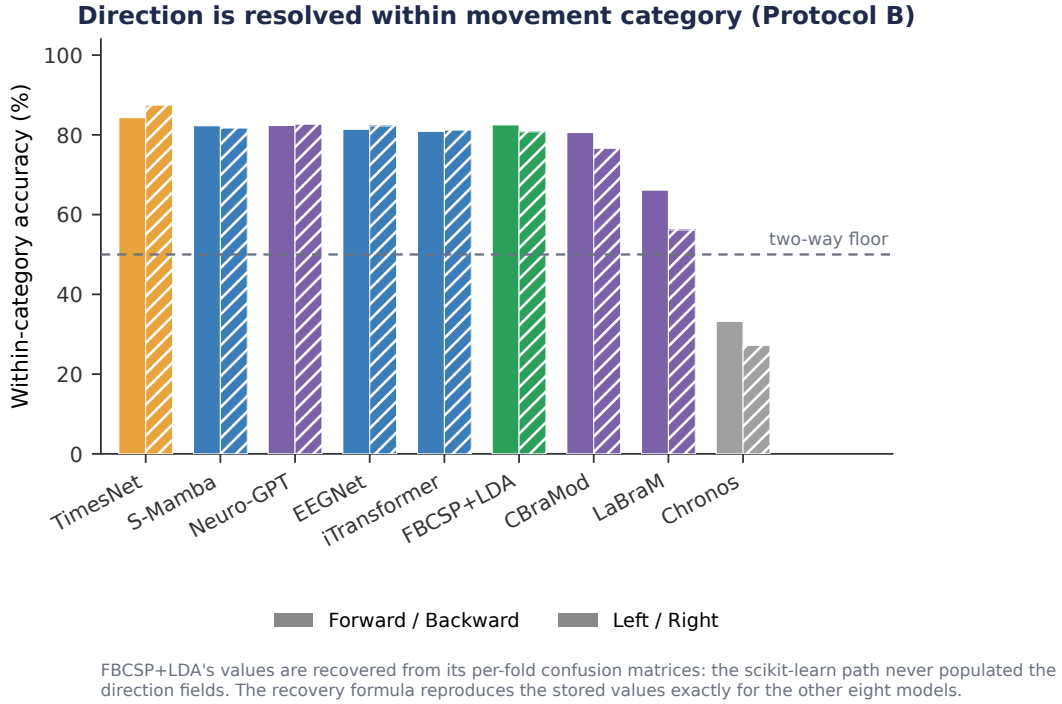


Figure 5.3: Directional resolution within each movement category under Protocol B. Every EEG decoder scores above the two-way floor, whereas the non-physiological control falls below it. This distinction separates directional decoding from the detection of motion.

Direction decoding under sensorimotor-band restriction

A control experiment evaluates whether directional decoding is maintained when the input is restricted to the μ and β sensorimotor rhythms [57]. This manipulation alters only the passband, and channel count, montage, windowing and labels remain unchanged, so, unlike a spatial ablation, the comparison is not confounded by changes in input dimensionality. A spatial ablation restricting the montage to subsets of electrodes was also conducted, but it is excluded here because the accompanying reduction in channel count confounds the hypothesis under test. Section 6.6 details the limits of that experiment.

This control was executed with a reduced budget (Protocol B, a single fold, five seeds, with only the learning rate and weight decay carried over from the main search), so its absolute values are comparable only within Figure 5.4 and cannot be directly compared with the main results table. The results reveal a split across architectures. EEGNet retains 95% of its broadband direction accuracy under the restriction (78.8% to 74.5%), and TimesNet retains 84%, with both remaining well above the two-way floor. S-Mamba and iTransformer lose their directional signal: their restricted direction accuracy falls to 38.3% and 35.6%, dropping below the floor from broadband values that were well above it. For the three FMs, the broadband values in this reduced-budget setting already fall below the floor, so the restriction is uninformative for them and no retention percentages are quoted. This split establishes that for the two highest-performing from-scratch architectures a usable proportion of the directional signal resides within the sensorimotor band, whereas the two forecasting architectures that decode direction from broadband input rely on spectral content outside this band. The implications of this dependence, and the limits of what it can establish regarding signal origin, are discussed in Chapter 6.

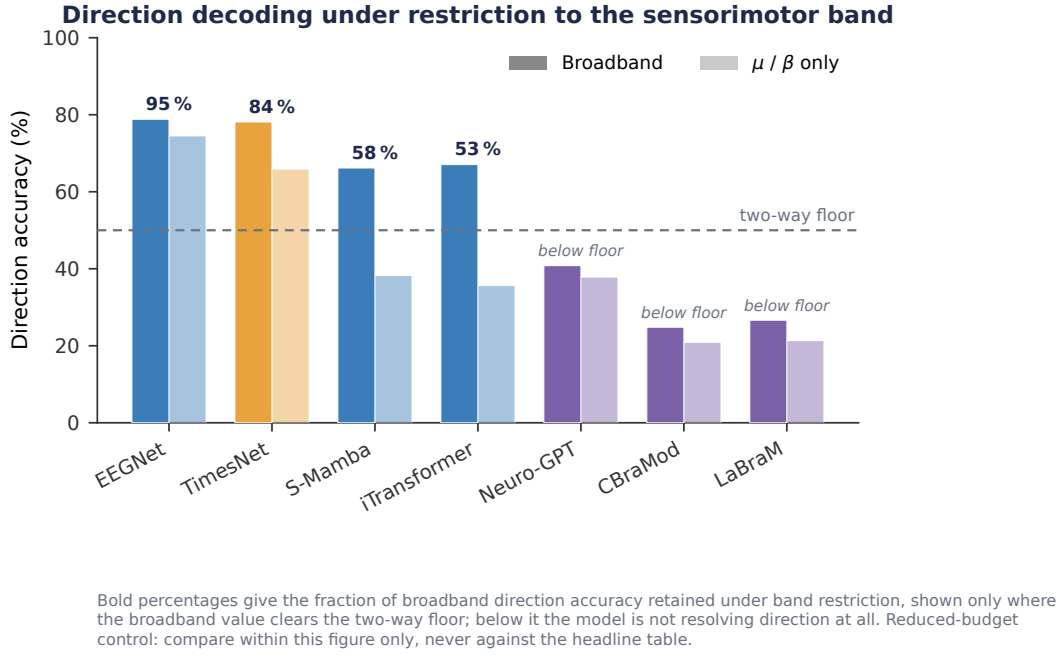


Figure 5.4: Direction accuracy with the passband restricted to the μ and β sensorimotor rhythms. Because the channel count remains unchanged by this manipulation, the results are not confounded by montage size, unlike a spatial ablation.

5.3 Pretrained models compared with decoders trained from scratch

This section addresses RQ1, the benchmark’s principal question. A fair evaluation required granting the three EEG FMs every advantage the comparison framework permits: they are evaluated using their native preprocessing with fully fine-tuned encoders. The performance gain derived from this adaptation is substantial, and the position the models then occupy constitutes the result.

Table 5.5 and Figure 5.5 separate the two factors that could account for this performance. With a frozen encoder, all three models perform near the chance level under Protocol A: 29.5 %, 29.6 % and 30.2 % against a chance level of 20 %. Full fine-tuning substantially alters this outcome. Unfreezing the encoder increases Neuro-GPT’s accuracy by 52.6 percentage points, CBraMod’s by 53.2 and LaBraM’s by 38.0. Fine-tuning, and not the native preprocessing, recovers these models, since under that regime with a frozen encoder they remain near chance, which indicates that the pretrained representations offer minimal utility to a frozen linear probe on this task.

LoRA [33] lies between these two regimes and yields inconsistent results: it produces negligible gains for Neuro-GPT and CBraMod, yet improves LaBraM to 61.4 %, most of the way to its fully fine-tuned result. This discrepancy is explained by adapter coverage rather than by architectural suitability. The adapter target lists named modules that the wrappers do not define: a patch-projection module for LaBraM and Neuro-GPT, which neither model contains, and attention and feed-forward names for CBraMod that match none of its layers. The adaptation each model received therefore depended on which of the remaining targets matched. LaBraM’s feed-forward layers matched, rendering 3.22 % of its parameters trainable. Neuro-GPT’s feed-forward layers lie inside unnamed container modules that the matching routine could not address, and none of CBraMod’s targets matched, which limited their trainable parameters to 0.04 % and 0.05 %, respectively. Because the low-rank arm was more restricted than specified, its results bound the adaptation those

two models received rather than the capability of the technique, and Appendix A records how that target list came to be written. The regime comparison nonetheless supports a consistent conclusion: parameter coverage, rather than the choice of adaptation technique, determined whether the low-cost route succeeded.

One methodological limitation applies to the interpretation of Table 5.5. The Protocol-B column for the frozen and low-rank configurations relies on a single fold, whereas only the full fine-tuning regime was evaluated across all five folds. Protocol A, which uses five seeds for every regime, therefore provides the more reliable basis for comparing adaptation strategies, and only five-fold evaluations enter the main results table. The data illustrate the necessity of this distinction: the identical fine-tuned Neuro-GPT configuration achieves 86.4% on fold 0 but averages 83.5% across all five. Ranking a single-fold figure against five-fold figures would have placed an FM at the top of the benchmark, an ordering the five-fold evaluation does not support.

On this basis, the answer to RQ1 is negative by a narrow margin. Optimally adapted, the highest-performing FM (Neuro-GPT) reaches 83.9% under the deployment scenario, compared with TimesNet’s 85.5%, with CBraMod at 80.1% and LaBraM at 65.4%. Pre-training brings a model into the leading group and does not place it above the others. The effect size between TimesNet and the best FM is large by convention [16] (Cohen’s $d = 1.597$, with a bootstrap interval [23] of [0.889, 6.476]). Against the remainder of the leading group, the point estimates range from medium against EEGNet ($d = 0.688$ [0.176, 2.027]) to large against S-Mamba, iTransformer and the classical baseline ($d = 1.505$, 1.387 and 1.542). Every associated interval is wide, those at fold level overlap, and the statistical ambiguity is bidirectional: TimesNet has the highest mean and does not separate from the cluster beneath it, and Neuro-GPT, one of its members, is not separated from TimesNet either. The benchmark does not indicate that pretraining degrades performance. It establishes that pretraining is not better on a task where a purpose-built convolutional network and a classical spatial-filter baseline from 2008 [4] already occupy the leading group.

Two of the three asymmetries recorded in Section 6.6 qualify the result in favour of the FMs. First, only Neuro-GPT received an HPO under the full fine-tuning regime in which it was finally evaluated, whereas CBraMod and LaBraM reuse hyperparameters found while their encoders were frozen, so their figures are lower bounds. Second, LaBraM’s checkpoint is the community re-release [11] rather than the weights published with the paper, a consequential distinction, because two of the checkpoint’s parameter tables were found untrained (Section 6.2). Neither asymmetry is sufficient to close the gap to the leading model, but both imply that the foundation-model results represent a lower bound rather than a performance maximum.

A final observation clarifies why the frozen-encoder results cannot be interpreted as measurements of model capability. On the shared evaluation harness, CBraMod’s frozen probe achieves a recall of 0.79 for Idle while the figure for Forward is 0.03 (per-class recalls of the shared-harness frozen run). This pattern indicates a near-total failure to decode the movement classes, with the aggregate score sustained by the over-represented Idle class. Comparative analyses relying on frozen probes would therefore have measured this artefact rather than the models’ representations.

5.4 The non-EEG control and the role of the pretraining corpus

Section 5.3 established that pretraining does not place a decoder above architectures trained from scratch. RQ2 investigates whether the domain-specific *content* of the pretraining corpus yields a measurable advantage. The Chronos model establishes a lower bound for this comparison. It is a large sequence model pretrained on tens of billions of observations drawn from domains such as energy, finance, transport and weather, and it contains no physiological data [5]. As the largest and slowest decoder in the benchmark, its performance limitations cannot be attributed to a lack of capacity.

Table 5.5: Adaptation regimes for the three EEG foundation models, all under their own native preprocessing. Balanced accuracy (%).

Foundation model	Protocol	Frozen probe	LoRA	Full fine-tune	Lift
Neuro-GPT	A	29.5	29.3	82.1	+52.6
	B	41.4	42.6	86.4	+44.9
CBraMod	A	29.6	35.6	82.8	+53.2
	B	31.1	32.4	80.8	+49.7
LaBraM	A	30.2	61.4	68.2	+38.0
	B	30.3	62.4	67.0	+36.7

Full fine-tuning, not preprocessing, is what recovers these models: frozen on native preprocessing they sit near chance. LoRA is insufficient for Neuro-GPT and CBraMod, which adapt 0.04–0.05 % of their parameters, but lifts LaBraM substantially at 3.22 % trainable. Target coverage rather than the technique is what decides whether cheap adaptation works. **Why coverage differed:** the adapter target lists named modules the wrappers do not define (a patch-projection module for LaBraM and Neuro-GPT, and attention and feed-forward names for CBraMod that match none of its layers), so the adaptation each model received depended on which remaining targets matched. LaBraM’s feed-forward layers matched, Neuro-GPT’s lie inside unnamed container modules the matcher cannot address, and none of CBraMod’s targets matched (corrected 2026-08-27 from `src/checkpoints.py:377-395`). The low-rank arm was therefore narrower than specified, and its results are read accordingly rather than as a verdict on low-rank adaptation itself. **LaBraM’s checkpoint** is the braindecode community release, not the weights published with the paper, which are behind an academic-licence request, and the file carries a four-class classification layer that is dropped on load, so it is a fine-tuned artefact rather than the pretrained model as released. Its position in this table should be read with that in mind. **Protocol B here is fold 0 only:** the frozen and LoRA regimes were never extended to five folds, so Protocol A, where all three regimes have five seeds, is the sounder axis for comparing them. Only full fine-tuning was bridged to five folds. That bridge supplies the 80.1 and 65.4 of Table 5.1 directly. For Neuro-GPT it yields 83.5, and the headline row instead reports the 83.9 of the model’s own native fine-tuning search, which outperformed the bridge. **Tuning is not equal across the three:** Neuro-GPT received its own hyperparameter search under native fine-tuning, whereas CBraMod and LaBraM reuse the configuration found for them on Protocol A, a search conducted with the encoder frozen, where their validation scores were 0.42 and 0.39. Two of the three are therefore fine-tuned on hyperparameters chosen for a different regime, and their numbers should be read as lower bounds.

RQ2 is addressed not by the margin between Chronos and the benchmark leader but by its distance from the EEG-pretrained models. Under Protocol B, Chronos achieves 39.3 % against a chance level of 20 %. The lowest-performing EEG-pretrained model exceeds this score by 26.1 percentage points, and the strongest by 44.6. Because both margins are measured against the capped control configuration of Section 4.5, they reflect the training restriction as well as the pretraining corpus. The configuration-matched comparison is Protocol A, where the control received the full hyperparameter search and training budget. In that regime, the weakest EEG-pretrained model still exceeds the control by approximately 17 points, and the strongest by more than 30. The training cap therefore influences the magnitude of the performance gap but does not alter the conclusion: domain-matched pretraining confers a substantial advantage over its unrelated counterpart. Section 5.2 adds context to this finding. Chronos’s within-category direction accuracy is 30.2 %, *below* the two-way floor, which indicates a complete failure to resolve direction rather than diminished performance.

The benefits of EEG-specific pretraining do not extend beyond this advantage over the non-physiological control. The classical FBCSP+LDA pipeline, which uses no pretraining, achieves 82.8 %, above CBraMod and 17.4 points above LaBraM. EEG-specific pretraining exceeds its general-purpose time-series counterpart by a wide margin, and it does not exceed a classical spatial-filter pipeline that has none. The volume of pretraining data does not determine the ordering of the outcomes either: CBraMod was pretrained on over 9 000 of the 27 000 hours in a clinical archive [70] and achieves 80.1 %, Neuro-GPT used roughly 5 700 [17] and scores higher at 83.9 %, and LaBraM, pretrained on approximately 2 500 [35], reaches 65.4 %. Corpus size alone does not determine the efficacy of pretraining for this task.

A methodological limitation regarding this control must be acknowledged. Chronos is

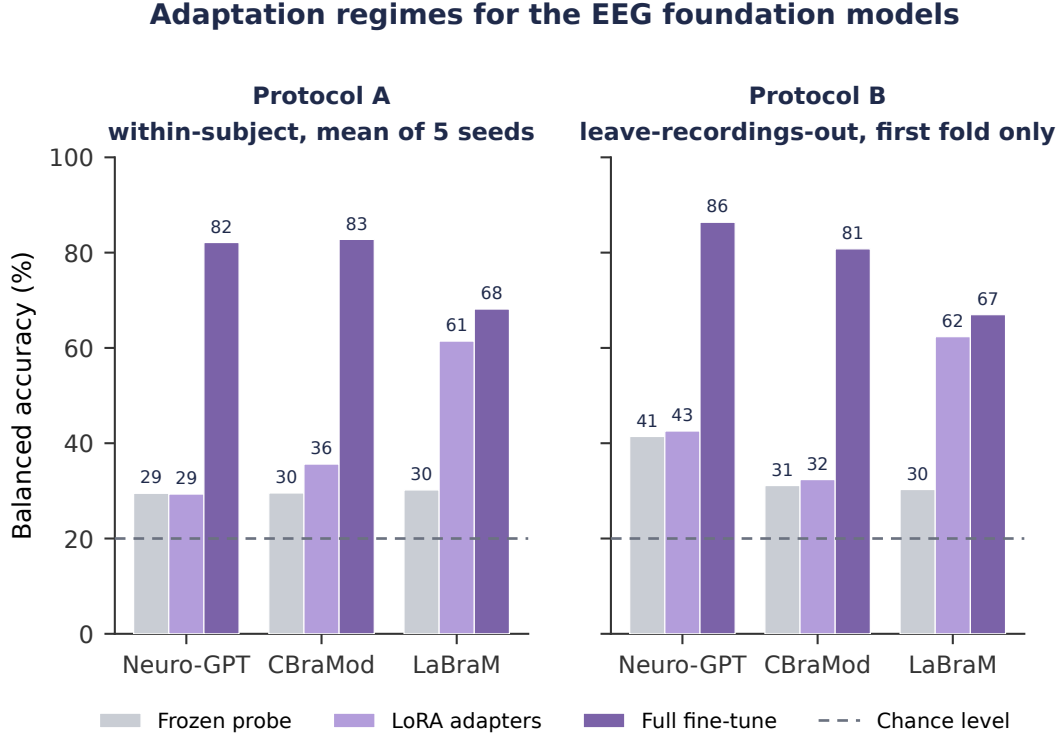


Figure 5.5: Balanced accuracy of the three EEG FMs under three adaptation regimes: encoder frozen (grey, the left bar in each group), low-rank adapters [33] (light purple, middle) and encoder fully fine-tuned (purple, right). The dashed line denotes the 20% chance level. Left: Protocol A, averaged over five seeds. Right: Protocol B, first fold only. Frozen encoders perform near chance, whereas full fine-tuning brings all three models into the leading group. Low-rank adapters yield a substantial improvement only for LaBraM, the one model in which the adaptation targets reached a sufficient share of the parameters.

one architecture pretrained on a single general-purpose corpus, and it is the sole non-EEG-pretrained model in the benchmark. Its deployment configurations were trained under the three-epoch cap described in Section 4.5, a restriction on the number of training epochs and not on the hyperparameter search, so these results establish a conservative lower bound of uncertain tightness rather than a performance maximum, and the configuration-matched comparison relies on Protocol A for that reason. The control model therefore *bounds* the expected performance of non-physiological pretraining on this task. It does not set the category of general time-series pretraining against that of EEG, and no such comparison is claimed. Its inclusion in the experimental framework served as a methodological control: a model expected to underperform every EEG-native decoder, where a failure to do so would have implicated the evaluation framework rather than the models. The empirical results align with this methodological intent.

5.5 Sample efficiency in the low-data calibration regime

The preceding results assume a decoder trained on the complete recorded pool. In clinical applications, this data requirement represents a primary constraint. Calibration data collection is limited by patient tolerance for prolonged headset wear, and models requiring extensive calibration sessions are unlikely to achieve clinical adoption. Addressing RQ3, Table 5.6 and Figure 5.6 evaluate the number of calibration trials per class required for each decoder in the analysis to reach usable accuracy.

Table 5.6: Balanced accuracy (%) against the number of calibration trials per class.

Decoder	5	10	20	30	50	75	100	150
EEGNet	63.2	66.7	72.9	75.2	81.7	82.3	82.3	82.3
Neuro-GPT	49.1	56.4	66.5	70.9	81.3	84.9	84.9	84.9
TimesNet	46.2	56.4	62.8	67.0	73.9	78.2	78.2	78.2
CBraMod	40.4	39.9	58.2	64.2	76.5	81.0	81.0	81.0
S-Mamba	36.5	44.7	53.8	61.8	68.6	71.3	70.7	72.2
iTransformer	34.3	42.3	51.5	59.8	68.4	71.7	71.7	71.7
LaBraM	38.1	42.0	49.1	52.0	61.2	64.8	64.8	64.8

Every decoder is in its benchmark configuration, so the seven curves are directly comparable. A single representative fold with five seeds, so absolute values sit below the five-fold deployment numbers: the shape is the result, not the level. At five trials per class EEGNet reaches 63.2% while the recommended decoder reaches 46.2%: the model that scores highest is not the model that becomes usable soonest.

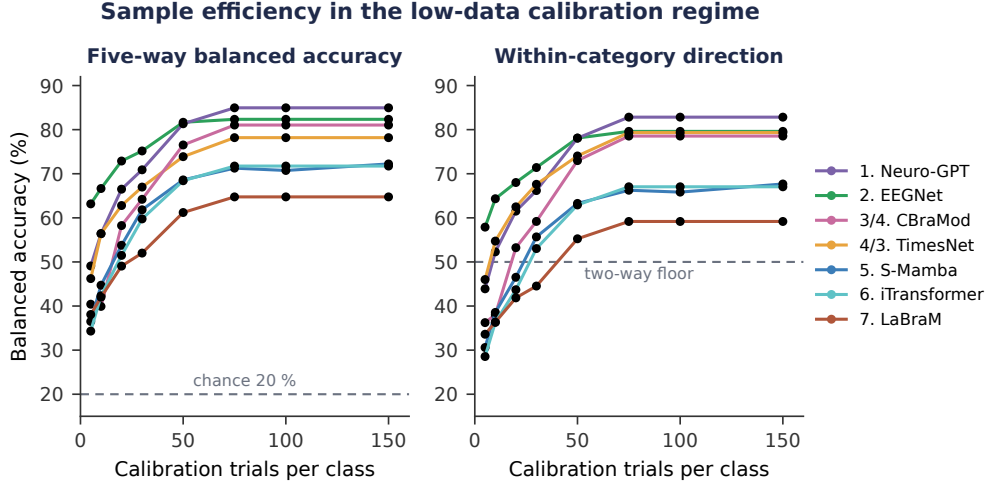
Decoder rankings in the low-data regime diverge from those of the main results table. At five trials per class, EEGNet reaches 63.2%, whereas TimesNet, the decoder with the highest overall mean, reaches 46.2%, a gap of 16.9 percentage points within this clinically relevant low-data regime. The rankings at the high-data end diverge as well. At 75 trials per class and beyond, six of the seven curves are flat, because that threshold already uses the complete pool, and at this plateau Neuro-GPT achieves 84.9%, EEGNet reaches 82.3%, TimesNet settles at 78.2%, and CBraMod lies between those two. TimesNet is therefore preceded by two decoders at five trials per class and by three at the plateau. Although TimesNet was identified as the optimal model under full-data conditions, one to three other decoders are ahead of it at every evaluated point along the calibration curve. Section 6.5 details the rationale for prioritising full-data performance in the final model selection despite these early-curve discrepancies.

The directional accuracy evaluation adds context to these findings and links this analysis to Section 5.2. At five trials per class, EEGNet is the only decoder whose within-category direction accuracy exceeds the two-way floor, scoring 57.9%. All other decoders fall below the 50% threshold (TimesNet at 46.0%, with the remainder scoring lower). At this restricted data threshold, the other decoders detect that movement is occurring but fail to resolve its direction. This failure mode is distinct from general classification inaccuracy and is masked by aggregate five-way performance metrics. EEGNet’s early advantage is also narrower than the aggregate metric suggests: it is driven predominantly by the left–right axis (65.2%), while its forward–backward accuracy at that threshold remains at the two-way floor (50.6%).

Three methodological constraints apply to this learning-curve analysis. First, it was computed on a single representative fold across five seeds, so its absolute accuracy values are lower than the five-fold deployment figures (TimesNet plateaus here at 78.2% compared with 85.5% under Protocol B), and the analytical focus must remain on the trajectory of the curve rather than on its absolute plateau. Second, because every decoder operates in its established benchmark configuration, the curves remain comparable with each other within this section, even though direct numerical comparison with Table 5.1 is precluded. Third, the analysis covers seven of the nine decoders. FBCSP+LDA is excluded by design, because its deterministic fit produces no spread across seeds to average, and the non-physiological control contributes no point at any calibration size, so no sample-efficiency finding reported here applies to either of them.

5.6 Deployment envelope: latency, memory and calibration

RQ4 investigates the operational criteria, beyond classification accuracy, that a decoder must satisfy to drive the platform in real time. Live operation imposes three further constraints: inference must complete before the next data window is acquired, the model must



Single representative fold with five seeds, so absolute values sit below the five-fold deployment numbers: read the shape, not the level.

Figure 5.6: Model accuracy plotted against the number of calibration trials per class, evaluated on the five-way classification axis and on the within-category direction axis. The rankings at the low-data end of these curves diverge from the full-data rankings of the main results table.

fit the hardware available in a clinic, and the probability estimates evaluated by the confidence gate must be well calibrated.

The latency constraint is satisfied with a substantial margin. The deployment policy enforces one hard bound, the decode cadence of 0.5s, and Table 5.7 shows that every decoder completes inference at least an order of magnitude below this threshold. These latencies are host-side inference times rather than measurements with the platform, and the classical pipeline’s latency is the single-window predict time recorded on the central processing unit (CPU) by the evaluation loop, since the GPU harness cannot time a scikit-learn fit: 5.3ms, a margin of $94.3\times$ against the cadence. The two slowest models, the Chronos control at $11.5\times$ and TimesNet at $16.7\times$ the cadence, still exceed the bound by more than an order of magnitude, whereas EEGNet (0.14ms) exceeds it by a factor of 3546. Figure 5.7 plots accuracy against latency and shows that no meaningful accuracy–cost trade-off exists for this task: the highest-performing decoder operates well within the latency budget, and selecting a less expensive model would yield latency reductions of no functional benefit to the pipeline. A methodological caveat applies to Table 5.7. The computational costs of the three FMs were measured in their frozen-encoder configurations, whereas their reported accuracies derive from the fully fine-tuned forms, so these latency and memory figures describe a different regime from the accuracy they accompany.

Calibration is the constraint that could have governed the choice, and the recommended TimesNet decoder demonstrates good, though not the best, calibration. Figure 5.8 presents the expected calibration error [28] across the evaluated architectures. TimesNet achieves a value of 0.031, the third lowest: iTransformer (0.025) and EEGNet (0.027) achieve lower errors, and all three occupy a narrow band. TimesNet achieves the best Brier score [12] among all decoders. LaBraM, by contrast, yields an expected calibration error (0.263) an order of magnitude higher than the leading models. This miscalibration is more detrimental to deployment than its accuracy deficit, because a model that produces confident yet incorrect predictions presents the one failure mode a confidence gate cannot mitigate.

The operational effect of the confidence gate can be quantified. The decision layer accepts a prediction window only if the maximum class probability reaches 0.80, and otherwise it forces an Idle state. Applied to TimesNet, this threshold retains 76.3% of the incoming windows and raises the balanced accuracy on that retained subset from 85.5% to 94.1%. The gating mechanism therefore yields an 8.6-point accuracy improvement at the cost of

Table 5.7: Measured inference cost for a single two-second window at batch size one, and the training time of one Protocol-B fold.

Decoder	Parameters (M)	Latency (ms)	Peak VRAM (MB)	Clears 0.5 s	Training (s)
TimesNet	28.513	30.0	134.7	yes	4 266
S-Mamba	0.249	1.9	10.3	yes	133
Neuro-GPT	1.268	1.6	18.3	yes	107
EEGNet	0.004	0.1	9.9	yes	29
iTransformer	0.201	0.7	10.1	yes	105
FBCSP+LDA	0.000	5.3	–	yes	43
CBraMod	4.925	3.0	28.5	yes	258
LaBraM	5.847	2.5	32.3	yes	138
Chronos	46.157	43.5	218.3	yes	777

The deployment policy is accuracy-first: cost is reported, not gated. The only hard bound is the 0.5 s decode cadence, and every decoder clears it by at least an order of magnitude: the two slowest, the control at 11.5 $\times$ and the recommended decoder at 16.7 $\times$, and the lightest network at over three thousand. For the three foundation models these figures were measured in the frozen-probe form while their accuracies are the fine-tuned form, so the cost and accuracy columns describe different regimes for those three rows. FBCSP+LDA is a scikit-learn fit that the GPU harness cannot time, so its latency is the single-window predict time recorded on the CPU by the evaluation loop (median of 20 calls, averaged over the five Protocol-B folds), and its memory is not measured. The evaluation loop recorded the same single-window timing for every run of every decoder, and those values and the harness agree to within a factor of one and a half in either direction, so the table reports the dedicated harness for the networks throughout. Floating-point operation counts were not measured for any decoder. Training time is the wall-clock time of one Protocol-B fold under the 80-epoch budget with early stopping, averaged over the runs that produced the accuracies: the control’s is under its three-epoch cap, and the classical fit runs on the CPU. The envelope reports what was measured: parameters, wall-clock latency, peak memory and training time.

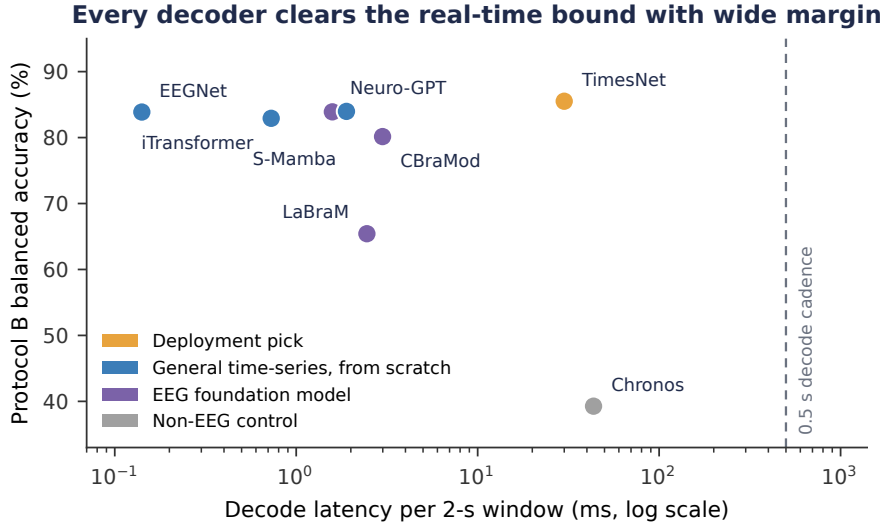
discarding 23.7% of the data stream. Because a discarded window causes the gait trainer to do nothing rather than to execute an erroneous movement, this trade-off favours safety, and Section 5.7 further justifies this design choice. These gating figures are reported for TimesNet alone, because the underlying threshold sweep was executed on the shared evaluation harness, where the FMs operate in their frozen forms, so their gating curves would describe a configuration this thesis does not report.

Two elements of the safety layer remain unvalidated within the current experimental framework, and Section 3.8 states both as limitations of the layer itself: the artefact-rejection threshold is a default rather than a value calibrated against this EEG montage. Nothing in the corpus was captured while the machine was in motion either, so the artefact environment of a live session is uncharacterised. The implications of both are addressed in Chapter 6.

5.7 Per-class recall and confusion structure

An aggregate balanced-accuracy metric quantifies overall decoder correctness, but it obscures both which commands are misclassified and what the incorrect outputs are. In the context of robotic gait rehabilitation, these distinct failure modes carry disparate clinical consequences. Table 5.8 and Figure 5.10 present the per-class recall, and Figure 5.9 gives the confusion matrix of the recommended decoder under Protocol B, the grid that counts, for each true class, the classes to which its windows were assigned.

The per-class analysis provides stronger support for the deployment model selection than the aggregate benchmark results. TimesNet is the only decoder whose recall never falls below 0.840 on any of the five commands, and it is the only model for which the Idle class does not constitute its highest recall score. Every other decoder in the leading group achieves its maximum recall on the Idle class, which indicates that their aggregate scores are supported disproportionately by the easiest class. A decoder demonstrating uniform competence across all five commands is a better foundation for a controller than a model with an identical mean and a weak class, a property that the aggregate metrics of Table 5.1 obscure.



FBCSP+LDA is omitted: it is a scikit-learn fit with no comparable GPU latency measurement. Cost is measured on the shared deployment harness, so for the three foundation models it is the frozen-probe form while their accuracies are the fine-tuned form, so the two axes describe different regimes for those models.

Figure 5.7: Protocol B balanced accuracy plotted against measured decode latency for a single two-second window. The dashed line denotes the 0.5 s decode cadence, the only hard bound imposed by the deployment policy.

The confusion structure exhibits a consistent pattern across the leading group: misclassifications occur predominantly *within* a movement axis. For the recommended decoder, 10.3 % of true-Forward windows are misclassified as Backward, and 7.9 % of true-Backward ones as Forward. These within-axis error rates exceed any single cross-axis confusion between the forward-backward and left-right axes by a factor of roughly two to six, whereas lateral confusions are lower and nearly symmetric at 3.6 %. Every decoder in the leading group except the classical baseline displays this dominance of forward-backward confusion, which suggests a property of the recorded signal rather than of any architecture. Below that group, CBraMod and LaBraM do not follow this pattern, as detailed below.

Recall is one of the three per-class metrics defined in Section 4.8, and specificity and precision are derived from the same stored confusion matrices (Tables C.2 and C.3, Appendix C). Specificity measures how rarely windows of other classes are assigned to a given command. For TimesNet this metric never falls below 0.951 on any class, so fewer than one in twenty non-matching windows is mislabelled into any command. Precision measures the proportion of issued commands that match the true class. For the recommended decoder, the lowest precision occurs on the Backward class at 0.817, which is the forward-backward reversal of the confusion analysis viewed from the predicted class. Backward constitutes the weakest precision column for five of the nine decoders. The lowest precision value among the EEG decoders is LaBraM’s for Left: its issued commands in that class are correct in only 0.523 of cases, which approaches the 0.5 expected from a random choice between the two lateral classes and reflects the directional deficit identified in Section 5.2.

These metrics carry direct safety implications for deployment. A Forward-for-Backward misclassification constitutes a command reversal, in which the gait trainer moves in direct opposition to the user’s intent. This is both the most severe operational error and the most frequently occurring single confusion. The most benign failure mode, the misclassification of a movement window as Idle, occurs comparatively rarely. The most common error therefore lies in the unsafe direction, which establishes the confidence gate of Section 5.6 as a safety prerequisite rather than an optional refinement. Whether a reversal is intercepted depends

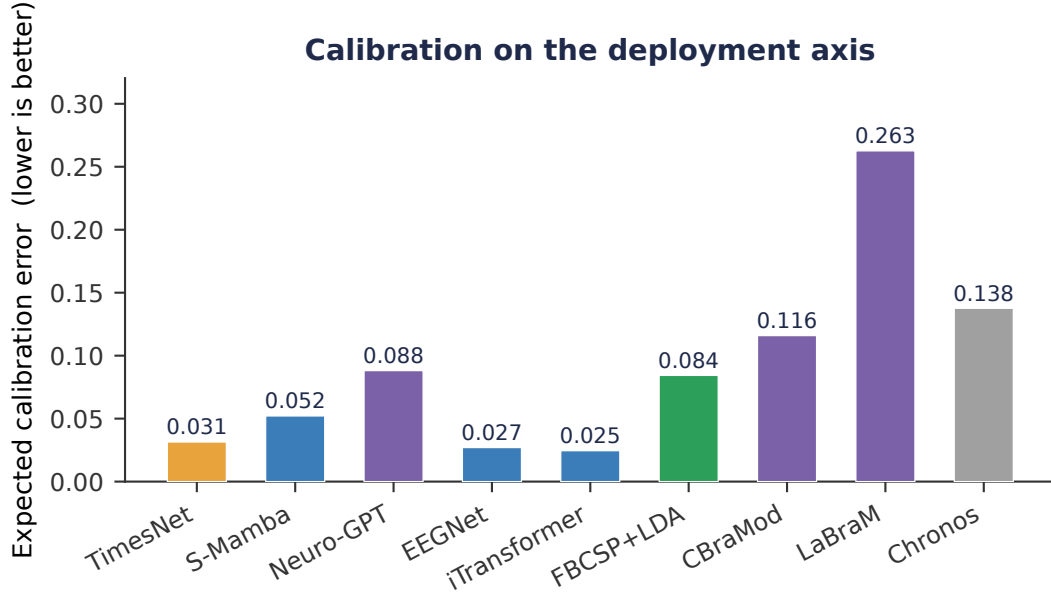


Figure 5.8: Expected calibration error for each evaluated decoder. Because the decision layer rejects predictions below a confidence threshold, a decoder without trustworthy probability estimates cannot be safely gated, regardless of its average accuracy.

Table 5.8: Per-class recall on Protocol B, averaged over folds and seeds.

Decoder	Idle	Forward	Backward	Left	Right
TimesNet	0.840	0.843	0.843	0.872	0.877
S-Mamba	0.919	0.813	0.832	0.809	0.826
Neuro-GPT	0.894	0.821	0.826	0.800	0.853
EEGNet	0.921	0.807	0.821	0.825	0.820
iTransformer	0.904	0.820	0.798	0.804	0.820
FBCSP+LDA	0.875	0.825	0.825	0.803	0.812
CBraMod	0.863	0.822	0.790	0.718	0.814
LaBraM	0.825	0.655	0.668	0.497	0.626
Chronos	0.755	0.075	0.590	0.080	0.465

Computed from the archived per-decoder rows, the same source the headline benchmark selected from, so the three foundation models appear here in their fine-tuned form. TimesNet is the only decoder whose recall never falls below 0.84 on any class, and the only one whose Idle recall is not its highest: it does not depend on the easiest class. Left and Right are the data floor at roughly one hundred recordings each.

on the decoder being *uncertain* when it reverses, which is the practical reason for reporting calibration alongside accuracy.

Two decoders exhibit a distinct failure pattern. CBraMod and LaBraM perform worse on the left–right axis than on the forward–backward one (LaBraM’s Left recall is the lowest score of any EEG decoder on any class), which is the confusion-matrix manifestation of the directional deficit established in Section 5.2. The classical FBCSP+LDA baseline is the only decoder in the leading group with comparable error rates across the left–right and forward–backward axes, which indicates that the spatial-filter pipeline distributes its errors differently from the deep networks despite lying in the same leading group. Chronos exhibits a general rather than a specific failure: it misclassifies approximately two in five true-Forward windows as Backward, a pattern characteristic of a model that detects gross motion but fails to resolve its direction.

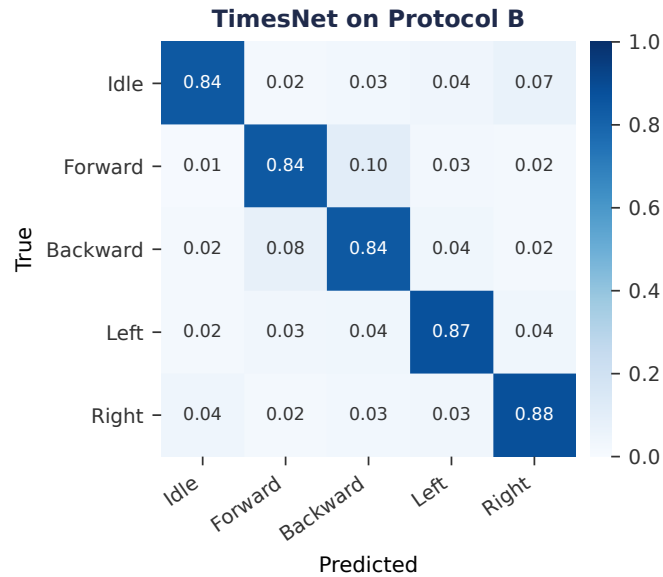


Figure 5.9: Row-normalised confusion matrix of the recommended decoder under Protocol B, averaged across all folds and seeds. Rows denote the true class.

5.8 Cross-person transfer to a second participant

All previously reported results evaluate within-participant performance. Protocol C addresses a question central to clinical scalability: whether a decoder trained on one individual generalises to another. The protocol trains the model on the primary participant and evaluates it on the second, restricting the problem to the two classes the latter recorded,

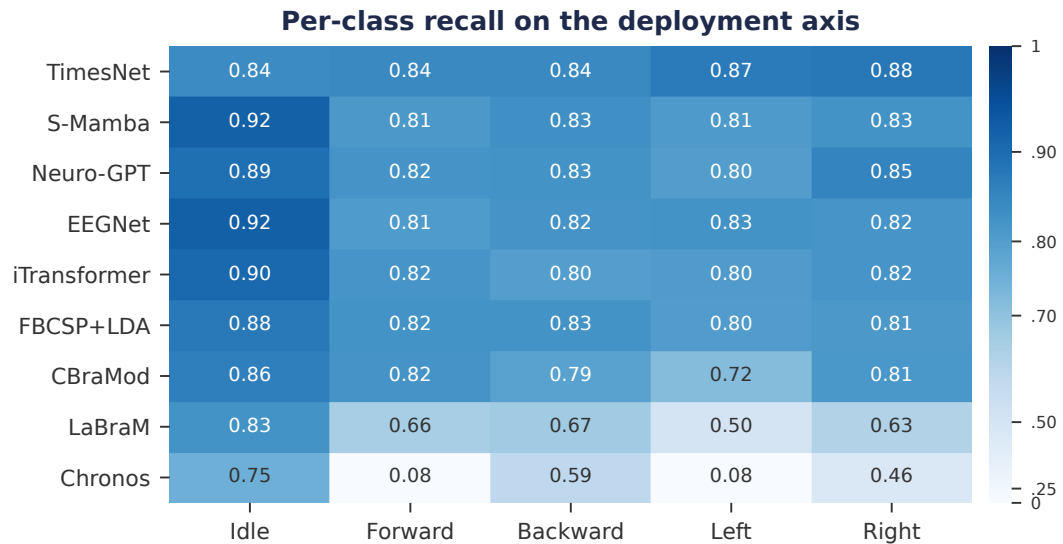


Figure 5.10: Per-class recall under Protocol B. An aggregate accuracy metric obscures how unevenly some decoders distribute their errors across the five commands. The colour scale is intentionally non-linear, expanded towards the upper end so that differences among the leading decoders (which recall most classes between roughly 0.65 and 0.92) remain visible rather than compressed into a single shade. The tick spacing of the colour bar illustrates this compression, and each cell displays its own value.

which sets the chance level at 50 %.

Cross-person transfer fails for all evaluated decoders. Table 5.9 and Figure 5.11 present the transfer accuracy alongside a matched within-participant reference, which provides the context required to interpret the data. Evaluated within participant, the EEG decoders in their benchmark configurations achieve approximately 80 to 90 % on this two-class problem, whereas under cross-person evaluation all but one degrade to between 49.1 % and 56.2 %, close to the 50 % chance level. The performance deficit associated with the change of participant is 31.4 percentage points for the recommended decoder and approximately 30 to 37 for the remaining EEG architectures, with the exception of Neuro-GPT, whose larger degradation is detailed below.

Evaluating transfer performance in isolation would distort these conclusions. Under the shared harness, LaBraM exhibits a transfer accuracy only about three percentage points below its reference score, ostensibly the smallest transfer loss in the benchmark, but only because its within-participant reference barely exceeds the chance level. A decoder that is weak within participant has little accuracy to lose across people. The reference is therefore an essential baseline rather than contextual information.

The most salient finding regarding the FMs contradicts the expectation that pretraining aids transfer. Native fine-tuning substantially improves within-participant performance (Neuro-GPT’s reference increases from approximately 67 % to 87.1 %) but provides no benefit to cross-person transfer. The adaptation is detrimental: Neuro-GPT’s transfer accuracy drops to 43.0 %, below the two-class chance level, which produces the largest inter-participant performance deficit in the benchmark at 44.1 percentage points. Below-chance performance on a two-class problem indicates more than a failure to generalise: the feature mapping optimised for the primary participant yields systematically incorrect classifications when applied to the second participant. Fine-tuning therefore achieved within-participant accuracy at the expense of whatever cross-person generalisability the pretrained representation possessed.

Two methodological constraints limit the interpretation of these findings. First, confidence intervals were not computed for Protocol C, and because every decoder performs near chance, this protocol cannot be used to rank model architectures: the differences between the transfer figures in Table 5.9 are not claimed to be meaningful. Second, because that participant recorded only two movement classes, the resulting problem cannot be quantitatively compared with the five-class benchmark of the preceding sections. Protocol C establishes an operational boundary rather than a model ordering: for this task and electrode montage, per-participant calibration constitutes a prerequisite for the system as built rather than a procedural convenience.

From a clinical deployment perspective, this lack of generalisability does not preclude multi-user adoption. It dictates that a decoder cannot be transferred between users without modification. Each new user records a calibration session, and the decoder is then trained on that user’s own recordings, as was done for the primary participant. The confidence gate does not change this requirement: at the deployment threshold the recommended decoder accepts 79.7 % of the second participant’s windows and is correct on 53.8 % of them (Section 6.5). The accuracy attainable at each calibration size is reported in Section 5.5. This transfer failure was not the factor that prevented a measured test with the platform, and the reasons for that limitation are detailed in Section 3.7. A qualitative observation regarding the practical implications of this participant boundary is discussed separately in Section 6.5.

Table 5.9: Cross-person transfer. A decoder trained on the primary participant, evaluated on a second participant over the two classes that participant recorded.

Decoder	Harness	Within-person reference	Cross-person transfer	Person cost
TimesNet	shared	87.6	56.2	31.4
S-Mamba	shared	87.1	53.1	34.0
Neuro-GPT	native	87.1	43.0	44.1
	shared	67.3	49.6	17.8
EEGNet	shared	89.4	55.9	33.5
iTransformer	shared	85.4	51.0	34.4
FBCSP+LDA	shared	87.2	50.2	37.0
CBraMod	native	81.5	51.9	29.6
	shared	57.3	49.6	7.7
LaBraM	native	81.3	49.1	32.1
	shared	56.8	53.6	3.2
Chronos	shared	60.7	48.1	12.6

Chance is 50 %. The reference column is essential: without it the foundation models' smaller apparent drop would read as better transfer, when it is an artefact of their weaker reference. Transfer is near chance for every decoder, so Protocol C cannot rank models: it establishes that the decoder must be calibrated per person. ROC-AUC is omitted because the evaluation loop stored no value for the two-class task.

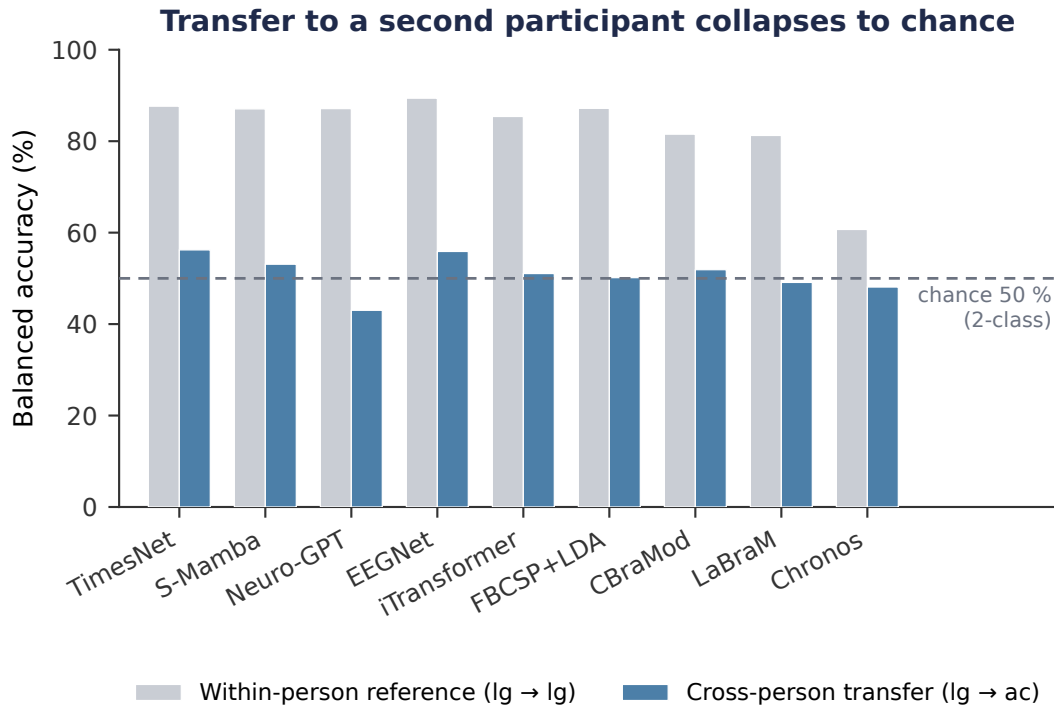


Figure 5.11: Transfer accuracy evaluated on the second participant, shown against the within-participant reference score. The reference is methodologically essential, because evaluating transfer performance in isolation would present a weak model's smaller absolute drop as better cross-person transfer.

Chapter 6

Discussion

6.1 Synthesis of the findings

The precondition and the four research questions of Section 1.4 share one overall problem. The thesis set out to determine whether a usable directional decoder can be built on a dry-electrode headset [73]. The evidence shows that it can, with per-participant calibration: the leading architectures reach between 82.8 % and 85.5 % balanced accuracy under Protocol B against a chance level of 20 %, resolve direction beyond the detection of movement alone, and run inside the latency budget with a wide margin. No large pretrained model is required for that level of performance. On this task, with this montage and this quantity of data, the decoders that attain it include small networks trained from scratch and a spatial-filter pipeline nearly two decades old.

The precondition of the benchmark holds: the decoders resolve direction and not only the presence of movement. Every EEG decoder except LaBraM resolves direction within a movement category well above the two-way floor, LaBraM exceeds it by a narrow margin, and the non-physiological control falls below it, so the benchmark ranks direction decoders rather than movement detectors. Separate evidence was required because overall accuracy cannot distinguish the two cases: a decoder that identified the movement category without resolving direction inside it would already score three times the five-way chance level.

Pretraining is not required for a place in the leading group (RQ1), because decoders trained from scratch already occupy it. Given native preprocessing and a fully fine-tuned encoder, the most favourable configuration the comparison framework permits, the best EEG FM joins that group without leading it. Its members lie within three points of one another under Protocol B and comprise a purpose-built convolutional network of 3 541 parameters [42], three general time-series architectures trained from scratch, an optimally adapted FM, and the classical filter-bank pipeline [4]. At five folds none of them separates from the leader statistically. The summary the statistics support is that pretraining brings a model into the leading group and does not place it ahead of the decoders already there. The highest mean belongs to the architecture that the Time Series Library’s own leaderboard places first on classification over its public benchmark archive [75], so its position here agrees with its record outside EEG.

The content of the pretraining corpus matters within limits (RQ2). EEG-specific pretraining exceeds a model trained on economic, industrial and environmental series [5] by between 26 and 45 points, so the domain of the corpus contributes substantially. It does not exceed the classical pipeline, which is fitted from scratch on the task data alone, and its volume does not order the outcomes, because the model with the largest corpus is not the best of the three.

RQ3 and RQ4 concern deployment, and their answers qualify the ranking in opposite directions. The architecture with the highest mean is not the one that becomes usable soonest: at five calibration trials per class it ranks third, and only one of the seven decoders the curve covers resolves direction at all at that point. The accuracy leader nevertheless satisfies both operational requirements. It clears the real-time budget by more than an order of magnitude, and its stated confidence tracks its realised accuracy closely enough for the gate to be worth applying, as Section 5.6 quantifies. Taken together, the answers fix the terms on which the feasibility verdict holds: calibration data recorded from each user, a decoder chosen for accuracy rather than for how little of that data it needs, and a confidence gate on its output.

6.2 Explanation of the pretraining result

Four observations, two from Section 5.3 and a further pair from the model cards of Appendix B, constrain the explanation of the negative result and rule out several candidate causes. A fifth observation concerns the non-physiological control, a hypothesis about the task then accounts for what the four leave open, and a last one addresses the checkpoint that does not measure its model.

Deficient adaptation does not explain the result. Each of the pretrained models received native preprocessing and full fine-tuning, and the resulting lift was large, between 38 and 53 points on Protocol A. A gain of that size on unfreezing rules out a misconfigured harness. In the frozen condition the encoders supplied a linear probe with nothing useful for this task, which locates the problem in the learned representation.

The size of the pretraining corpus does not explain the ordering. CBraMod, Neuro-GPT and LaBraM were pretrained on over 9000, roughly 5700 and about 2500 hours respectively [70, 17, 35], and their Protocol-B results do not follow that sequence. The contribution of pretraining on this task therefore does not accumulate with hours of data.

Two of the three models carry a structural mismatch in the shared harness, though not in the condition their accuracies come from. CBraMod and LaBraM were pretrained at 200 Hz [70, 35], where their fixed 200-sample patch, the chunk of samples that becomes a single input token, spans exactly one second. On the shared deployment harness, which runs at 300 Hz, the same patch covers two thirds of a second, so every temporal relationship the pretraining encoded is compressed into two thirds of the interval it was learned on. The patch size is not a free parameter, because the pretrained weights require it at load.

That compression bounds the deployed system, and the shared-harness rows, including the inference costs of Section 5.6, are measured under it. The accuracies above are not. The native-preprocessing concession of Section 4.1 resamples each model to its own pretraining rate, so in the runs behind those figures every patch spanned the interval it was pretrained on, for all three models. The mismatch is a deployment cost rather than an explanation of the ranking.

Neuro-GPT’s embedding, the layer through which every pretrained feature must reach the classifier, is narrow. The model [17] is loaded encoder-only, and its embedding is 40 dimensions wide, an order of magnitude narrower than that of any other decoder. This bottleneck is a plausible partial explanation for the collapse of its frozen probe, and a reason for caution before reading its recovered accuracy as evidence that the pretrained features, rather than the fine-tuning, account for the result.

These four observations concern the EEG-pretrained models. The non-physiological control fails for a different and simpler reason, and its own authors supply most of it. Chronos was pretrained on a univariate forecasting corpus spanning domains such as energy, transport, healthcare, retail, web, weather and finance, and its authors note that their evaluation does not include enough datasets sampled at 15-minute resolution or faster [5]. Its healthcare entries are administrative daily and monthly counts rather than physiological recordings, so no part of the corpus resembles a biosignal. EEG at 300 Hz is orders of magnitude outside that range, and the corpus is univariate where this task has 19 simultaneous channels. The objective is forecasting rather than classification, a transfer for which the authors offer a hypothesis and no evidence.

The wider benchmark literature supports the same reading. On a 23-dataset forecasting suite, task-trained deep models exceed pretrained ones at higher sampling frequencies and on series with high entropy and little trend [2]. Where a pretrained time-series encoder has been set against one trained from scratch on physiological signals, under the same frozen-encoder protocol used here, the scratch encoder is the more accurate on one task and falls narrowly short on the other, while a model carrying the wrong prior altogether drops to single figures [41]. Chronos serves in this benchmark as a control, and its failure is the expected behaviour of a capable model applied to data outside its pretraining domain. That account rests on the full-budget evidence: the control’s deployment rows carry the

three-epoch cap of Section 4.5, but its Protocol-A row does not, and there, fully searched and trained, it remains more than 37 points behind the leader.

A property of the task itself underlies all four observations. These models were pretrained on clinical EEG, two of them on the Temple University Hospital corpus [52], which holds diagnostic recordings of patients at rest, where the signal of interest is spontaneous rhythm and pathology. The corpus recorded here is one of executed gait: a walking body, a dry montage, and a signal in which movement-related potentials (the Bereitschaftspotential family [65] of Section 2.2, the slow cortical potentials that precede and accompany voluntary movement) and motion contamination form part of the phenomenon rather than noise around it. Pretraining transfers what its corpus held, and recordings of stationary patients contain comparatively little about a walking one. This hypothesis is undemonstrated, but it is consistent with all four observations, and it predicts the finding the next section examines: the decoders that succeed here should be those built to locate structure in this signal directly, rather than those carrying the most prior knowledge of a different signal.

One row in the foundation-model column additionally does not measure its model, because LaBraM’s checkpoint carries two untrained parameter tables. LaBraM [35] adds a learned electrode embedding to every token, a table with one entry per channel that encodes which electrode the token came from, and a second table that encodes patch order. In the community checkpoint both tables are statistically indistinguishable from their random initialisation: element-wise SDs of 0.0200 and 0.0192 against a seeded value of 0.02, and cosine similarities between electrode entries scattered about zero at the chance spread for vectors of this width. The remainder of the file is trained, with normalisation gains well off unity and heavy-tailed attention weights, so the deficiency is confined to those two tables and not to the checkpoint as a whole. CBraMod [70], which derives its positional information from a pretrained convolution over the channel-by-patch grid instead of from a lookup table, carries no equivalent gap.

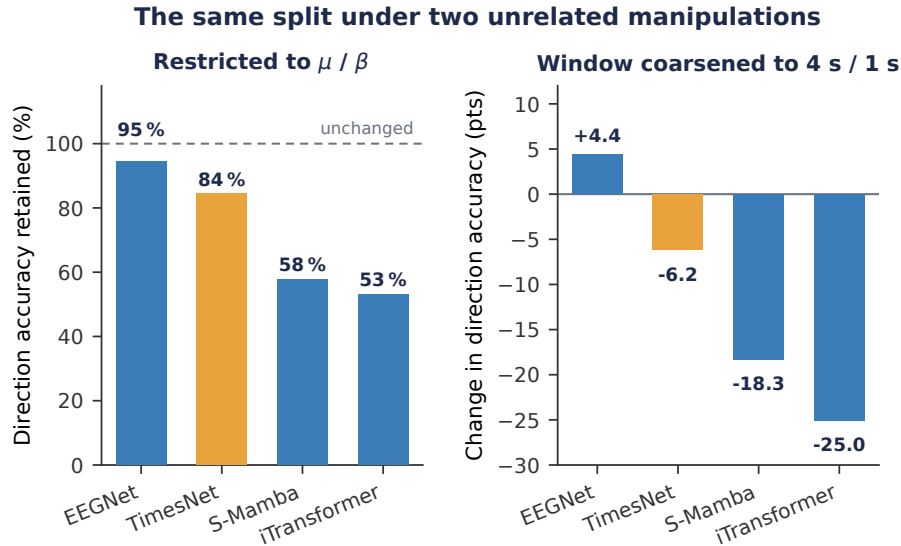
The cost of the missing tables can be bounded, because LoRA [33] freezes every base parameter: LaBraM ran with the random tables intact and still reached 62.4 % on the first Protocol-B fold, against 67.0 % on the same one once full fine-tuning could learn them. Learning the spatial code is therefore worth a few points and not the 15 that separate LaBraM from CBraMod, and no cause identified here accounts for the remaining deficit. The evidence supports a bound rather than a cause: LaBraM’s row characterises a degraded artefact and not the model its paper describes, which is why it is read as a lower bound throughout. The shortfall also falls where a missing spatial code would place it, on the one contrast that requires knowing which electrode lies over which hemisphere (Left/Right 56.1 % against CBraMod’s 76.6 %). Appendix B records the measurement.

6.3 A hypothesis on the signal used by the from-scratch decoders

The mechanism proposed in this section is consistent with two observations and is not demonstrated. Every later use of it is labelled as a hypothesis.

Four decoders trained from scratch belong to the leading group, and they do not behave alike when the input is perturbed. Two supporting studies, run separately and for different reasons, produce the same division between them.

Under restriction to the μ and β sensorimotor rhythms (Section 5.2), EEGNet [42] retains almost all of its direction accuracy and TimesNet [74] most of it, while S-Mamba [72] and iTransformer [44] fall below the two-way floor: they stop resolving direction altogether. Under a different manipulation, coarsening the analysis window from two seconds at a half-second stride to four seconds at a one-second stride, EEGNet improves and TimesNet loses about six points of direction accuracy, while the other two fall by 18 and 25 points. Figure 6.1 places the two studies side by side. Neither study alone would be strong evidence, because both are reduced-budget controls and each admits an explanation specific to its own manipulation. Taken together they are more informative, since the two manipulations share nothing except the decoders they disrupt.



Two reduced-budget controls, run separately and for different reasons. Compare within each panel only, never against the headline table. The three foundation models are omitted: they sit below the two-way floor at both settings of both studies, so any ratio or difference computed on them would report noise.

Figure 6.1: Two unrelated disturbances of the input, and the same division between decoders. Restricting the passband to the sensorimotor rhythms leaves EEGNet and TimesNet largely intact and moves S-Mamba and iTransformer below the two-way floor, and coarsening the window produces the same division. Both are reduced-budget controls and are comparable within each panel only.

The division follows a difference in architecture rather than in capacity or accuracy. EEGNet applies a temporal convolution whose kernel spans half a second before combining channels spatially. That length is neither arbitrary nor a band selector: it is half the sampling rate, chosen so that the kernel can resolve frequencies down to 2 Hz, and its authors halve it when the input is high-passed higher [42]. What the kernel selects within that range is learned. Its authors describe the temporal convolution as a bank of narrow-band frequency filters and the depthwise one as their band-specific spatial counterpart, so the oscillatory reading is the architecture’s stated design intent. TimesNet identifies dominant periods by Fourier transform and folds the signal into a period-by-phase representation before convolving, so periodicity selection is its first step. Both architectures therefore carry an inductive bias toward oscillatory structure.

iTransformer and S-Mamba carry no such bias. Both are channel-token architectures that invert the usual arrangement and treat each electrode as a single token whose embedding is its entire two-second series projected into a fixed width. Neither performs an explicit frequency decomposition, and the iTransformer paper states that its variate tokens require no position embedding [44], so their identity conveys no electrode location either.

The hypothesis is that the two groups of architectures use different components of the signal. An architecture with an oscillatory bias is constrained to find rhythmic structure, and the sensorimotor rhythms survive band restriction by construction. An architecture that embeds a whole window as one token can draw on anything that covaries across that window, and in executed gait many components do. Movement-related slow potentials [64] and step cadence lie below 8 Hz, and band restriction removes that range along with everything above 30 Hz. If the channel-token models were relying on slow, window-spanning structure rather than on band-limited rhythm, restriction should remove their signal while leaving that of the oscillatory models intact, which is what the study observes.

The same account predicts the window result. For a model that tokenises the whole

window, altering its length changes the token wholesale: a four-second embedding is a different object from a two-second one, and the learned relationships do not carry across. For a model whose first operation is a half-second kernel or a period decomposition, a longer window supplies additional evidence of the same kind. The prediction and the observation agree both in direction and in which decoders are affected.

Three points remain unestablished by this hypothesis. It makes no claim about the anatomical origin of the discriminative signal: an oscillatory feature surviving band restriction is not thereby cortical, and Section 6.6 records why the control built to address that question cannot answer it. It does not show that the channel-token models rely on artefact rather than on neural activity, since slow cortical potentials are as consistent with the observation as motion is. It also remains unexamined on the decoder best placed to test it: FBCSP+LDA is a five-band filter bank by construction, so the hypothesis predicts it should be the most robust of all to band restriction, and it was never included in that study. That experiment is the direct next step, and its cost is a single deterministic run.

A published result also runs against the hypothesis, and it comes from the same paper that supplies two of the architectures it concerns [13]. In a 14-dataset comparison of long-term forecasting models, the only one on which the roster separates is an EEG MI set, and the highest-scoring model on it is S-Mamba, a channel-token state-space architecture from the group this section describes as lacking the relevant bias. The same authors tested the general form of the argument, asking whether models that mix across variates perform better when those series are more similar, and found no relationship [13]. Neither finding can be set aside.

The difference between those results and this one lies in the task: theirs is forecasting an EEG series, this is classifying a direction from it, and the two do not rank models alike. The TimesNet paper demonstrates the point directly, because DLinear [78] is a competent forecaster and then falls to the bottom of a classification comparison, since a single temporal projection that suffices for autoregression degrades markedly when the task requires a higher-level representation [74]. The reading this section supports is therefore narrow: the architectural split observed here concerns which models extract a *discriminative* representation from band-limited neural data, and it carries no claim about those architectures in general. A channel-token model that forecasts EEG well is not evidence against it, and the S-Mamba result does not count for it either.

6.4 Lessons for benchmark design

Four findings from the construction of this benchmark generalise beyond it, and three of them are errors caught during development. Each is recorded below together with the evidence that exposed it, since the final numbers alone would leave them invisible.

A grouping variable can be confounded with the label, and the failure that follows is difficult to distinguish from a genuine finding. The first version of the deployment protocol grouped folds by recording date. Because recordings were made one movement block at a time, holding out a date meant removing most of a class, and the numbers that followed were some 30 points below the corrected ones (Section 4.4). Nothing about the architectures changed between those two columns. Had the flawed split not been caught, it would have supported a confident and false account: a failure of generalisation across sessions, a class that could not be recovered, and the conclusion that no decoder reached usable accuracy. The general lesson is that a grouping variable must be checked for association with the label before it is used, and that a dramatic result from a new split warrants more suspicion than an unremarkable one.

Model selection on a single fold fails in both directions. Two selection errors narrowly avoided in this project came from evaluating on one fold. A fine-tuned FM reached 86.4% on fold 0 against 83.5% when the same configuration ran all five, and ranking that single-fold figure against five-fold figures would have placed it first. Separately, an HPO run on one fold found configurations that exceeded the frozen ones on that fold's validation split and

then fell behind across the full cross-validation for three of four models. The first inflates a model and the second a configuration, and both are the same error, that a single fold's estimate is not one of the mean. The remedy here was a hard gate requiring five folds before any result can enter the comparison, a low-cost safeguard that any benchmark should adopt by default.

A remedy fitted to a defect becomes a liability once that defect is removed. Several design choices in this project were made to counter behaviour that the broken split had produced: a parameter cap and a latency ceiling justified by models that appeared to fail, and targets set against numbers that turned out to be artefacts of the grouping. When the split was repaired the behaviour disappeared, the remedies remained and constrained the design for no reason, and all of them were retired. The general point is that a fix should be tied to the evidence that motivated it, so that when that support is withdrawn the fix is reconsidered rather than inherited.

Five folds cannot support pairwise significance claims. With five matched blocks the smallest attainable two-sided p -value is above conventional thresholds before any correction, so no pairwise comparison in this benchmark survives the Holm procedure [32], including the leader against a control it exceeds by more than 40 points (a gap that also carries the control's capped budget, whereas the configuration-matched Protocol-A gap is nearer 38 points). This is a property of the design rather than of the models, and it has a practical consequence: at this scale the appropriate output of a benchmark is a rank, an interval and an explicit statement of which differences are ties. Reporting a single winner would have been simple and superficially defensible, and the statistics do not support it. A benchmark of this scale should decide in advance that it will report effect sizes and intervals, because a decision taken afterwards is exposed to the result it is meant to describe.

6.5 Deployment implications for a clinical system

Per-participant calibration is a requirement rather than an optimisation. Cross-person transfer fails for every decoder, and for the best-adapted FM it falls below chance: the mapping learned from one person yields systematically incorrect classifications on another. The result bounds this system rather than the task. A decoder trained on one person has never encountered the variation between people and cannot have learned to discount it, and whether a decoder trained on many people would perform better on a new one is a question this single-participant corpus cannot answer. For the system as built, every deployment begins with a recording from the patient, and that step belongs in the design as a precondition for every new user, healthy or patient, rather than as a setup cost to be minimised.

Given that requirement, accuracy governs the choice of decoder, and this thesis adopts that position. The reasoning is clinical before it is statistical. Calibration is a bounded cost, incurred once per patient at enrolment, whereas classification quality is paid on every window, at every session, for the life of the deployment, and nothing downstream can recover what the decoder failed to extract. The recommendation is therefore TimesNet, which holds the highest mean on the within-subject ceiling and under Protocol B alike. Those two protocols probe different failure modes, so their agreement on the same decoder is evidence about the model rather than about either split, and it carries the statistical caveat of Section 5.3 unchanged: highest mean, not separated from the cluster beneath it. The learning curve of Section 5.5 remains relevant as a fallback, because a clinic whose binding constraint is how little calibration a patient can tolerate would select EEGNet, and the curve is reported so that such a clinic knows what that constraint gains and costs.

The considerations that argued against the leader are engineering variables rather than properties a clinic must accept. Every decoder exceeds the half-second cadence by at least an order of magnitude on a consumer GPU or, for the classical pipeline, on the CPU. A deployment that hosts inference on a server, reached by the bedside device through an API call, removes the model's size from that device entirely, with the cadence leaving generous

room for a network round trip. Latency, memory and hardware cost can all be designed around, whereas no component placed beside the decoder can supply accuracy it did not produce.

A reading of Table 5.7 by cost band, grouping the decoders by parameter budget, quantifies what the constrained alternative would cost. Under a budget of half a million parameters the accuracy optimum is S-Mamba at 84.0%. Relaxing that budget twentyfold, to 10 million, leaves the optimum unchanged, because the decoders admitted between those sizes score lower and the nearest of them, Neuro-GPT, lies within a tenth of a point of S-Mamba and is a statistical tie with it. Only the unconstrained band improves on it, where TimesNet’s 85.5% costs 28.5 million parameters, so a clinic that cannot host or serve that band gives up about a point and a half of mean accuracy and keeps the ability to decode.

Training time is the other cost the choice carries, and it is paid at every calibration: one Protocol-B fold trains EEGNet in 29 s and TimesNet in 4 266 s under the same budget (Table 5.7), about 146 times longer. A calibration session with the recommended decoder therefore ends roughly 70 minutes of computation after one with the lightest, a cost a clinic can schedule but should know in advance.

Protocol C does not enter the choice of decoder. Cross-person transfer is a boundary every candidate must be checked against, and here it is a property the decoders share: all near chance, none usable across people. A protocol on which every model fails cannot select between them, and with per-participant calibration established as a requirement, transfer becomes a property verified downstream of a strong within-participant decoder rather than a criterion that selects one. High Protocol-A and Protocol-B accuracy is the precondition, and whatever cross-person robustness later work achieves will be built on decoders that solve the within-participant problem first.

The operating point must be gated, and the gate’s cost is modest. In every decoder of the leading group the largest single confusion is between Forward and Backward, a command reversal and the most damaging error a gait trainer can make. The confidence gate addresses it: at the deployed threshold it discards under a quarter of windows and raises accuracy on the rest by more than eight points, and a discarded window leaves the device stationary rather than moving incorrectly. That trade-off is acceptable in a rehabilitation context, where a missed intention costs a repetition and a reversed one the patient’s trust or a fall. The reliability of the probability estimates therefore becomes a primary requirement, since a gate reading untrustworthy numbers is worse than none at all, and the values it reads are raw softmax outputs (Section 4.8).

The deployed checkpoint is a separate retrain rather than a benchmark row. The decoder carried into the live pipeline is a separate training run of the recommended setup (TimesNet under its Protocol-A configuration) on the within-participant pool, which holds out a grouped validation slice of one tenth of the recordings for early stopping and no test set at all (10 800 training windows and 1 200 for validation). It ran the live pipeline in the author’s self-test reported below, which is the first occasion on which the recommended decoder drove the decision path. Its expected live accuracy is the cross-validated deployment mean of 85.5%, since the run’s own validation score is measured on data the training was stopped against and is not a performance estimate. Retraining on the full pool follows from the calibration requirement above: once per-participant data is a precondition, the deployed model should use all of it, and the benchmark rows serve as the estimate of what that model will do.

The safety of the components that act on the patient is not yet established. The safety layer described in Section 3.8 is sound in construction (default to idle, asymmetric hysteresis, two independent watchdogs), but a pair of its parameters have never been calibrated against data recorded in the condition they will operate in, and none was recorded while the machine was moving. A live session on a person with the moving platform is therefore not a smaller version of what was measured here but a different measurement, and it should be treated as the next experiment rather than as deployment. One further exclusion belongs with the choice of decoder above. A supplementary analysis in the archive recommends a different

decoder under a constrained cost envelope, and it does not inform the choice made here, because its constraints were set against the superseded numbers discussed in Section 6.4.

One observation from the live pipeline is qualitative rather than measured, and it is reported with that status. During the project, two visitors considering an internship in the laboratory volunteered to try the system briefly. Neither had worn the headset before, and both are distinct from the second participant of Section 3.4, whose recordings were only ever used as held-out test data and never drove the live system. Each wore the headset in turn while the pipeline of Section 3.7 ran unchanged: EEGNet, trained solely on the primary participant, driving the wheeled platform in the room where the training data had been recorded. EEGNet was used because the recommended decoder’s deployment checkpoint did not yet exist, so the demonstration and the recorded session planned with that checkpoint concern different decoders. At the deployment threshold of the confidence gate no window passed and the platform stayed idle, so the threshold was lowered below its deployment value for the demonstration.

Asked to perform the directional movements of the recording protocol, both visitors elicited what a calibration-free decoder should mostly produce, windows the gate rejected. The windows it accepted behaved differently. When the decoder was confident, the platform moved in the direction being performed in most cases, for both visitors, with occasional errors, and in the author’s account the wrong inferences mostly fell below the lowered threshold and left the platform idle. In the clearest instance one visitor walked the length of the room forward, the gate rejected most of the windows along the way, and the platform started with a substantial delay and then moved forward. That visitor’s turns were followed as well: when they pivoted right the platform did the same, a response for which no offline test exists, because the second participant recorded Forward and Backward only. Nothing was recorded or controlled, the session was short, the environment was the one the decoder was trained in, and the observer was the author, so no number can be attached to any of it.

The archive holds an offline counterpart, and the two disagree. The Protocol C models stored a probability for every window of the second participant, so the confidence gate can be applied to them after the fact. For EEGNet, the decoder the demonstration ran, the gate at the deployment threshold accepts 84.0 % of that participant’s windows and is correct on 53.5 % of those, against 93.2 % on the primary participant’s own held-out windows. For the recommended decoder the figures are 79.7 % and 53.8 %, against 92.9 %. Neither decoder is uncertain on the new person, and both are wrong with as much confidence as they are right. Applying the start rule of Section 3.8 to each of the second participant’s recordings in turn, three consecutive accepted windows agreeing, gives EEGNet a correct start in 52.9 % of recordings and a wrong one in 46.5 %, and the recommended decoder 53.5 % and 43.3 %. Lowering the threshold to 0.6 admits more of the windows and changes none of the accuracy and start figures by more than two points.

On this evidence the gate cannot be assumed to substitute for calibration, and a decoder that has never seen a person should not drive the platform for them on the strength of the gate alone. The offline figure bounds one quantity and is not a verdict on the observation above. It comes from the two-class Protocol C models, which must answer Forward or Backward on every window and cannot stay idle, whereas the demonstration ran a five-class decoder, for which Idle is the class the gate defaults to. The two differ in decoder and in participant, and which of them a recorded session with a new person would reproduce is not established.

Reported with the caveats above, the visitors’ observation still warrants inclusion, because it is the only cross-person experience of the live system that exists and it points the opposite way from Protocol C. The two findings are reconcilable. Protocol C is a two-class test, because the second participant recorded two classes, so it says nothing about the Left and Right responses the demonstration produced, and the scarcity of its data is a limit of the test before it is a property of the decoders. It received no HPO of its own, by design (Section 4.5), so it measures how decoders optimised for the within-participant protocols respond to a new person, not what a transfer-optimised decoder could achieve. The data

behind it is one session from a single second participant. The live pipeline also does not act on single windows, because the start hysteresis of Section 3.8 moves the platform only after three consecutive agreeing windows, so the demonstration was judged on filtered behaviour whereas Protocol C scores every window in isolation.

The first and third of those four differences are consequences of the participant and time constraints recorded in Section 6.6, the second is a deliberate design restriction, and the fourth separates the unit each result is scored on. Together they mean that the protocol bounds offline transfer under unfavourable conditions rather than practical usability behind a confidence gate. The observation does not overturn the quantitative result, since per-person calibration remains a requirement on the evidence this thesis can offer, and a qualitative observation establishes nothing about what signal the decoder was reading. It changes the expected return of the follow-up experiment instead. If a gated decoder trained on one person can respond useably to another even some of the time, the distance between the transfer numbers and a deployable system may be smaller than those figures alone suggest, and that is a measurable question, taken up in Section 7.2. Section 6.6 lists the candidate explanations for the disagreement between the observation and the offline figure.

A second qualitative observation is the author’s own self-test, and it has the same status: one unrecorded session, the training participant as both subject and observer, and no number. The pipeline ran with TimesNet, the recommended decoder, in the laboratory the training data had been recorded in, with the laptop carried in hand and the machine endpoint in the simulation mode of Section 3.7, which prints the velocity each accepted window would send, so no platform moved. The confidence gate stood at its deployment value of 0.80 and the start hysteresis was unchanged. Three TimesNet checkpoints were tried in turn, the best Protocol-A seed, the strongest Protocol-B fold and the deployment retrain above, and they behaved alike, which is recorded as an observation and supports no inference. The artefact-rejection stage of Section 3.8 was set aside for the session, because at its configured threshold it forced Idle too often for the decoder to be observed, so the confidence gate and the start latch were the remaining checks on the decision path, and in the author’s account the decoder’s predictions then remained stable despite sudden spikes in the trace.

In the author’s notes from the self-test, the Left and Right classes reached the command state during every turn, whether performed on the spot or within a small radius. Similarly, Forward reached that state on nearly every straight walk across the room in the training direction. Idle was recognised frequently and accurately while the participant was sitting or standing, passing the confidence gate, yet it never reached a command state, because nothing issues the safe one. A direction other than the one performed was rarely commanded, although small preparatory movements appeared to trigger unexpected but accurate readings, such as a Backward command during a step back taken before a turn. This behaviour was not verified further, but the classification was clearly observed during these preparatory actions.

Correct detection required a movement to be performed cleanly and with a single intent, and detection was judged to have failed when several small movements were combined. Backward never reached the command state during a clean performance of the basic task, because the attempted windows either stayed below the gate or switched rapidly between other classes with none dominant. Turning was recognised when the participant looked ahead but not while gazing down at the laptop displaying the live inference, whereas Forward reached the command state on the straight walk even while the participant watched it. The participant’s own analysis of the system’s behaviour suggested that eye movements, re-centring of the gaze, small facial gestures and certain timed blinks may contribute to the decoded label, which aligns with the limitation in Section 6.6, although that session provided no definitive conclusions.

The Backward result contrasts with the offline confusion structure only qualitatively: there, the dominant error of the recommended decoder is a confusion within the forward–backward axis in both directions (Section 5.7), whereas live, attempted Backward produced no command at all and no single class took its place, not even Forward. The cause is not es-

tablished, and several candidates remain equally plausible: the decoder itself, the Backward recordings (whose task types may not centre on the plain backward walk performed live), confidence spread just under the gate, the three-window latch never seeing three agreeing windows, a live movement unlike the training clips, and artefacts. The trial is the calibrated case the deployment protocol estimates, and the participant’s feedback is consistent with a pipeline that reaches the command state for three directions on nearly every attempt by that user but not for the fourth, and a measured session would be needed to isolate why.

6.6 Limitations and threats to validity

The conclusions presented above must be interpreted within the context of their limitations. Accordingly, this section details the constraints of the study and clarifies which claims fall outside its scope.

One participant supplies every training recording. All Protocol-A and Protocol-B results describe how well one person’s movements can be decoded, and Protocol C establishes that they do not transfer. No result here is a claim about a population. The second participant contributed two classes in a single sitting, which suffices to demonstrate that transfer fails and not to characterise how it does so. One consequence is stated separately: whether the decoder that is optimal on one participant’s data remains so when recordings from many are pooled is not measured by this work, and no ordering reported here should be projected to that setting.

Both participants are considered healthy. The clinical target is SCI and stroke [63, 67], and those populations differ in the respects that matter: altered cortical organisation after injury, different and often asymmetric gait, medication, and fatigue. Executed movement is also not available to a patient who cannot perform it, so the paradigm this thesis validates is not straightforwardly the one a patient would use. Extension to that population is not established by this work and is not claimed.

The recording environment is constant. Every session was recorded in the same room with the same equipment. Environmental noise and electromagnetic interference are therefore common to training and test, which may make the decoders look more consistent than they would be in a different room, an optimistic bias of unknown size.

The signal contains variation that the protocol did not control (Section 3.3), and none of it was removed, because ICA assumes spatially stationary sources, an assumption the sway and head movement of executed gait violate (Section 3.6). A decoder that performs well here has done so in the presence of blinks, jaw clenches, postural sway and environmental noise, which is the right condition for a clinical system, but it also means the decoders may be using some of that variation, and nothing in this thesis separates the contributions. The gaze observations of the author’s self-test (Section 6.5) point the same way and are equally unmeasured.

The spatial ablation cannot answer the question of signal origin. A spatial ablation restricting the montage to frontal and then to motor electrodes was run. It supports no claim about the origin of the discriminative signal, because its condition is confounded with channel count (19, 7 and 3 electrodes) and because even a channel-matched version could not separate a cortical source from muscle and motion contamination, both of which are distributed across the montage. The thesis therefore claims direction decoding, evidenced by within-category accuracy, and does not claim to have localised the signal to cortex, and Section 6.3 is a hypothesis about architecture rather than about anatomy.

The non-physiological control ran under a reduced budget on the deployment protocols. Chronos’s Protocol-B and C rows were trained under the three-epoch cap of Section 4.5, so every deployment margin quoted against the control is measured against a floor, and its Protocol-A training length indicates that the floor is a loose one. The configuration-matched statement of what non-physiological pretraining delivers here is the Protocol-A result, and Section 5.4 states its margins so that the conclusion survives the move. The Protocol-C row is additionally moot, because every decoder lies near chance on cross-person transfer.

Artefacts from the moving device are unmeasured. No training data was recorded while the platform was in motion, so the artefact environment of a live session with the moving platform is unknown, and the artefact-rejection threshold is an uncalibrated default. At its current setting it would reject 21.1% of the corpus’s windows, mostly during movement, and because each rejection forces Idle and resets the start counter, the platform would halt on roughly one window in five, more often while walking, and need three clean consecutive ones to move again (Section 3.8). In the author’s self-test (Section 6.5) the stage reverted to Idle too often at its configured threshold to be usable, so the session ran without it, which leaves the calibration of that threshold, or its replacement, as open work.

The corpus includes two groups of recordings that should be handled differently in any future replication of this experiment.

The first consists of an estimated 5 to 10% of the recordings, during which the participant deliberately blinked or clenched their jaw to verify that the system was operational. These clips were later renamed to the base task type of their respective directional movement, as the intentional artefacts were produced while the lower body performed the targeted gait motion. This retention was intended to introduce realistic variability for model generalisation across all classes, most notably in Forward and Backward (Section 3.5). However, because these specific recordings are not flagged in the archive, their impact on the final results could not be measured.

The second group consists of nine of the 500 pooled training recordings, which are early files of the `test` task type that had been left outside the class folders and that the loader admitted because their names carry a class token. They are identifiable, so an ablation without them was feasible, yet it was declined for a procedural reason: removing them changes the seeded selection, so the sample would be drawn afresh rather than reduced by nine, and every protocol would have to be rerun on a different subset. They account for 216 of the total 12 000 windows (under 2%), all Forward or Idle, and the six in the first class are also among the recordings Protocol C draws on, so their effect on the results is likewise unmeasured. A repetition of this work should record checks under a task type of their own and keep set-aside recordings out of the loading path.

Protocol A’s test set is one fixed draw. The within-subject split is cut once, over a seeded recording order that pins one representative of every task type to its front, so the test set is a single sample of roughly 10 recordings per class and a task type with one recording in the pool is never tested under it (Section 4.3). Neither property affects a ranking claim, because all of them rest on Protocol B, where each recording is tested in exactly one fold. A repetition of this work should shuffle the full pool before the cut and repeat the split over several draws, so that the ceiling, too, carries fold variance and every recording can reach its test set.

Randomising membership at the level of the individual window would not serve the same purpose, because consecutive windows share three quarters of their samples, so a window-level draw fills the test set with near-duplicates of training examples (Section 4.4). The nearest valid form of that refinement assigns contiguous segments of each recording, with the one-window gap discarded at every internal boundary, and would let within-recording material be tested at a cost of about one sixth of a cut recording’s windows.

One preprocessing step the evidence recommends was not applied. Bad-channel interpolation is one of only two steps Delorme’s comparison found to improve outcomes reliably [18], and this pipeline does not perform it: the detection and interpolation routines exist in the codebase but are reachable only through a preprocessing function nothing calls. The same is true of the common average reference, the re-referencing that subtracts the mean of all channels from each of them. A repetition of this work should wire the interpolation in and measure what it contributes on dry-electrode data. Section 3.6 records the operational grounds for the omission, and concedes that they were never put to a test. The same repetition should probe the other side of the 1 Hz high-pass corner: that section concedes that the cutoff deliberately excludes the movement-related-potential band [64], and a low-frequency variant of the band-restriction control would measure what that exclusion costs.

The classical family has a single representative, and a second was warranted. The filter-bank spatial-pattern pipeline is the only non-deep decoder in the benchmark, and the other established classical family, Riemannian classification of spatial covariance matrices, was not evaluated. Tangent-space classification, which maps covariance matrices onto a flat space where linear classifiers apply, has been reported to exceed CSP [10] with LDA by around five points on a standard MI benchmark, the BCI Competition IV data [68, 6]. Because FBCSP+LDA finishes inside the leading group here, the claim that classical methods remain competitive on this task rests on a single instance of one family, and a Riemannian decoder might have placed higher still. The conclusion that pretraining contributes little would not be weakened by that result and might be strengthened, but the set of decoders as reported is narrower than the argument it supports.

The classical baseline was measured under a normalisation its features were not designed for. Per-epoch z-scoring reached FBCSP+LDA along with every other decoder, rescaling each channel’s broadband amplitude before the filter bank, and Section 3.6 discloses the divergence between the code’s recorded intent and the pipeline as executed. The direction of the effect is unknown, and no reported conclusion depends on the baseline being unharmed: it finishes inside the leading group as measured, so a handicap, if one exists, means the classical anchor is understated rather than flattered. A single refit without the per-epoch scaling is the direct check, costs one deterministic run, and belongs in any repetition of this work beside the band-restriction slot Section 6.3 already reserves for this model.

Five folds underpower every comparison. No pairwise test survives correction, so the benchmark supports a ranking with intervals and explicit ties and does not support a claim that any decoder is better than another. Protocol C additionally carries no intervals at all, so it establishes a boundary and cannot rank.

Three asymmetries remain between the decoders, all disclosed where they arise. Two of the three FMs were fine-tuned on hyperparameters chosen under a frozen encoder, so their results are lower bounds. The LoRA arm was narrower than specified, so it bounds what those models received rather than what the technique can achieve. LaBraM’s checkpoint is also a community re-release carrying a discarded classification head, not the weights published with the paper, two of whose parameter tables arrived untrained (Section 6.2), so that row is a bound on the artefact, and the model its paper describes remains unmeasured. If it were re-run on the licensed weights and reached the leading group alongside the other two, the argument of this chapter would gain a further case, one more pretrained model reaching the leading group without exceeding it.

Two further gaps are not asymmetries. One searched hyperparameter, head dropout, never reached any model, and every run trained with the head’s default of zero (Section 4.5), a condition shared by all nine decoders. No ablation of training without augmentation exists either, so the contribution of that library to any row is not measured.

The discrepancy between the live observation and the offline transfer analysis remains unexplained. Section 6.5 reports a gated decoder responding mostly correctly to two uncalibrated visitors, and the same section reports the archived two-class transfer models being confidently wrong on the second participant at nearly the rate they are right. The candidate explanations are all untested: the five-class decoder of the demonstration with Idle as its default class against the two-class models of the archive, the absence of any Left or Right recording from a second person, the lowered threshold, the training room, the visitors’ movement style, the operator’s judgement of what counted as correct, the shortness of the session, and chance. A recorded session with a new person under the five-class system, scored per window and per start, is the experiment that separates them, and until it exists the observation is a reason for that experiment and not evidence about its outcome.

The platform was exercised but not measured. The live pipeline ran in two unrecorded sessions, a demonstration to a pair of visitors on the wheeled surrogate and the author’s self-test with the machine endpoint in simulation, and no measured result with the platform exists, for the reasons stated in Section 3.7. The inference latency of Section 5.6 is a property of the decoder on the host rather than a measurement with the machine, and no claim in this thesis rests on either session.

Chapter 7

Conclusion and Outlook

7.1 Conclusions

This thesis asked whether a directional intent interface for a mobile, gravity-supported gait trainer is feasible on a dry-electrode headset with the quantity of data a single participant can realistically supply, and which family of decoder should drive it. It answered both questions by benchmarking nine decoders across three evaluation protocols under a contract that grants each model its own optimal configuration and holds everything else fixed.

The work comprised five parts. It recorded a corpus of fifteen-second single-movement EEG recordings on a dry-electrode headset from one participant, with a second person’s Forward and Backward recordings held out for the transfer test, and built the acquisition, conditioning and control pipeline that turns a decoded class into a platform command. It benchmarked nine decoders, from a classical filter-bank pipeline to three pretrained EEG FMs and a non-physiological control, under a comparison contract with a matched hyperparameter search across the three evaluation protocols. It defined the within-category direction metric that separates such decoding from movement detection. It measured the deployment envelope of latency, memory, confidence reliability and gate behaviour. It exercised the live pipeline in two unrecorded sessions, with EEGNet on the wheeled surrogate for two visitors and with TimesNet by the training participant, without a measured test of the full system, which is the experiment that follows this work.

The first finding concerns feasibility, the second aim of Section 1.2. A usable directional decoder can be built on a dry-electrode headset with per-participant calibration: several architectures reach between 82.8 % and 85.5 % balanced accuracy against a chance level of 20 % and run inside the latency budget. Every EEG decoder resolves direction within a movement category above the two-way floor, LaBraM only narrowly, which is the precondition that makes any ranking meaningful. Reaching it requires no large pretrained model: decoders that learn from the participant’s own recordings alone suffice. The feasibility established is that of the decoding stage. The live system was exercised in two unrecorded sessions and not measured, so the second aim is met for the decoder and remains open for the system around it.

The second finding answers RQ1 and RQ2. EEG FMs, given native preprocessing and fully fine-tuned encoders, reach the leading group at best and do not head it: the strongest of them reaches 83.9 % balanced accuracy on the deployment axis against 85.5 % for a general time-series architecture trained from scratch, with a classical spatial-filter baseline at 82.8 %. At five folds none of the leading decoders separates statistically from any other, so the defensible statement is that large-scale EEG pretraining admits a model to the leading group, which already contains a purpose-built convolutional network and a spatial-filter method first published in 2008, and adds nothing once inside. Pretraining on a non-physiological corpus, by contrast, is worth 46.2 points less than the leader under the control’s capped deployment budget, and more than 37 points less on the configuration-matched Protocol A, where it still fails to resolve direction at all, so the domain of a corpus affects the result even where pretraining itself does not.

Three further findings stand, with two qualitative observations beside them. Cross-person transfer does not survive for any decoder offline, so per-participant calibration is a requirement of the system rather than a convenience. In one short unrecorded session EEGNet drove the surrogate for two uncalibrated visitors and the platform moved mostly in the performed direction, a qualitative observation that the offline two-class proxy does not

reproduce and that only a recorded session with a new person can settle. In the author’s own unrecorded trial, TimesNet reached the command state for Forward, Left and Right on nearly every attempt and for Backward on none, an observation of the calibrated case that a recorded session would turn into a number. The decoder with the highest mean is not the one that becomes usable with least calibration: at five trials per class EEGNet leads the learning curve and is the only decoder on it that resolves direction at all. The discussion resolves that tension in favour of accuracy, calibration being a cost paid once per patient while accuracy is at stake on every window the system ever decodes. The accuracy leader also runs more than an order of magnitude faster than the cadence requires, and its confidence estimates are reliable enough to gate on.

Four methodological findings stand beside the empirical ones. A grouping variable confounded with the class label produced a 30-point artefact that would have supported a confident and false account of the field. Single-fold selection nearly produced two further wrong conclusions, arriving from opposite directions. Design constraints introduced against the first artefact outlived it and had to be retired. A five-fold design also cannot support a claim that one decoder is better than another, whatever the point estimates suggest.

7.2 Outlook

Seven lines of work follow, roughly in order of what they would settle per unit of effort.

Test the hypothesis of Section 6.3 on the decoder that would settle it. FBCSP+LDA is an explicit five-band filter bank, so if oscillatory inductive bias is what confers robustness to band restriction, it should be the most robust decoder in the roster. It was not included in that study. Adding it costs one run of an already-written control and either strengthens the hypothesis or falsifies it, which is the largest gain per run available here. A second one-run check on the same decoder belongs beside it: refitting the baseline without the per-epoch scaling that Section 3.6 discloses reached it, which would measure a distortion whose direction is currently unknown.

Add the second classical baseline. A Riemannian decoder (covariance matrices classified on their manifold, or projected into a tangent space and separated linearly [6]) is the one well-established family this benchmark left out, and it is the second-cheapest experiment on this list: mature open implementations exist, the method needs little hyperparameter search, and it would slot into the existing harness unchanged. Given that a 2008 spatial-filter pipeline already finishes inside the leading group, the question of whether a better classical method finishes at the top of it is answerable in a single run.

Re-run LaBraM on the licensed weights, and audit checkpoints by their statistics rather than their key names. The checkpoint benchmarked here is a community re-release whose electrode and patch-order embeddings arrived untrained (Section 6.2), so that row bounds a damaged artefact rather than the published model. Requesting the official weights and re-running one decoder across the three protocols is a bounded experiment and the only one that would move that number. The checkpoint audit itself costs little to apply. The loaders here verify that a checkpoint populates at least 90 % of the wrapper’s parameters, which is a test of presence and not of content: a tensor that is present and empty passes it. Comparing each loaded tensor’s dispersion against its initialisation is a few lines, it would have caught this before the first run, and it belongs in any pipeline that treats public checkpoints as inputs.

Repeat the benchmark with the corrections this work identified. Section 6.6 records a set of design choices that a repetition should change before any new data is recorded, and all of them are bounded. On the evaluation side, Protocol A should shuffle the full pool before the cut and repeat the split over several draws, so that the ceiling carries fold variance and every recording can reach a test set, and any within-recording split should assign contiguous segments with a discarded gap rather than individual windows. On the data side, chain checks should be recorded under a task type of their own, set-aside recordings should be kept outside the loader’s reach, a second person should record all five

classes so that the transfer test covers the turning directions, and the deployment retrain should stratify its validation slice on label and task type together, so that no gait style is absent from training. On the preprocessing side, bad-channel interpolation and the common average reference exist in the codebase and should be wired in and measured on dry-electrode data, and a low-frequency variant of the band-restriction control would measure what the 1 Hz high-pass corner costs. On the fairness side, CBraMod and LaBraM should receive the same search under full fine-tuning that Neuro-GPT received, the LoRA arm should target the layers each model defines, head dropout should be searched as intended, an ablation without augmentation should be run, and the non-physiological control should be trained under the full budget on the deployment protocols so that its margin is measured against a like-for-like comparator. None of these changes the question the benchmark asks. Together they remove every asymmetry the comparison contract was unable to close, and they turn the reported lower bounds on the pretrained models into measurements.

Record more participants. The single-participant design is the binding limitation on everything this thesis can claim, and the cross-person result establishes that adding participants changes what can be asked and not only how much data there is. A multi-participant corpus would permit the question this one cannot: whether a decoder pretrained across people calibrates faster on a new one than a decoder trained from scratch, which is the form in which pretraining might still turn out to help, and which this benchmark did not test. The same corpus would also measure what Protocol C could only bound: how much calibration data a new person must supply, and whether alignment or few-shot adaptation methods reduce it, so that the calibration requirement becomes a measured cost rather than a boundary. It is also what would turn the qualitative observation of Section 6.5 into a measurement: whether the five-class gated system responds usably to an uncalibrated person is, at present, something this project has seen in one short unrecorded session with EEGNet and measured only through a two-class offline proxy that points the other way.

Run the pipeline on a person with the moving platform, and measure it as an experiment rather than a demonstration. The safety layer is built and the latency budget is met with a wide margin, but two of its parameters have never been calibrated against data recorded while the machine moves, because none exists. Recording it is a prerequisite: the artefact environment of a live session with the moving platform is a different measurement from anything reported here. Artefact handling must be settled first, since the rejection stage at its configured threshold forced Idle too often and the author’s self-test ran without it (Section 6.5). The Backward result of that trial is the first question such a measurement should resolve, and two additions the trial suggested would sharpen it: an eye-movement recording alongside the EEG, by eye tracking or an electro-oculogram, to cross-validate the decoded labels against gaze, and a headset more sensitive to small cortical signals.

Patients last. Every step above uses healthy participants performing movements they can execute. The clinical target population differs in cortical organisation, in gait, and, most consequentially, in whether executed movement is available at all. That work requires a separate study under formal ethics approval, and the paradigm it would use may not be the one validated here. What this thesis offers it is a characterised decoder, a clear account of what it does and does not read, and a benchmark design whose failure modes have already been found. One idea from this project’s early notes belongs to that stage rather than this one. The microstate readout of Section 3.2, used here only as a signal-quality monitor, was originally conceived as a recovery marker: recording a patient’s topographic patterns across a course of rehabilitation and asking whether they converge toward those of able-bodied participants performing the same movement. Nothing in this thesis tests that, but a patient study equipped with the same headset would obtain the measurement at almost no cost.

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Appendices

Appendix A

Declaration of Artificial Intelligence Use

Provided in fulfilment of Version 2 of the declaration of originality, which requires the exact designation of each technology used and the prompts employed.

A.1 Purpose and structure of this appendix

Version 2 of the declaration requires that every element taken from an artificial intelligence be declared, together with the exact designation of the technology and the prompts used. This appendix provides that record. It covers the use of artificial intelligence over the thesis period stated below, both in the project work (the design, the code, the experiments and the analysis) and in the writing of the thesis document itself. It first separates the two unrelated senses in which artificial intelligence appears in this thesis, then states the division of responsibility between me and the tools, names each technology, and then works through the project area by area, giving for each the nature of the assistance and representative prompts. It states explicitly what artificial intelligence was not used for, records what the version-control history does and does not show, and closes with my statement of responsibility.

A.2 Two distinct senses of artificial intelligence in this thesis

The term carries two unrelated meanings in this project, and this declaration concerns only the second of them.

The first is the subject of the research. The thesis is a benchmark of nine decoders for five-class directional movement intent, four of which are pretrained models loaded from public checkpoints: the EEG foundation models Neuro-GPT, CBraMod and LaBraM, and Amazon Chronos as a deliberate non-EEG negative control. These models are the objects under study. Their architectures, pretraining corpora, weight sources and reference papers are documented in the project’s model documentation and cited in the thesis. They are not aids used to produce this work, and nothing they generated enters its text.

The second is the use of artificial intelligence assistants as a working aid during development and writing. That use is what is declared below.

A.3 Division of responsibility

The research question, the experimental design and the interpretation of the results are mine throughout. In particular, I formulated the thesis question and the four research questions it decomposes into; selected the nine models and the tier structure that organises them; designed the three nested evaluation protocols and the reasoning behind each; chose the five-class label set including Idle and the accuracy-first deployment criterion with its confidence gate; defined the recording protocol and personally carried out all data collection with the two consenting participants (myself included); fixed the signal conditioning and the leakage rules; diagnosed the flaw in the earlier session-grouped cross-validation split and specified its replacement; and verified every reported number against the archived, provenance-stamped result artefacts.

I used artificial intelligence assistants throughout, and my working method was consistent enough to state once. For any question of consequence I formed a position first,

from published work and from earlier experience, and then put that position to an assistant to be tested rather than executed: to be argued against, examined from angles I had not considered, and corrected where it was wrong. The decision that followed was mine, taken in the light of that exchange. I then set out the logic, the approach and the methodology to be followed, and an assistant produced the implementation. Where a matter was routine I would instead ask which approach was standard practice and have the assistant apply it.

I read every change before accepting it and checked it against what it was expected to do. Review of this kind establishes that a change does what it was asked to do; it is not proof against every defect, and this appendix records two that survived it, a tuned parameter that reached no model and a low-rank adaptation target that matched no module, both found by a later audit of the codebase and disclosed below. Where a result was unsatisfactory it was reworked, frequently by directing an assistant again rather than by editing the file by hand. Because an assistant could carry one change across several files at once, integration was often done that way as well; I checked the outcome rather than performing each edit myself. Inside the decisions set out at the head of this section, the assistants made routine implementation choices of their own.

A.4 A note on the record of prompts

I used the tools interactively across the thesis period, which began in March 2026, largely inside a coding environment where instructions were short, iterative, and issued in sequence against a live codebase. No complete verbatim record of that use exists, and none is presented here as complete.

A substantial partial record does survive, and I have recovered everything I was able to. On the primary workstation the coding assistant’s own logs cover 14 June to 4 August 2026: three sessions, 246 prompts as recorded in the tool’s own history file and 295 counted from the transcripts themselves, 4 720 assistant replies, 41 background sub-agent transcripts and 2 931 tool invocations, with a SHA-256 checksum recorded for every file. On the second machine, the laptop used as the acquisition bridge, they cover 6 July to 4 August 2026: four further sessions, a dozen or so prompts, five sub-agent transcripts and roughly 200 tool invocations, together with the assistant’s retained memory notes, its one surviving implementation plan and its configuration. Since 4 August 2026 the work has continued on a third machine, a personal desktop computer at home, on which the thesis document has been written and revised. Checked on the day this declaration was signed, the assistant’s local store on that machine holds 27 sessions begun on or after 7 August 2026, together with 72 sub-agent transcripts, all of them on Claude Opus 5 and Claude Fable 5. They have not yet been added to the archived record above, and they will be, on the same terms. A few further prompts may still follow between the signing of this declaration and the submission of the thesis, for the final corrections, and any such prompt is recorded in the same store and falls within the uses declared here. To these are added six exported conversations from the literature-review tool, totalling 502 pages, and 11 exported conversations from the chat interface. I retain this record and will make it available to the examiners on request.

The gaps are set out here so that the surviving record is not mistaken for a complete one. The largest was not a matter of choice. The coding assistant deletes its own local transcripts once they pass a retention window of about 30 days, without prompting the user. On 4 August 2026, while this disclosure was being assembled, that cleanup ran on the laptop and destroyed five sessions dated 10 to 15 June 2026, covering early robot-control and deployment work, together with the implementation plans belonging to them. I attempted recovery and failed.

Two further sessions, one on each machine, were lost earlier still, and each is known only because a short note the assistant wrote during it survived, so others may have gone without leaving any trace. I have since disabled the automatic cleanup on both machines.

Beyond that, I deliberately switched conversation history off in most cases, so that the exchanges would not be retained and used for the further training of the providers’ models.

For Google Gemini this was so throughout, and nothing was ever stored that could be exported. I also kept sessions short and reset them often, because a context window that fills up degrades the quality of an assistant's answers and raises the rate of confident but incorrect output, so the record is fragmented across many short exchanges rather than held in a few long ones. Work predating the retained logs falls outside them, the tool having offered no means of export at the start of the project. Some material was lost through changes of operating system and tool version, and some through one routine clearing of history before its relevance to this thesis was apparent.

In fulfilment of the requirement to indicate the prompts used, this appendix therefore describes, area by area, the nature of the instructions I gave and provides representative examples. The examples are of two kinds, and each is labelled where it appears.

Most are faithful reconstructions of the kind of prompt I used rather than quotations. They are written to show what I asked for and why, which a fragment lifted from the middle of a working conversation often does not, and they cover the areas where no log survives at all. My instructions ordinarily began from something of mine, an existing design, a base implementation or a written specification, and named the elements, parts and structures the result was to include. A draft from nothing was the exception, asked for where I did not know how, and it was then reworked with the assistant over as many rounds as it took to match the behaviour and appearance I intended. The reconstructions are written in that form.

A smaller number are quoted verbatim from the retained log, reproduced exactly as typed, including their errors. These are supplied because the declaration asks for the prompts I employed, and where the real ones survive it seemed right to show them rather than only to describe them. They are less tidy than the reconstructions and are meant to be: they are evidence rather than exposition, each is checkable against the record, and they carry things a reconstruction cannot, such as who noticed a problem and when. The retained log is weighted towards the later phase of the work, covering the benchmark re-runs, the analysis and the final documentation, so verbatim examples appear only in those sections. Much of the earlier implementation described below falls in the periods named above, where nothing survives.

A.5 Technologies used

Claude Code (Anthropic)

The primary tool, accounting for most of the assisted work described below. Claude Code is a command-line coding assistant; I used it in the Ubuntu terminal and inside Visual Studio Code, and from August 2026 on the home desktop named above, running the models Claude Opus 4.6, 4.7, 4.8 and 5 as they were released over the project period, and latterly Claude Fable 5 for work on the thesis document. Background sub-agents ran on Claude Haiku 4.5 as well as on Claude Opus 4.8.

I supplied reference material to the assistants myself: links, files, and copied pages from official documentation and from technical forums, pasted into the exchange so that the work proceeded from the correct information rather than from what a model might recall. Whether the tools also consulted sources of their own cannot be established from my side for the sessions whose logs are lost. The coding assistant's client recorded no web-search request on the primary workstation. On the home desktop the retained logs show it fetching bibliographic index records and running web searches, chiefly while the reference list was being verified under my rule that no bibliographic field be taken from memory, and the bibliography source marks the records fetched in that way with their index and date. The literature-review tool returns answers carrying numbered source citations, so it plainly retrieved material as part of its ordinary operation, and any assistant may in addition have drawn on documentation held in its training. My practice of supplying material myself was a means of ensuring the information used was correct. It is not a claim that nothing else was consulted.

The assistant’s conduct in the repository is governed by an instruction file, CLAUDE.md, which records the technical conventions it was required to follow. An earlier and longer version of the same file, preserved in the retained record, set out standing instructions on research conduct: that the recordings and the pretrained weights were read-only and never to be modified, that a completed run’s results and checkpoints were never to be overwritten, that no metric or datum was ever to be fabricated and anything missing or uncomputable was to be reported as such, that performance claims required statistical backing, and that all results were to be framed as within-subject. I later had the file shortened, on the view that dense directives constrained the assistant more than they helped. Several source files carry comments referring back to that file as the authority of record.

Claude (Anthropic), chat interface

The same model family, latterly Claude Opus 5 and Claude Fable 5, used chiefly for analysis and decision support. The use was conversational. I supplied the information and used the exchange to discuss and analyse it, turning a question over across several rounds in order to break it down, see it from different angles and then decide. A second use was feedback on material I had already written, including a critique of the prompts used elsewhere in the project. A third was the rapid trial of alternatives: several candidate wordings for the same piece of information, structures and visual ideas for the graphical interface, and titling and naming conventions tested in quick succession rather than one at a time. The written material worked on in this way includes the results write-up, the figure captions, the presentation, the recording-protocol and data-description documents, the project title, and the structure and wording of this declaration.

Two uses fall outside that description and are recorded here because the account above would otherwise be too narrow. On one occasion I had this tool rewrite a source file directly, the live microstate visualiser, to correct a scalp topography display that would not hold steady. Its proposal was adopted. Separately, I used the tool to turn my rough notes into structured briefs that I then gave to the coding assistant, so a number of the instructions reaching that tool were composed here from material I supplied.

Perplexity, running ChatGPT-5.6 Terra (OpenAI)

Used chiefly to review and synthesise published work. I passed papers I had read, and other documents relevant to the project, to this tool to be summarised and broken down, so that the technologies and methods described or referenced in them could be extracted and explained. The exact designation is Perplexity, used in both its Search and its Deep Research modes, with ChatGPT-5.6 Terra (OpenAI) selected as the underlying model. I did not use ChatGPT on its own at any point and reached it only through Perplexity. The same tool was also used in writing the thesis document, which is described under “Writing the thesis document” below.

I exported and retained six exchanges with this tool. Four were literature synthesis, methodological consultation and an early exploratory pipeline description. Two were code sessions, and between them they contributed considerably more than review.

The first reviewed the three acquisition files as they then stood. The restructuring it proposed, a settings hierarchy and a set of output formatter classes, was not adopted and appears nowhere in the project. Two things from it were adopted. It supplied the code from which I built the first version of the live microstate module, as I state within the exchange itself. It also produced the revision that carried the configuration module from the seven-channel headset to the 24-channel one, including the 19-channel list, the auxiliary-channel list and the old-to-modern electrode rename map used for montage lookup. Those three constants are in the configuration module today.

The second session began with the acquisition bridge script, where three corrections were adopted and are marked in the committed source. It then extended to the streaming module, where several refactors were taken, among them moving the PowerShell command

builder out of the entry function; that builder was later relocated again by the coding assistant into a shared helper file. Finally, I used Perplexity to generate a full rewrite of the live microstate module, explicitly guiding the output with documentation and directives I supplied myself. The module in the project today derives from that assisted draft, sharing most of its principal identifiers. The parts that differ are the template-fitting and cluster-analysis functions, which the coding assistant added to the module afterwards and which are recorded as such in its own project notes. The surviving record does not fix the date of that addition. It falls after 19 March 2026, the date of first access to that tool on this account; before then I worked only in the chat interfaces and with the other assistants named above.

Of the remaining four exchanges, two were methodological consultation, on whether the intended classical and deep-learning approaches were reasonable, and on whether the class imbalance in my own recordings would be a problem. One was literature synthesis on EEG preprocessing. The last was an early exploratory request for a plain-language description of the intended pipeline. The design it returned is not the pipeline that was built, and no part of it reached the thesis.

No protocol, statistic, figure or reported number originated in these exchanges. The channel constants are the one place where code from them sits upstream of the benchmark, since the preprocessing path reads the configuration module. They encode the headset's own channel names and the standard electrode relabelling rather than any analytical choice, but the code expressing them came from that exchange and is declared here rather than left to be found.

Google Gemini

Used for error diagnosis and debugging support, typically reading a stack trace and suggesting a probable cause, and to draft short single-purpose test scripts and individual functions. Long or many-part instructions produced unreliable output from this tool, so I gave it only short prompts with a single objective, such as locating one bug or writing one small function. No record of these exchanges survives, so what became of their output cannot be established. My recollection is that the scripts were throwaway diagnostics rather than project code, but a debugging tool used over months will have influenced the code it was pointed at, and some of its suggestions were very likely acted on even where no trace of them now remains. I do not claim that nothing from this tool reached the project. The exact designation is Google Gemini 3.1 Pro. I used it for the thesis document as well, together with its image model Nano Banana 2 on photographs, and the section on writing that document records how.

A.6 How the assistant was run

The reason for working this way was mostly an engineering one. Several parts of the project began as my own code and grew steadily more complex as features were layered onto them, until fitting a new behaviour in by hand meant threading it through many nested conditions and several outside calls. That is where manual editing is most likely to introduce a fault that is not immediately visible, and it is where the assistant was most useful: it could carry one change through every place that needed it and leave the execution path coherent, while I read the result rather than performing each edit. The same reasoning applied to work that spanned files. Convenience was secondary to this.

Two features of the working setup bear on the division of responsibility. The permission configuration pre-approved no tools, so proposed edits and commands were presented to me for approval rather than carried out automatically. Separately, I ran all work on the training machine myself. The assistant supplied command blocks and I executed them and pasted the output back, a rule the assistant recorded in its own notes after I declined to let it run commands on that machine directly.

The assistant maintained a small set of persistent project notes across sessions, 15 of them on the primary machine, recording decisions already taken so that they survived the resetting of conversations. Those notes record my decisions rather than the assistant’s: that the deployment choice was to be accuracy-first with cost reported and not gated, that the corrected split superseded the earlier one, and that commands were to be run by me.

The assistant also delegated work to background sub-agents, 41 on the primary machine and five on the second, running Claude Haiku 4.5 and Claude Opus 4.8. Their tasks were almost entirely read-only: tracing execution paths through the code, mapping loader and runner internals, and auditing artefacts before the final archive was assembled. They were launched by the assistant rather than prompted by me, and their transcripts form part of the retained record.

A.7 Areas of use and representative prompts

Acquisition, recording and the live dashboard

I defined the recording protocol, the five-label scheme, the file-naming convention and the majority-label rule used to discard ambiguous recordings. I wrote the original streaming module myself, including the terminal interface that selects between the live dashboard, recording and system-status modes for the DSI-24 stream and dispatches to the corresponding handler. It was then extended over time with assistance, mode by mode and layer by layer. The assistant produced the groundwork for the live overview dashboard built on PyQt6 and pyqtgraph: the scrolling 19-channel time-domain panel, the scalp topography rendered through MNE, and the per-electrode power-spectrum panel. The microstate panel is the exception: it lives in its own module, built from code supplied by the literature-review tool as described above, and I integrated it with the rest of the dashboard. I designed the keyboard-labelled recorder that writes MNE annotations into EDF, setting out its behaviour as guidelines from which the assistant built it; the offline recording viewer, with its autoscale behaviour and topographic channel ordering, was drafted the same way. I specified panel behaviour and refresh rates and corrected the threading. The streaming module later received a set of refactors from the literature-review tool, described above, which I reviewed and kept.

Representative prompts

“Using my design for a terminal menu, please clean it and adapt it so it connects to the different routines I’ve added, there should currently be 3 options which are: live view, record and system status for the EEG headset.”

“Here is the design I’d like to have for my overview dashboard drawn by hand. Please implement and sketch it accordingly using the latest version of pyqt. On the left there is a multi-channel plot for each electrode of the EEG headset, bottom right is the spectrum of each channel that can be selected and on the top right there is the scalp topography with the microstate decision bar graph next to it.”

“I’d like to implement a simple recorder to read the data streamed to the terminal and build it into saved .edf files using my keys to define the label assigned to those samples in real-time. If I press a key down and hold it, the release should unbind the label and revert to Idle and if I do a single quick press then the label gets locked until another key is explicitly pressed.”

The WSL-to-Windows bridge

The headset SDK runs only on Windows while the rest of the project runs under WSL2 Ubuntu. I established that split and worked the bridging out by hand first, performing the

whole sequence manually before writing the script that automates it. I then brought the assistant in to help run that script in the background from the WSL terminal, reading its standard output without blocking and keeping the bridge alive across mode switches and successive recording sessions, and the code was improved over many iterations the same way. The same pattern was later reused for the machine-control agent.

Representative prompts

“From a Python script running in WSL Ubuntu, how do I start a Windows PowerShell process in the background and keep reading its output without blocking?”

“How can I keep a Windows-side acquisition process alive across multiple runs while driving it from WSL, and restart it only when the mode actually requires it?”

“Here is the bridge sequence I’ve worked out and run manually, and the script I started from it. Please wrap the Windows DLL entry point I’m using with ctypes and stream the samples out as CSV rows, following my sequence exactly.”

Preprocessing and the data pipeline

I fixed the signal conditioning and the anti-leakage rules, and wrote the filters themselves in the first instance. They were later revised with assistance onto a common signature and style, a paired design-and-apply form shared across the finite-impulse-response, infinite-impulse-response and notch families, and extended the same way to further filtering and preprocessing methods. The assistant wrote base forms of the windowing and dataloader path against my specification, including the single filter-construction function shared by the offline pipeline and the online preprocessor so that the two cannot drift apart, the NumPy epoch cache keyed on a hash of every preprocessing parameter, and the per-worker random-number reseeding that decorrelates augmentation noise across PyTorch DataLoader worker processes.

Representative prompts

“Take the bandpass and notch filters I’ve written and build the pipeline around them following my specification: cut the signal into fixed windows with a stride, z-score per channel and per epoch, and yield PyTorch dataloaders.”

“Please make one function that holds the bandpass coefficients, so the offline pipeline and the live preprocessor can’t end up different.”

“Please cache the preprocessed epochs to npz and key the filename on a hash of every preprocessing parameter, so changing any value makes it redo the cache.”

“Adjacent samples in a batch are getting the same augmentation noise. Please reseed each dataloader worker’s NumPy generator, including the ones cached inside the transforms.”

External dataset adapters

I selected four public datasets as candidate comparison material and defined how their labels would map onto this project’s five classes. The assistant drafted the adapters that convert each source into the project’s format. In the end neither the external datasets nor these adapters were used: the thesis rests entirely on data I recorded with the two consenting participants. The adapter code remains in the repository, but no result reported in this work derives from it.

Representative prompts

“Please write an adapter that converts this public EEG dataset into the same windowed EDF format as my own recordings, resampling to 300 Hz.”

“Please load this .mat file with scipy.io, work out the trial structure, and map the four cue classes onto my label set.”

Data augmentation

I decided which perturbations are physiologically defensible for EEG, for example that a CutMix variant must mask a contiguous slice of time rather than mixing across channels, which has no biological meaning for scalp EEG. I wrote the first of the transforms myself, the simplest and most obviously defensible ones, and the library was extended from there with assistance. The assistant implemented the remainder and the registry that exposes most of those transforms behind a single uniform magnitude parameter so that a hyperparameter search can sample them consistently, together with the random-augmentation policy, the MixUp and CutMix collator, the loss wrapper that accepts either hard labels or soft mixed labels, and the targeted wrapper that applies augmentation only to under-represented classes.

Representative prompts

“Please wrap these transforms behind a common interface with a magnitude in $[0,1]$ so a search can sample them uniformly.”

“Please write a MixUp and CutMix collator for EEG where CutMix masks a contiguous time slice, plus a loss wrapper that accepts either hard labels or a soft $(y_a, y_b, \text{lambda})$ triple.”

Model integration and the classification head

I selected the nine models, the tier structure and the common interface every model had to satisfy. The assistant wrote the scaffolding that loads each model from its source library or repository and standardises it behind that interface, with one wrapper file per model, including the non-strict loading of local pretrained checkpoints for the three EEG foundation models, the lazy load for the Chronos backbone and the encoder-only treatment of NeuroGPT. It also drafted the classification head with its linear and multi-layer variants, and the import-isolation loader that resolves a package-name collision between the supervisor’s model library and the project’s own package.

Representative prompts

“I’ve defined the common interface every model has to satisfy, encode and forward behind one API. Please write the wrapper class that loads this model from its library and standardises it behind that interface, one wrapper file per model like the others.”

“Please write a simple classification head module with a couple of head variants, linear and a small MLP, that I can attach on top of an encoder output.”

“Two packages in this project are both called models and the imports collide. Load the library one from an explicit file path under a different module key and restore sys.path afterwards.”

“Loading these pretrained weights fails on missing keys, can you load them non-strictly and tell me which tensors didn’t match.”

Benchmark protocols and the training harness

I designed the three nested evaluation protocols and their rationale, which the thesis sets out. I also diagnosed the flaw in the earlier session-grouped split, whose grouping variable was confounded with the class label, and specified the recording-grouped replacement. The assistant implemented and integrated the protocols: the recording-grouped splitter, the cross-subject split together with the audit that asserts the held-out participant never reaches training or validation, the task-aware file selector, and the shared training routine used by every protocol. It also wrote the results writers: the flat per-run table, the nested results file carrying the winning hyperparameter configurations, and the per-run manifest recording the command line, the environment and the git commit.

Representative prompts

“Following the protocol design I’ve laid out, please implement the benchmark runner so every model trains under the same splits and logs balanced accuracy, macro F1 and kappa per seed.”

“Please set up a recording-grouped cross-validation split with StratifiedGroupKFold so that no recording appears in more than one of train, validation and test, and every fold still carries all five classes.”

“Please add a check that makes sure the held-out subject never appears in train or validation, and fails loudly if it does.”

“Please write the results table after every model finishes, so a run that is interrupted partway through does not lose everything that completed before it.”

Quoted verbatim from the retained log

“To update you on the state of the project. I’ve found out that protocol B was incorrect and had to reframe it. Before moving on I’d like you to understand and tell me what each protocol is and does. Then once we agree on the protocols and they are supposed to be. We’ll move on to the next step.”

Quoted verbatim from a retained chat export, my own notes specifying the protocols

“protocol A: me protocol B: me but more realistically protocol C: train only on me, but test on someone else (generalizes across patients)”

“make sure that the protocol A and B never used ‘ac’ data in train or val, and it was strictly test for both A and B”

Hyperparameter optimisation

The Optuna framework was suggested to me by my supervisor. I researched it, decided to adopt it, and specified the search: which parameters were allowed to vary, the budget for each model, and the rule governing when a winning configuration is reused across protocols. The specification and its rationale are set out in the thesis. One point belongs here, because it concerns the limits of my review of assisted code: a later audit found that the head-dropout value, though sampled and carried through the configuration, was consumed by no model and so never influenced training. It went unnoticed at review and is recorded here so that the search is not overstated.

The framework was new to this project. I supplied the assistant with the framework’s own documentation, pasted directly into the exchange, together with worked examples, and the assistant arranged the study code to fit the project’s existing configuration structures. That work covered the Bayesian tree-structured Parzen estimator sampler, the choice of a no-op pruner, whose reasoning is documented in the source, the conditional parameter

groups in which the augmentation regime governs which further parameters are sampled, and the exception handling that logs a single unusable configuration as a failed trial instead of ending the study. I checked the resulting search space and the frozen winners.

Representative prompts

“Here is how Optuna studies are written in these examples. Please set up the same kind of study for my runner so it uses my existing config structures and gives back the winning trial’s config.”

“Please set up conditional suggestions so the augmentation-regime choice controls which further parameters get sampled.”

“Which pruner makes sense if my objective never calls trial.report? Explain the reasoning rather than just picking one.”

“Please make one bad sampled config log as a failed trial instead of killing the study, and fall back to a no-augmentation config if every trial fails.”

Quoted verbatim from the retained log

“Is this standardized though? Because I recall we did some 120 optuno episodes for the A HPO, here it’s different? Is it alright the way it is now? The important part is to ensure there are reasons behind our choices and to standardize everything across all values and variables, so it makes sense, it is easily reproducible and that it is scientifically robust?”

The foundation-model adaptation study

I designed the fairness study that re-reads each pretrained foundation model under the preprocessing it was itself pretrained on, and specified the isolation rules that keep its output away from the main results. The assistant implemented the per-model native preprocessing table, extended the epoch-cache key to include it so these variants never reuse or overwrite the default cache, and built the low-rank adaptation path using the PEFT library against the targets I specified. A later audit found that one of those targets, the patch projection, matched no module in Neuro-GPT or LaBraM, so for those two models it adapted nothing and the low-rank arm was narrower than specified. This too went unnoticed at review, and it bears on how those results are read.

Representative prompts

“Please add a per-model preprocessing override table so each foundation model can be fed the sampling rate, band and amplitude scaling it was pretrained on, and make the epoch cache key include it.”

“Please add PEFT LoRA adapters on the target modules I’ve listed for each model and tell me which modules were adapted and which were skipped, and why.”

“Please write these results to their own directory and make it impossible for this study to overwrite the main results file.”

Quoted verbatim from the retained log

“This is great, but you’ve misjudged the pretrained models, as they were trained and tested on a harness that wasn’t their native setup when building their respective pretraining. Also, S-Mamba and iTransformer, maybe other models too, they performed fairly okay, what is the real probability of it not out shining TimesNet and EEGNet?”

Analysis, statistics and figures

I chose the statistical approach and, for each protocol, the matched unit on which the tests operate. The assistant wrote the implementation: the significance tests and their correction for multiple comparisons, the bootstrap confidence intervals and effect sizes, the metric panel, the calibration measures, the inference-cost measurement used in the cost table, and the Matplotlib figures. I specified what each figure had to show, and checked every reported number against the archived, provenance-stamped result artefacts rather than against any assistant's summary.

Representative prompts

“Which non-parametric tests should I use to compare several models across folds, and how do I apply a correction for multiple comparisons?”

“Please do Friedman plus pairwise Wilcoxon with Holm correction, and use fold means as the matched unit for Protocol B rather than raw fold-by-seed pairs. Explain the reason in a comment.”

“Please write the plotting code for a grouped bar chart of balanced accuracy per model with bootstrap 95% confidence intervals over seeds.”

“Please add expected calibration error, Brier score and negative log-likelihood to the metric panel, and plot a reliability diagram.”

Machine control integration

I was given an existing program that already drove the wheeled prototype machine, and established the velocity contract against that working reference driver as ground truth. I passed it to the assistant to adapt to this project and connect to it, so that the project environment could drive the machine. The assistant, under my guidance, merged that material into the project environment and mapped the five decoded direction labels onto the machine's movement commands: the translation of each label into a body velocity triple, the fixed-rate command thread, the stale-command watchdog that returns the machine to idle if no fresh command arrives, the Windows-side EtherCAT agent built on the pysoem library, and the simulation mode that prints the velocity triple instead of driving hardware so the mapping can be verified without powering the machine. I set the calibration constants and speed limits and specified the confidence gate that rejects low-confidence predictions to the Idle command.

Representative prompts

“Here is the working control script I was given for the machine. Please merge it into my project environment and map my five decoded direction labels onto its movement commands as I've specified.”

“Please add a watchdog that zeroes the velocity if no fresh command has arrived within the last few hundred milliseconds.”

“Give me a simulation mode that prints the velocity triple each key would send, so I can verify the mapping without powering the machine.”

Engineering support, testing and environment setup

The assistant wrote smoke tests and short throwaway scripts, contributed to the unit-test suite, and built the reduced-budget smoke modes that exercise the whole pipeline in minutes, including the mode that runs each model in its own subprocess so that a crashing model cannot block the rest. It also wrote the idempotent environment installer. I used it throughout for error diagnosis, on problems such as out-of-memory conditions,

dataloader worker behaviour under the Windows process-spawn model, an optional CUDA kernel dependency that has no official Windows wheel, and mismatched tensor shapes. I also used it to keep the repository README current as the project changed.

Representative prompts

“Please write a quick smoke test that runs one training step on a tiny synthetic batch and checks the output shape.”

“I’m getting this error, can you explain the likely cause and suggest a fix.”

“Please add a smoke mode that runs every model for one epoch in its own subprocess, so a crashing model cannot block the rest.”

“Please update the README so it matches the new benchmark command and folder layout.”

Quoted verbatim from the retained log

“Yes please apply. I’d like to avoid going myself in the code and making changes that are untracked by you. So set the necessary env vars as belt and suspenders, and the incremental results saving.”

“did the smoke test work properly? are we ready to launch the full pipeline of all the final steps? please give me again the 3 main commands to run the final processes, monitor them and quick sanity check them.”

“Can you please investigate the readme and tell me if what you’ve mentioned is all correct, as I believe what you said to be the best wording for it, but I’m neither sure it is the reality or if it is properly written as such in the readme?”

Documentation, presentation and this declaration

The assistant helped draft and refine the project’s written material: the model, data and preprocessing documentation, the recording-protocol and data-description documents, the results write-up and figure captions, the presentation, the project title, and the structure and wording of this appendix. A participant information and informed-consent form was also drafted with assistance, by reworking the recording protocol into a single participant-facing document. It was never used, and no part of it was put to anyone. It is recorded here only because the draft survives in the retained record. In each case I supplied the content and the claims; the assistant worked on organisation and phrasing, and I reviewed and rewrote the result.

Representative prompts

“Help me tighten the wording of this section and keep it accurate to what I actually did.”

“Please draft an appendix that honestly lists where AI was used in this project, with representative prompts for each area.”

Writing the thesis document

The thesis document was written over the same period as this declaration and after most of the work described above was complete. I used artificial intelligence assistants in writing it, and that use is declared here so that this appendix covers the whole of the work and not only the project behind it. The models are Claude Opus 5 and Claude Fable 5 (Anthropic), used both through the chat interface and through Claude Code, which I run inside Visual Studio Code alongside the LaTeX source of the thesis, so that an assistant can read and edit

that source directly. Two of the tools named above are used here as well: Perplexity, with ChatGPT-5.6 Terra (OpenAI) as the underlying model and including its Deep Research mode, and Google Gemini 3.1 Pro.

My use of the Claude tools is of three kinds. The first is grouping and consolidating material I have gathered from several sources into a single ordered account, so that I can see what I have before writing from it. The second is feedback and analysis: I put a section, an argument or a piece of my own reasoning to an assistant to be criticised, and to have angles I had not considered raised against it. The third is drafting and redrafting, in which I supply the content, the claims and the structure and use the assistant to produce and then clean successive versions of the text.

A fourth use is specific to Claude Code. Over the course of the project I had produced a body of working documents and analysis files recording the results and the reasoning behind them. I used Claude Code to carry that material into the thesis source cleanly and in the right places, so that those results and other established information could be shaped into the document faster than by retyping and reformatting them by hand. The material was already mine; what the tool contributed was placement and presentation. I double-checked and proofread what it produced against the documents it came from.

Most of the text was drafted from material I had prepared for the purpose throughout the project, my notes, analyses and working documents, under direction from me that set out in detail the design, the structure and the content each part was to follow, and then read, checked and where necessary rewritten by me. The figures were produced the same way. The drawn figures and plots were generated by scripts the coding assistant built from the archived records of my own recorded data, directed by a markdown record of those results and by the same notes and documents. The remainder are my own screenshots of the terminal and the dashboard software, and photographs taken by me or with the help of colleagues at the laboratory, prepared for print by removing their backgrounds, one with Google Gemini’s image model Nano Banana 2, run through Gemini 3.1 Pro, and edited manually in GIMP for further cleaning and the layering of assets, and one by a scripted whitening rule I defined. What the assistants contributed at this stage was drafting, placement and presentation; the substance was mine before the drafting began.

Perplexity’s Search mode, with ChatGPT-5.6 Terra (OpenAI) as the underlying model, was used to analyse sources. I also used its Deep Research mode to find sources more rapidly, to read through them, and for help with the structure of my citations and with the quality of the reference list.

Google Gemini 3.1 Pro, run with extended thinking, was used to correct the thesis text part by part: lines, sentences, words, paragraphs and whole sections. In the revision rounds that followed the drafting I reread and corrected every chapter, section, paragraph and line of the document myself, working alongside this tool to revise and refine what I rewrote, and returning detailed feedback to the Claude tools for the finer edits, so that no claim passed into the final text unexamined or unconfirmed by me. This is the same short, single-objective pattern described for that tool above, applied to prose rather than to code; as stated there, no record of these exchanges survives.

The content, the claims, the results and the conclusions of the thesis are mine in every case. No assistant supplies a fact, a number, a citation or a finding that I have not verified against the underlying material, and nothing generated by one enters the thesis without being read, checked against the underlying material and, where necessary, rewritten by me.

Representative prompts

“Can you please read through the analysis documents I’ve written and use the results in them to write this part of my thesis report. Please don’t type any number yourself, take them from the results files.”

“Please summarise these papers and tell me what each one actually measured, so I can see if it really backs the point I’m making.”

“Correct this text, make it easier to read, rework it minimally.”

Quoted verbatim from the retained log

“Before anything this is a scientific research and a report worthy of high quality science papers, this therefore means that it is NOT a story, though it can be structured following a red line”

“About the ‘thing you nearly got wrong’, you mention frozen probe scores, and make it sound like that was the highest accuracy, but wasn’t the best accuracy achieved with a fully fine tuned version of the model?”

“What do you recommend is the best practice for a clean and high quality scientific research paper we are trying to write? What is the most optimal course of action to be fully transparent and as professional as possible in regards with how we report each part and aspect?”

This part of the declaration is signed before the thesis is submitted, so it describes the use as it stands, with only the final corrections before submission remaining ahead of it. I am keeping a record of it on the same terms as the record described above, and will make it available to the examiners on request.

A.8 What artificial intelligence was not used for

The design and interpretation of the research

No assistant proposed the research question, chose the models or protocols, decided what to measure, or determined what the results mean. The scientific conclusions follow from the data and from my reading of it.

The data

No artificial intelligence took part in the recording of the EEG data. All recordings are genuine acquisitions from the dry-electrode headset with two consenting participants. No data was generated, synthesised or altered by an artificial intelligence at any point. I supplied no recording, participant identifier or other personal datum to an assistant deliberately. The coding assistants nevertheless worked inside the project directory, which holds the corpus, and in the course of ordinary work they read file names and, to check how a label had been written, individual headers. The signal itself was not something that work required and I have no evidence that any was read, but the material was within reach throughout, and for the sessions whose logs did not survive I cannot establish which files were opened. The recordings carry no identifier beyond the two-letter participant codes of Section 3.4, and on every account used I had disabled the retention of my content and its use for training. The augmentation applied during training is classical signal transformation implemented with NumPy and PyTorch, not generative modelling.

The project’s runtime behaviour

The project code calls no language model or external artificial-intelligence service at run time. The environment file contains one unused key, for a third-party documentation-lookup service rather than for any model provider, and no project code reads it.

Unverified numbers or citations

No result, number, figure or citation was taken on an assistant’s word: every result, number and figure is traceable to the archived, provenance-stamped result artefacts, and every citation to the work it names.

Automated tooling that is not a generative assistant

For completeness, the project also uses conventional automated machine-learning tooling that is not an artificial-intelligence writing or coding aid: the Optuna hyperparameter search described above, the PEFT low-rank adaptation used in the foundation-model study, a teacher-to-student knowledge distillation step in the supplementary improvement track, the automated inference-cost measurement described above, shell scripts that ran the benchmark unattended over several days, and the scripts that build the thesis document itself, which recompute every reported figure from the archived per-run records, generate the tables and figures from those records rather than from transcription, and run a verification pass that fails if any of them drifts out of agreement. These are disclosed here because they are automated, not because they authored anything.

A.9 A note on the version-control record

The repository’s earliest commits predate the thesis and belong to an earlier internship in the same laboratory; no part of the thesis proper was produced then. The thesis itself began in March 2026 and runs to September 2026. All work reported in the thesis, and all of the artificial-intelligence assistance declared here, falls within that period.

Two commits originally carried a trailer naming the coding assistant as co-author. When I later had to fix both, I rewrote and recommitted them myself, so the history as it stands carries no such trailer; the original trailers are preserved in the archived record. In either state the commit log does not measure the extent of the assistance described here, and this appendix, not the commit log, is the accurate account.

A.10 My responsibility

The account above is offered as an accurate and good-faith record of how I used artificial intelligence in this work. All artificial-intelligence-assisted code and text was investigated, directed and reviewed by me and, where adopted, adapted by me. I take full responsibility for the final work.

Appendix B

Model Cards, Data Card and Provenance

B.1 Model cards

One card per decoder: what it is, what it contributes to the comparison, and the property most likely to matter when reading its result. Measured cost is in Table 5.7 and accuracy in Table 5.1, and neither is repeated here.

FBCSP+LDA The classical baseline, and the reference point the deep decoders have to beat. Five Butterworth sub-bands from theta to gamma, one-versus-rest CSP at four components per band, log-variance features, LDA with automatic shrinkage. It has no learned parameters beyond its spatial filters and is deterministic, so a single fit per fold suffices and its Protocol-A SD is zero by construction rather than by luck, while the spread reported for Protocol B is fold-to-fold variation. It is trained on the same per-epoch normalised windows as every other decoder, and an intended exemption recorded in the code was never implemented (Section 3.6). *Limitation:* it cannot be run in the deployment pipeline, which requires a model with a tensor inference path.

EEGNet The purpose-built comparison: a compact convolutional network designed for EEG rather than adapted to it. A temporal convolution whose kernel spans half a second, a depthwise spatial one across the 19 channels, then a separable block. Implemented from the paper rather than taken from a repository. At 3 541 parameters it is the smallest decoder in the field, nearly 60 times lighter than the next in size, and the fastest by a factor of about five, and it is the model that becomes usable on the least calibration data (Section 5.5).

TimesNet A general time-series architecture whose benchmark suite includes classification, and the decoder with the highest mean on the deployment axis. It identifies dominant periods by Fourier transform and folds the series into a period-by-phase representation for two-dimensional convolution. *Note:* its 28.5 million parameters are a configuration consequence, not a reimplement. Its embedding is the sequence length times the model width, and all of it feeds one linear head. Its Fourier transform cannot run in half precision on a 600-sample axis, so that operation stays in fp32 when the rest of the model is cast.

iTransformer A forecasting transformer that inverts the usual arrangement: each electrode is a token whose embedding is its entire two-second series. *Limitation, stated by its own documentation:* no spatial inductive bias, since token identity carries no electrode position. It has the lowest expected calibration error of the nine decoders. Section 6.3 argues that its tokenisation is also why it is the decoder most disturbed by changes to the input.

S-Mamba A state-space forecasting model sharing iTransformer’s inverted tokenisation, with a bidirectional Mamba encoder in place of attention. It reaches within one and a half points of the leader at a 114th of its parameter count, which is the strongest efficiency result in the benchmark.

Neuro-GPT An EEG foundation model pretrained on some 5 700 hours of clinical recordings, and the best of the three here. Loaded encoder-only: the causal decoder and unembedder are discarded at load, since language-model features serve no purpose in classification. *Limitation:* its embedding is 40 dimensions wide, an order of magnitude narrower than any other decoder’s, so everything the pretraining learned must pass through that bottleneck. Its input expects 22 channels, so the 19 recorded ones are zero-padded.

CBraMod An EEG foundation model pretrained on over 9 000 of the 27 000 hours in a clinical archive, the largest corpus in the roster, and it does not finish best of the three, which is part of the RQ2 finding. Its encoder splits each layer between attention across channels and the same operation over time patches. *Limitation:* pretrained at 200 Hz, where its fixed patch spans one second, whereas at the deployment rate the same patch spans two thirds of a second, a compression fine-tuning cannot undo. It applies on the shared harness: the native-preprocessing runs behind its reported accuracy resample to 200 Hz and do not carry it.

LaBraM An EEG foundation model pretrained on roughly 2 500 hours, and the weakest EEG decoder in the benchmark. *The limitation that matters:* the checkpoint used is the community re-release rather than the weights published with the paper, which are behind an academic-licence request. It carries a four-class classification layer discarded on load, evidence that it is itself a fine-tuned artefact rather than the pretrained model as released. Two of its parameter tables arrived untrained: the electrode embedding (65 by 200) and the patch-order embedding (9 by 200) match a random initialisation on every test applied, with element-wise SDs of 0.0200 and 0.0192 against the 0.02 they were seeded with, a largest absolute value of 0.0811 (4.1 SDs, the expected maximum of that many Gaussian draws), and pairwise cosine similarity between electrode entries of 0.000 ± 0.071 against a chance spread of 0.071. The transformer body, by contrast, is trained (normalisation gains of 0.91 and 1.22 against 1.00 at initialisation). Measured directly from `labram-base.pth` in the archive and read in Section 6.2. It has CBraMod’s sampling-rate mismatch on the shared harness. A second deviation belongs to this port rather than to the checkpoint: the wrapper assigns the first 19 entries of the electrode table to the 19 recorded channels in device order, with no mapping from their names to electrode entries. With the table untrained this changes nothing, but it would have to be corrected before any re-run on weights whose table is trained. It is also the only decoder for which LoRA worked, because it is the one whose feed-forward layers the adapter targets actually matched.

Chronos The non-EEG negative control: a forecasting model pretrained on tens of billions of observations from domains such as energy, finance, transport and weather, and no physiological data at all. It encodes each channel sequentially as an independent univariate series, which is why it is the slowest decoder measured despite the timing budget being met. *Note:* the published model is usually described at around eight million parameters, whereas the loaded pipeline measures 46 million, and this thesis cites the measurement. It is the only decoder that can reach the network at run time, and it is deliberately kept offline. Its role is a methodology check: a model that should underperform every EEG-native decoder, and whose failure to do so would have indicted the evaluation rather than the models. Its Protocol-B and C rows were trained under a three-epoch cap (Section 4.5), while the Protocol-A row carries its full-budget, fully searched result.

B.2 Data card

A datasheet for the recorded dataset, in the spirit of Gebru and colleagues. Chapter 3 gives the narrative account, and this is the structured one.

Motivation Recorded specifically for this thesis. No public corpus covers five-class directional gait intent: the standard benchmarks (PhysioNet EEGMMIDB [62], BCI Competition IV-2a [68]) are limb-based imagery or execution, and the mobile locomotion sets [43] record visual paradigms while walking rather than gait direction. The task here is executed, full-body, directional movement, for which there is no off-the-shelf equivalent.

Instrument DSI-24 wireless dry-electrode headset, 19 scalp electrodes in the 10–20 arrangement and two ear-clip sensors applied as a linked-ears reference, at 300 Hz, 16-bit conversion, Bluetooth. The hardware reports four electrodes under pre-1991 labels, which a rename map in the configuration module translates.

Participants Two. The primary participant contributed all five classes and every training recording. The second contributed Forward and Backward only, is never trained on, and appears solely as the held-out person of Protocol C. Both are adults with no known neurological or motor condition, self-reported. Age, sex and clinical screening were not recorded.

Recordings Each is a fifteen-second clip of one intended movement, stored as EDF with the class in the file header. Two counts exist and are not interchangeable: 774 recordings were made and archived (664 and 110 by participant), all of which parse into the corpus, and 500 of the primary participant’s form the pool the benchmark trains on, after a cap of 100 per class. Protocols A and B evaluate within that pool, and Protocol C evaluates on all 110 recordings of the second participant. The corpus also holds two groups that Section 3.5 describes: an estimated 5 to 10 % of recordings made while the participant deliberately blinked or clenched the jaw, and 14 early files of the `test` task type, nine of which are in the pool. Table 3.1 gives the breakdown.

Label semantics Idle is standing or seated with no intended motion, and the four movement classes are physically performed walking or turning in the named direction. The critical framing: the participant *executes* the movement. This is movement execution, not motor imagery, so the recorded activity spans planning as well as the act itself, and the results are not comparable with hand-based MI benchmarks. Every claim in this thesis is framed within-subject.

Task types Twenty-one, listed in Section 3.5. The cap-100 selector takes one recording of every observed task type first and fills the remaining quota by seeded sampling, so no rare variant is lost, but a known imbalance survives: ordinary walking is 140 of the primary participant’s 211 Forward recordings, with 60 spread across 11 other gait styles and 11 early `test` files.

Collection protocol The recorder offers two modes: free-form, in which the operator marks the live stream from the keyboard and a majority rule then gives the file its single label, and timed trial, in which that value and the task type are set before the recording starts and the file is checked by duration. The free-form mode was built first, in the belief that live labelling was the right design, and it proved needlessly complicated in practice: the headset’s Bluetooth range was enough to cross the room and return to the laptop. Each recording was therefore started from the keyboard with its label and task type predefined, switched or repeated as the day’s recording went on. Every recording in the corpus is a timed trial, and the majority rule never ran on it. Section 3.3 gives the day’s routine and the physical protocol.

Splits All training draws from the primary participant. Protocol A is a per-class 80/10/10 split in a fixed seeded recording order, with a one-window gap at each boundary. Protocol B

is leave-recordings-out stratified group k -fold at $k = 5$, keyed on the individual recording, with an inner stratified-group split for early stopping. Protocol C trains on the primary participant and tests on the second plus a matched within-person reference, with an audit that raises if the held-out participant reaches training or validation. Fold integrity is verified: each carries all five classes, 2 400 test epochs balanced across them, zero recording leakage.

Known biases Single training participant, so no cross-subject claim is made and Protocol C establishes that none could be. Movement execution rather than imagery, limiting comparability with published benchmarks. No clinical verification: both participants healthy, and the eventual target population out of scope. Constant recording environment, so noise and interference are common to training and test and may inflate apparent consistency. A single right-handed participant, so any cortical asymmetry may not transfer. Section 3.3 catalogues the uncontrolled variation within sessions.

Ethics and distribution The second participant consented verbally, before recording, to thesis use. The recordings are coded, not anonymous: the two-letter codes are the only identifiers, no other personal information was recorded, and the link to the two people is known to the author. The dataset is not released, because sharing would require re-consent. Section 3.4 states the ethics position in full, including that no formal approval was sought and why.

External data Four public datasets were adapted and none was used. No reported result derives from external data, and no external-data run may be merged into a results file with thesis runs.

B.3 Model provenance and checkpoint integrity

Four decoders load pretrained weights. Their provenance is recorded here because a benchmark of pretrained models is only as interpretable as its knowledge of which artefact was actually loaded.

Checkpoint	Source	Size
neurogpt_weights.pth	wenhuic/Neuro-GPT	303 MiB
cbramod_base.pth	weighting666/CBraMod	19 MiB
labram-base.pth	braindecode/Labram-Braindecode	22 MiB

SHA-256 digests for all three are recorded in the project’s model documentation and independently in the archive’s checksum manifest, and the two agree. Chronos is not listed because it carries no vendored checkpoint: its weights are fetched from a model hub on first use, cache-first, and it is the only decoder in the roster that can reach the network at run time.

Loading behaviour, and why it is checked All three are loaded non-strictly, because none of the wrappers is a byte-identical reconstruction of its reference implementation. Non-strict loading is convenient and dangerous: it will populate two per cent of a model and return without complaint, and the resulting decoder would produce plausible, meaningless numbers. Each loader therefore computes the fraction of wrapper parameters the checkpoint actually populated and raises if fewer than 90 % land. This is what turns a successful load into a checked property, and it is the mechanism that would have caught a silent architecture drift. One further silent failure was screened for by construction: LaBraM’s reference implementation applies its temporal embedding by in-place slice assignment, which corrupts automatic differentiation under bfloat16 arithmetic and leaves the model stuck at chance with no error raised, and the wrapper replaces it with an out-of-place equivalent. It

is the class of bug that produces a plausible number rather than a crash, which is why it is recorded here.

Pretraining corpora Neuro-GPT on roughly 5 700 hours of clinical EEG, CBraMod on over 9 000 of the 27 000 in a hospital archive, LaBraM on roughly 2 500 across some 20 datasets, and Chronos on tens of billions of observations from 28 non-physiological sources. Section 5.4 notes that this ordering does not predict the results.

Best checkpoints The archive ships 11 trained checkpoints, one per model variant, each paired with a pointer file. The pointer is the interpretive layer: it records which run produced that checkpoint (protocol, fold, seed, the validation and test scores, the full hyperparameter and architecture configurations, and the commit), without which a bare weights file is an anonymous tensor dump. Two notes apply. The selected best is not uniformly from the deployment protocol, since for several models the highest-scoring single run came from Protocol A, and the pointers carry an earlier commit than the canonical results files, together with absolute paths from the machine they were written on.

B.4 Deviations from published models and third-party code

Deviations Four decoders differ from the model as published, and each difference is stated where its result is reported as well as here. Neuro-GPT is loaded encoder-only, discarding the causal decoder and unembedder, with 19 channels zero-padded to the 22 its input expects. LaBraM uses the community checkpoint rather than the paper’s weights, with parameter names remapped to match it and a four-class classification layer discarded on load. CBraMod and LaBraM both run, on the shared harness, at a sampling rate half again their pretraining frequency, so their fixed patches span two thirds of the duration they were trained on there, whereas their reported accuracies come from the native rate. TimesNet’s parameter count follows from a configuration choice rather than a reimplementaion. The LoRA arm of Section 5.3 targeted a patch-projection module that neither LaBraM nor Neuro-GPT defines, and attention and feed-forward names that match none of CBraMod’s layers, so what each received was decided by which remaining targets matched.

Third-party code Of the nine decoders, two are original implementations: the classical baseline and EEGNet were each written from their papers rather than taken from a repository. The classification head and the harness around every model are original throughout.

Three model bodies (iTransformer, TimesNet and S-Mamba) are imported unmodified from a bundled library, the codebase behind the forecasting benchmark that supplies them [13], and only the configuration wrapper and the tensor reshape are original. That library carries an MIT licence. Three further bodies are ported: CBraMod’s encoder verbatim under Apache-2.0, and Neuro-GPT’s and LaBraM’s almost so, from their reference implementations. Chronos is a package dependency with no vendored source.

Two corrections to the project’s own documentation belong here, because this section is where a reader would check. The documentation states that two model files contain ported code, but it is three, LaBraM’s being recorded only in that file’s own docstring. It also records a licence for CBraMod alone, describing the bundled library as merely “used unmodified”, although it does ship an MIT licence, which is stated above. Three of the nine (Neuro-GPT, LaBraM and Chronos) have documented third-party provenance with no licence recorded anywhere in the project. Their upstream repositories carry licences, so the omission is in this project’s records, not in the upstream, and it is noted here rather than left to be discovered.

Appendix C

Supplementary Results

C.1 Protocol C by regime and the per-class tables

Chapter 5 reports the full metric panels for Protocols A and B, where they carry the argument. The breakdown of Protocol C by adaptation regime and the per-class tables are given here.

Table C.1 breaks Protocol C down by adaptation regime for the three FMs. It is the sharpest form of the finding in Section 5.8: as the within-person reference climbs from the frozen probe through LoRA to full fine-tuning, cross-person transfer does not follow it up, and for one model it moves down. Reading the transfer column alone would suggest the frozen probes carry over best, whereas reading it against the reference shows that they merely have less to lose.

Tables C.2 and C.3 complete the per-class view of the deployment axis whose recall member is Table 5.8. All three derive from the same stored confusion matrices, whose orientation the table generator verifies against the stored recall values before deriving anything. Section 5.7 reads the safety-relevant cells, and the full grids are here so that no per-class claim rests on an unpublished number.

C.2 Hyperparameter search space and winning configurations

Each of the eight searched decoders received 120 trials at 20 epochs, sampled by a tree-structured Parzen estimator against validation balanced accuracy, with no pruner. The space varied the learning rate over $[10^{-4}, 5 \times 10^{-3}]$ logarithmically, weight decay over $[10^{-5}, 10^{-2}]$ on the same scale, the mixing probabilities and their Beta parameter, the class-

Table C.1: Cross-person transfer for each adaptation regime, with the within-person reference the same run achieved.

Foundation model	Regime	Within-person reference	Cross-person transfer	Drop
Neuro-GPT	frozen probe	59.6	63.0	-3.5
	LoRA	59.2	63.0	-3.9
	full fine-tune	87.1	43.0	44.1
CBraMod	frozen probe	60.7	60.4	0.3
	LoRA	60.2	59.0	1.2
	full fine-tune	81.5	51.9	29.6
LaBraM	frozen probe	55.8	57.1	-1.4
	LoRA	71.7	54.9	16.8
	full fine-tune	81.3	49.1	32.1

Full fine-tuning is the best regime everywhere it matters, and by a wide margin: it is the only one that learns the task at all, reaching 87.1, 81.5 and 81.3 % within-person here against 59.6, 60.7 and 55.8 % for the frozen probe, and the same ordering holds on Protocols A and B (Table 5.5). The single column in which the frozen probe reads higher is cross-person transfer, and that is an artefact rather than a result, which is why the two accuracy columns must always be read together. The frozen probe sits near chance in *both* directions, 59.6 % within-person and 63.0 % across, so it has learned nothing that could fail to transfer. The fine-tuned model learns the participant well and then reaches 43.0 % on a new one, *below* the 50 % chance level: what it learned is not merely person-specific but actively misleading applied to someone else. That is a stronger statement of the person barrier than the frozen numbers could support. No regime achieves genuine cross-person generalisation.

Table C.2: Per-class specificity on Protocol B, averaged over folds and seeds.

Decoder	Idle	Forward	Backward	Left	Right
TimesNet	0.975	0.963	0.951	0.966	0.964
S-Mamba	0.959	0.959	0.950	0.969	0.963
Neuro-GPT	0.968	0.959	0.953	0.961	0.958
EEGNet	0.962	0.952	0.939	0.972	0.973
iTransformer	0.959	0.953	0.951	0.962	0.962
FBCSP+LDA	0.978	0.965	0.947	0.941	0.955
CBraMod	0.973	0.937	0.945	0.953	0.944
LaBraM	0.947	0.926	0.910	0.884	0.901
Chronos	0.791	0.945	0.762	0.943	0.800

Per-class specificity (how rarely a decoder issues this command when the truth is anything else), computed from the same stored confusion matrices as Table 5.8, whose orientation is verified against the stored recall values at build time. Complements the recall table: recall is usability, specificity is safety, precision is trust (Section 4.8).

Table C.3: Per-class precision on Protocol B, averaged over folds and seeds.

Decoder	Idle	Forward	Backward	Left	Right
TimesNet	0.896	0.854	0.817	0.866	0.861
S-Mamba	0.849	0.832	0.811	0.868	0.849
Neuro-GPT	0.874	0.837	0.816	0.840	0.838
EEGNet	0.863	0.815	0.783	0.883	0.886
iTransformer	0.847	0.815	0.806	0.841	0.844
FBCSP+LDA	0.908	0.855	0.800	0.776	0.820
CBraMod	0.889	0.772	0.791	0.794	0.787
LaBraM	0.798	0.691	0.651	0.523	0.614
Chronos	0.490	0.211	0.390	0.162	0.426

Per-class precision (how often an issued command names the true class), computed from the same stored confusion matrices as Table 5.8, whose orientation is verified against the stored recall values at build time. Complements the recall table: recall is usability, specificity is safety, precision is trust (Section 4.8).

imbalance strategy over four mutually exclusive options, and the augmentation regime over four, with conditional groups, so that choice governs which further parameters are sampled at all. Section C.3 gives the transforms those regimes draw on.

Table C.4 lists the winning configuration per decoder. Section 4.5 explains which of them also ran on the later protocols and which two decoders departed from that convention.

The table records where eight individual searches happened to terminate, which is not the same as what the search space rewarded, and reading it alone would suggest that augmentation is optional. Section 4.5 sets the distribution over the strongest trials against it and draws the consequence for how each decoder is tuned.

Head dropout was in the space and is deliberately absent from the table: it was sampled and carried through the configuration but never reached a model, so printing it as a tuned hyperparameter would misstate what the search controlled.

C.3 The augmentation library

Twenty-three per-sample transforms exist in the library, of which 18 are exposed to the search, and Table C.5 gives them with the range each magnitude value maps to. Five are withheld deliberately. They are diagnostics rather than plausible perturbations: segment permutation, for instance, destroys temporal dependence while preserving the spectrum, which makes it an instrument for asking what a model relies on rather than a way of making more training data.

Table C.4: Winning hyperparameter configuration per decoder, from 120 Optuna trials against the Protocol-A objective.

Decoder	Learning rate	Weight decay	Imbalance	Augmentation	MixUp p	CutMix p	α
TimesNet	1.78e−4	2.08e−3	none	multi lambda	0.37	0.24	0.48
S-Mamba	1.12e−3	3.83e−5	weighted loss	multi lambda	0.28	0.09	0.10
EEGNet	2.14e−3	7.17e−3	targeted aug	none	0.03	0.16	0.27
iTransformer	9.88e−4	3.84e−4	weighted loss	multi lambda	0.46	0.11	0.26
CBraMod	3.84e−4	8.86e−4	targeted aug	none	0.35	0.01	0.49
Neuro-GPT	4.98e−3	1.25e−5	balanced sampler	single	0.11	0.06	0.33
LaBraM	1.18e−3	8.81e−5	balanced sampler	single	0.26	0.04	0.35
Chronos	4.98e−3	3.47e−4	weighted loss	pairs	0.34	0.18	0.29

These are the Protocol-A winners, and for six of the eight decoders listed they are also the configuration that ran on Protocols B and C, by three routes. TimesNet, S-Mamba and iTransformer each received a Protocol-B search whose winner scored lower, so the Protocol-A configuration was kept. CBraMod and LaBraM had no Protocol-B search and reuse a configuration found with their encoders frozen. Chronos had none either and ran under a three-epoch training *cap* on the later protocols, a budget cap, not a reduced search. The two exceptions are EEGNet, which adopted the better configuration from its own Protocol-B search, and Neuro-GPT, which was searched again under the native fine-tuning harness. FBCSP+LDA is omitted from the table: it is a classical fit with no search. §4.5 gives every case in full. **Head dropout is deliberately not listed.** The search sampled it and the per-run records log it, but the value is never applied on the path these results came from (every reported model ran with the classifier head’s default of zero), so reporting it as tuned would misdescribe the search. The winners table should be read alongside the frequency analysis in §C.2: the winners split evenly across augmentation regimes, while the distribution over the strongest 10% of trials does not.

The design problem this table solves is that the search samples *one* magnitude and applies it to whichever transforms a regime selects, so the parameter must mean the same everywhere. In 16 rows it scales a perturbation upward from nothing. In two it cannot: a tighter amplitude clip and a smaller minimum crop are both stronger interventions as their parameter falls, so those mappings run backwards. Without that inversion the search would be sampling “stronger” in 16 dimensions and “weaker” in 2 simultaneously.

Section 4.7 gives the batch-level mixing and the argument that CutMix must mask a contiguous slice of time rather than mixing across channels.

C.4 Window-geometry probe

The two-second window at a half-second stride is an operating point, not a tuned parameter: it was chosen because it matches the decode cadence the live pipeline needs. This control asks whether that choice matters, by re-running every decoder at a coarser four-second window on a one-second stride.

As a robustness check the answer is favourable for EEGNet and TimesNet and unfavourable for S-Mamba and iTransformer, and Figure C.1 shows the split. EEGNet is slightly *better* at the coarser geometry and TimesNet only mildly worse, while S-Mamba and iTransformer lose enough to fall below the two-way direction floor. The operating point is therefore not a lucky choice for the leading decoders, but it is load-bearing for the two channel-token architectures.

That is a robustness result, and it would be the whole of this section were it not that the same split appears under an unrelated manipulation. Section 6.3 treats the two together, because disturbances with nothing in common that break the same pair of decoders say more about architecture than either says alone. A reduced-budget control: compare within the figure, not against the headline table.

C.5 Numerical precision: fp32 versus fp16

Half precision was applied after training, with no recalibration, and measured on the within-subject protocol. Table C.6 gives the result, which is mostly a negative one and useful for

Table C.5: The augmentation registry: every transform the hyperparameter search could sample, with the value its magnitude parameter maps to at each end of its range.

Key	Transform	At $m = 0$	At $m = 1$	Parameter
<code>gaussian</code>	Gaussian noise	0.02	0.20	std
<code>band</code>	Band-limited noise	0.02	0.18	std, 8–13 Hz
<code>fractal</code>	Fractal noise	0.02	0.15	std, $\alpha = 1$
<code>amp_scale</code>	Amplitude scaling	1.00	0.60–1.40	gain
<code>sign_inv</code>	Channel sign inversion	0.05	0.50	p per channel
<code>dc_shift</code>	DC shift	0.05	1.00	max shift
<code>tdrop</code>	Temporal dropout	0.05	0.20	max fraction, 1–3 segments
<code>cdrop</code>	Channel dropout	1	4	channels
<code>sat_clip</code>	Saturation clip	4.5	2.0	threshold (<i>inverted</i>)
<code>twarp</code>	Time warping	1.00	0.80–1.20	rate
<code>rcrop</code>	Random crop	0.95	0.65	min. fraction (<i>inverted</i>)
<code>freq_shift</code>	Frequency shift	0.3	3.0	Hz
<code>poly</code>	Polynomial approximation	0.05	0.50	mix, degree 8
<code>mag_warp</code>	Magnitude warp	0.05	0.30	σ , 4 knots
<code>time_jitter</code>	Time jitter	0.005	0.04	σ , 8 knots
<code>win_warp</code>	Window warp	0.10	0.30	window fraction
<code>tshift</code>	Time shift	0.02	0.15	max fraction
<code>mul_noise</code>	Multiplicative noise	0.02	0.12	σ

Magnitude is a single value on the unit interval, sampled by the search and shared across whichever transforms a regime selects, so it must mean the same thing in every row. In sixteen of the eighteen it scales a perturbation upward from nothing. In the two marked *inverted* the parameter falls as the perturbation strengthens (a tighter amplitude clip and a smaller minimum crop are both more aggressive), so their mappings run backwards and the convention holds throughout. A further five transforms exist in the library (ThresholdSilence, SegmentPermutation, SignalConvolution, ShapeDistortion, ActiveContourSmooth) and are deliberately withheld from the search: they are diagnostics rather than plausible perturbations of a recording. The registry is transcribed from the project’s augmentation library, which travels with the archive as a git bundle (Appendix D).

Table C.6: Single against half precision, applied after training.

Decoder	acc. fp32	acc. fp16	Δ acc.	lat. fp32	lat. fp16	VRAM fp32	VRAM fp16
EEGNet	84.1	84.1	-0.017	0.1	0.2	9.9	9.5
TimesNet	88.8	88.8	0.000	27.9	30.9	134.6	85.8
iTransformer	84.0	84.0	0.000	0.7	0.7	10.1	9.6
S-Mamba	84.7	84.7	-0.034	1.8	2.0	10.3	9.7

Half precision costs essentially no accuracy, the largest change being 0.034 of a point, but it does not buy latency: three of the four decoders become slower. At this model size kernel-dispatch overhead dominates and there are no large matrix multiplications to accelerate. The real gain is memory, and only on the large model. Accuracy is Protocol A, lg-only.

that reason: accuracy costs essentially nothing, and *latency does not improve*, with three of the four decoders tested becoming slower. At this model size the time goes to kernel dispatch rather than to arithmetic, and there are no large matrix multiplications for the reduced precision to accelerate. The gain is memory, and only on the largest model.

One architecture-specific constraint emerged and is worth recording, because it would surprise anyone repeating this. TimesNet’s Fourier transform runs over a 600-sample axis, and 600 is not a power of two, which the CUDA library rejects in half precision. That operation therefore stays in single precision while the rest of the model is cast, which is part of why its latency does not fall.

C.6 Confidence-gate coverage curve

Section 5.6 quotes the gate at its deployed threshold. Figure C.2 gives the whole trade-off, sweeping the threshold from accepting everything to almost nothing, so that the chosen operating point can be seen in the context of the alternatives rather than asserted.

The curve is smooth and has no elbow, which is itself the finding: there is no threshold

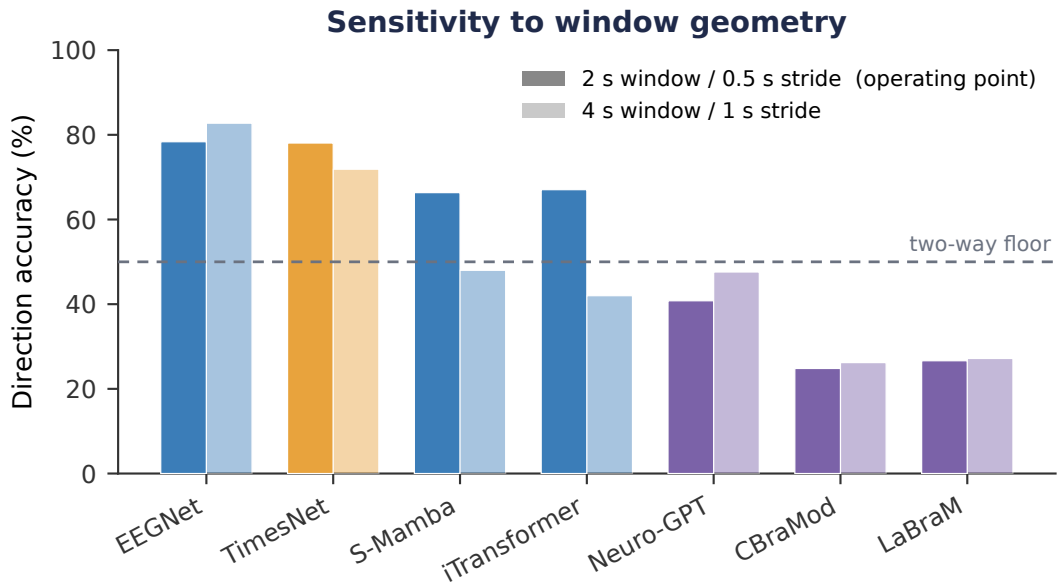


Figure C.1: Direction accuracy at the 2s/0.5s operating point against a coarser 4s/1s alternative. A reduced-budget control: compare within the figure, not against the headline table.

at which the gate suddenly starts or stops paying, so the operating point is a judgement about how much coverage a clinical application can afford to lose rather than a value the data selects. At the deployed threshold roughly three quarters of windows are accepted and accuracy on them rises by more than eight points. Tightening further continues to buy accuracy at a steepening cost in coverage. Since a rejected window means the device does nothing, coverage is not free: a system that rejects half of all windows cannot drive the device.

The curve is plotted for the recommended decoder alone, and the reason is provenance rather than preference. The underlying sweep was computed on the shared harness, where the three FMs appear as frozen probes, so their curves would describe a configuration this thesis does not report. The recommended decoder is not an FM, so its curve is unaffected.

C.7 The improvement track as a negative result

A separate track attempted to improve the decoders for deployment by applying four levers on top of the tuned configurations: a focal loss, class balancing, augmentation targeted at the Backward and Left classes, and temperature scaling of the output. It produced a decisive negative result. Every levered decoder was worse than its own baseline, and the higher-capacity models degraded severely: the loss was roughly monotone in how much regularisation a model could absorb. Calibration worsened too, despite temperature scaling being one of the levers.

The run's numbers are not cited here. Its outputs are quarantined in the archive and classified obsolete, for the reason that makes the result interesting rather than merely disappointing.

The levers were designed against the behaviour of the *broken* Protocol B: the apparent collapse of one class, and the severe imbalance that the degenerate session-grouped folds produced (Section 4.4). Both were artefacts of the grouping variable. The corrected recording-grouped protocol is balanced by construction (every fold carries all five classes in equal proportion), so remedies aimed at rescuing a starved class have nothing to rescue and over-regularise the model instead. Targeting augmentation at two specific classes on a

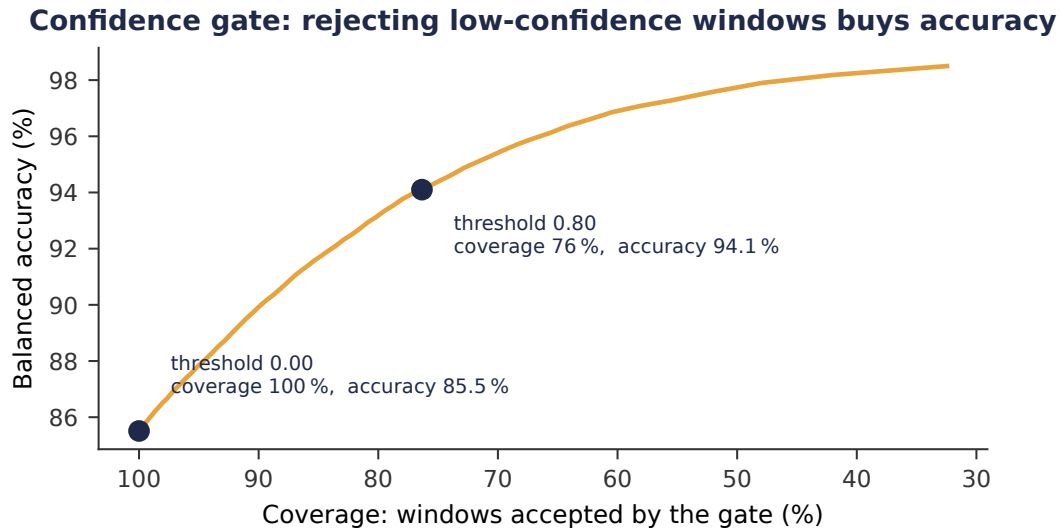


Figure C.2: Accuracy against coverage for the recommended decoder as the confidence threshold varies. Shown for the recommended decoder alone: the underlying sweep was computed on the shared harness, where the three FMs are frozen probes, and the recommended decoder is not one of them.

balanced split does not help them, and in the worst case does harm, while costing accuracy on the classes it was taking capacity from.

This is the clearest second-order consequence of the split repair in the project, and the reason Section 6.4 states it as a transferable lesson: a remedy fitted to an artefact becomes a liability once the fault is removed. It is also why the numbers stay quarantined. They are a valid measurement of a configuration that was designed for a protocol which no longer exists, and reporting them beside the corrected results, even as a comparison, would invite exactly the reading the repair was meant to prevent. What survives is the mechanism, which does not depend on the figures.

Appendix D

Reproducibility

D.1 The archived results record

Every number in this thesis comes from a single archive of 4060 files, organised so that a reader can go from a printed figure back to the run that produced it. Its curated portion holds 1186 files in 11 numbered directories: the assembled benchmark, the raw per-protocol training rows, the foundation-model adaptation studies, the characterisation studies (learning curve, spectral and spatial controls, window probe, precision, gate curve), the measured inference costs and the shipped checkpoint, the best checkpoint per model, 321 figures, the defence deck, the narrative documents, the reproduction sheets, and the 774 recordings themselves.

The single source of truth for any reported value is the file `final_benchmark.json` in `final/01_benchmark/`. It holds one block per decoder, each carrying the Protocol-A and Protocol-B means with their SDs, run and fold counts, confidence intervals and full metric panels, the Protocol-C transfer and reference under both harnesses, the measured cost and a comparability flag. For Protocol B it also holds every competing configuration that was considered, so that the reported figure can be seen to be the best of its candidates rather than a selection. Its header records the data version, the generating commit, the rule system, and a `pending` list which is empty: that emptiness is the completeness gate, and a non-empty list means the benchmark was assembled before every study had finished.

One property of the archive warrants stating because it looks like untidiness and is not. The archive retains its own superseded artefacts. A pre-repair results backup, an earlier unified table whose Protocol-C column predates the canonical run, analysis reports computed on the broken split, and a summary built on the frozen-probe harness are all still present, behind explicit banners. They are deliberately not cited anywhere in this thesis, and `docs_analysis_final/evidence_map.md` §7 is the authoritative list of what must never be cited, with the reason for each. Deleting them would have removed the evidence: a correction whose prior state has been erased cannot be checked. The same file records the retired framings (the “calibration cliff”, the collapsing class, the claim that no decoder reached usable accuracy), which arose from the confound described in Section 4.4 and must not be reintroduced.

D.2 Environment, hardware and version stamps

Every invocation of the benchmark wrote a run manifest, and 119 of them are archived. Each records the command line, the working directory, a timestamp, the full git state (commit, branch, the list of modified files and a diff summary), and a complete package freeze running to 234 entries. The environment is uniform across every run that produced a reported number:

Python	3.12.13	NumPy	2.4.4
PyTorch	2.11.0+cu128	SciPy	1.17.1
CUDA	12.8	scikit-learn	1.8.0
cuDNN	9.19	MNE	1.12.1
Matplotlib	3.10.9	Optuna	4.8.0

All training and inference ran on one machine: an NVIDIA RTX 4080 with 16 GB of memory and compute capability 8.9, 32 logical CPU cores, 32.8 GB of system RAM, under Linux. Two gaps remain. The manifests record the CPU core count but leave the processor model empty, so no CPU is identified anywhere in the archive, and a handful of early exploratory runs used a different CUDA build, though none of them produced a reported number.

Two stamps make a result row traceable. Every row carries the `git_commit` it was produced under: this holds without exception across every results file. The data version, `a=lg_only/b=cross_recording/stratgroup/k5/v1`, is carried once per file in its header rather than per row, and it names the Protocol-A test population, the Protocol-B grouping variable, the splitter, the fold count and a revision, so a file can be matched to the split convention that produced it and a pre-repair file is immediately distinguishable from a corrected one.

Four commits appear across the archive, and the reason is worth stating rather than smoothing over. The archive itself is sealed at `85d02652`, which is what `repo.bundle` carries. The assembled benchmark was generated at `0efea56`. The Protocol-B, Protocol-C and foundation-model rows were produced at `659a39d`, and the frozen Protocol-A anchor at `aa7f3f9`. One file, the merged rebuild, contains rows from three of those commits, because it is precisely a merge of the frozen anchor with later retrained rows. A single commit across the whole archive would have been cleaner but would have required re-running work that had not changed.

D.3 Reproducing the benchmark

The archive is self-contained: it carries the code as a git bundle alongside the data, so the project can be rebuilt from the archive alone with no network access and no other repository. The same codebase is also kept in the project’s repository [26].

```
git clone final/10_reproduce/repo.bundle EEGxAi2GaitRehab
cd EEGxAi2GaitRehab
mkdir -p data && cp -r ../final/11_raw_data/manual data/manual
source setup_env.sh
bash scripts/finalize_run.sh
```

The bundle carries the full history to `85d02652`, and all four of its refs sit at that commit. The second line is the reason the recordings are archived under `11_raw_data` as well as being checksummed: they are the one input that cannot be regenerated.

Beyond the recovery path, the reproduction sheets record the pipeline as an ordered sequence rather than a single command, because several stages take days. Training runs first, one study at a time, each writing incrementally so that an interrupted run resumes rather than restarts. A provenance guard is run after each, which exits non-zero if a results file is stale with respect to the code that claims to have produced it. The assembled benchmark is then rebuilt, and its `pending` list must come back empty. Analysis follows as a single pipeline of 11 steps (among them the merged rebuild, the within-category back-fill, the precision study, the statistics, the gate curve, the crown selection, the tables and the figures), none of which trains anything, so it can be re-run freely after any change. A final provenance sweep covers every results file at once.

Timings are recorded only where they were long enough to matter. The foundation-model five-fold bridge takes two to three hours. The full benchmark across all models takes 10 to 16 days on the hardware above, which is why CBraMod and LaBraM received no Protocol-B search of their own and reuse the configuration found for them on Protocol A (Section 4.5). Everything downstream of training is analysis-only and fast, and no wall-clock figure is recorded for it.

A note for anyone re-running this: the FMs and the control pull weights from a model hub on first load, and the reproduction sheets set that hub to offline mode throughout. This

is not a preference. With network access enabled and no cached weights, the runs hang rather than fail, which is a considerably worse failure mode.

D.4 How this document is built

This thesis claims that its tables match the archive, and this section describes the chain that makes the claim checkable: no benchmark value in this document was transcribed by hand.

Every table carrying a measured value, and every figure, is generated from the archive at build time by a set of read-only tools, which never modify anything they read. The chain has five links.

`canonical.py` computes each reported aggregate from the raw per-run rows using exact rational arithmetic, and rounds only once. This is not fastidiousness for its own sake: the archive stores its aggregates to two decimals, and deriving the one-decimal figure this thesis reports from that stored value would be a second rounding. Three cells change as a result, and are recorded as changed.

`build_tables.py` emits 17 of the 18 tables (the protocol overview is written by hand from design facts) together with a layer of macros, one per canonical quantity. Prose uses those macros rather than digits, so a sentence and the table beside it carry the same object, and a figure quoted in the text cannot drift from the table it came from because they are the same value.

`make_figures.py` emits 19 vector figures, and the raster images that are not generated are two photographs and the screenshots of the running software. Every plotted value is read from the canonical source at build time rather than transcribed, and where a value also exists in an independently stored form the figure asserts the two against each other as it draws (the ranking figure checks its exact means against the assembled benchmark, and the direction figures are built on a formula validated against every model whose stored values permit the check). A figure whose data has moved therefore fails the build rather than being published. The two schematics carry no measured values by design.

`verify.py` then works backwards: it re-parses the emitted $\text{\LaTeX}$ and compares the cells back against the archive, checks that every cross-reference resolves, and confirms that no asset in the document comes from one of the superseded artefacts of Section D.1.

`safety_pass.py` is the adversarial pass, and it assumes nothing. It accepts neither the assembled benchmark, nor the generated tables, nor any document in the project as given. For every reported value it returns to the per-run rows, recomputes the aggregate, and compares along the whole chain, raw runs to assembled benchmark to printed table, reporting any disagreement. Around that sit checks on the claims that are not numbers: that chance levels are stated correctly, that a decoder’s row describes one harness rather than two, that no retired framing has crept back, that the corrections found by earlier audits have not reappeared, that the declaration page transcribed in the front matter still matches the signed clause, and that no reported percentage in the prose was typed in place of its macro. The number of checks is printed when it runs and is not quoted here, because it grows whenever a new class of error is found and a stale count in this appendix would be precisely the drift the pass exists to catch.

One tool exists for a specific correction. The archive’s own Protocol-A statistical report was computed before the FMs were re-evaluated under their native harness, so it describes the frozen-probe field. `build_native_a_stats.py` rebuilds those statistics on the fair field, and its output, not the archived report, is what this thesis cites.

D.5 Artefact integrity

Two records support verification of the archive. A checksum manifest covers 4060 files (everything in the archive except itself) as SHA-256 digests against archive-relative paths, in the standard format, so the whole archive can be checked with one command. Separately,

`final/MANIFEST.json` records 401 entries for the curated portion, and each carries not only the size and digest but the *source path the file was copied from* in the working repository, together with its data-version stamp where it has one. That source field is what makes the curated archive traceable rather than merely intact: a reader can see which working artefact each archived one came from. Two of the 401 entries are directory aggregates rather than files (the hyperparameter analysis and the 774 recordings), whose per-file digests live in the checksum manifest instead. The list of artefacts that were expected but absent at assembly time is empty.

One discrepancy is stated here rather than left to be discovered. Verifying the archive against its checksum manifest today reports 24 entries that cannot be read. All 24 point into a directory of reference literature that was relocated one level up after the archive was sealed, and the files still exist under the new path. The remaining 4 036 verify in place, and there are zero checksum mismatches: no archived file’s contents have changed. No result, figure, checkpoint, statistic or recording is affected, and the curated portion is intact in full. The manifest was deliberately left as sealed rather than regenerated, so that the discrepancy remains visible and this paragraph can be checked against it.

D.6 Project chronology

The thesis ran from March to September 2026. The phases below are recorded as provenance (which part of the archive came from when), not as an account of the work, which Chapters 3 to 5 give in the order the argument requires rather than the sequence events occurred in.

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|----------|--|
| 1 | Recovery of the acquisition prototype built during a preceding internship in the same laboratory: the headset link, the Bluetooth bridge and the streaming code brought back into working order. |
| 2 | The acquisition software proper (the interface, the recorder, the live dashboard) and the first implementations of the model wrappers and the augmentation library. |
| 3 | Recording. The protocol, the task types and the naming convention were settled during this phase rather than before it, and the dashboard and recorder were revised repeatedly in response to what recording sessions exposed. |
| 4 | Model training. The augmentation library was completed, the external public datasets were adapted and then set aside, and the full Protocol-A benchmark was run: 12 days of continuous computation at 120 search trials per decoder. |
| 5 | Machine integration: the control scripts adapted to this codebase, the live decoding pipeline wired to the dashboard, and the whole system exercised end to end. It was during this phase that the original Protocol B was found to be contaminated (Section 4.4). |
| 6 | The repair and the analysis: Protocol B re-run under recording-grouped folds, then the statistics, the figures and the metric panels, to a stable and archivable state. |
| 7 | Writing this document. |
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Two observations are worth recording alongside that table. The first is that the project as built diverges substantially from the one proposed: the original plan named ICA, k -means segmentation, a neural compression codec, a time-series language model for explainable classification, and generative validation. None of those is in the final system, and the reasons are given where each would have appeared: ICA in Section 3.6, the external datasets in Section 3.5. The second is that the single largest correction, the Protocol-B repair, was found in phase five rather than four, which is to say after the benchmark had already been run once. The retired numbers in Section 4.4 are the output of that first run.